Outline

QuantERA Call 2025: Research Proposal Outline

Working Title: Scalable, Robust, and Explainable Quantum AI for Grand Scientific Challenges
Topic: Quantum Phenomena and Resources (QPR)

1. Consortium Structure & Synergistic Roles

Our consortium addresses the foundational bottlenecks of Quantum Machine Learning (QML)—Scalability, Trainability, and Robustness—through a "Full-Stack" algorithmic and theoretical co-design.

Role	Partner	Core Expertise & Contribution (QPR Focus)
Algorithm Theory & Design	**Korea (SNU/Yonsei)** (Prof. Cha & Prof. Yoo)	**Foundational QML:** Theory of Multi-chip Entanglement, Quantum Transformers, Quantum Diffusion Dynamics. **Scientific Testbeds:** Computational Neuroscience (fMRI) & High-Energy Physics (CERN).
Quantum Optimization Theory	**Italy (Naples)** (Prof. Acampora)	**Optimization Physics:** Theoretical bounds of Gradient-Free Quantum Optimization. **Hybrid Theory:** Mathematical frameworks for Fuzzy-Quantum logic integration. **Security Theory:** Quantum adversarial robustness & forensic analysis.

Robustness & Benchmarking	**Germany (Fraunhofer IKS)** (PD Dr. habil. Lorenz)	**Reliable Quantum AI:** Theory of noise resilience and generalization bounds in hybrid models. **Benchmarking:** Rigorous evaluation frameworks for quantum advantage in industrial/medical contexts. **Medical QML:** Specialized quantum algorithms for medical imaging analysis.

2. Scientific Goals & Methodology (Excellence: 50%)

Goal 1: Foundational Breakthroughs in Scalable Quantum Architectures

- **1.1. Theory of Multi-Chip Quantum Ensembles (Korea & Germany):**
  - **Objective:** Establish the theoretical limits of **distributed quantum entanglement**.
  - **Method:** Deriving entanglement entropy bounds for "Divide-and-Conquer" circuits. Dr. Lorenz will contribute by analyzing the **generalization bounds** of these ensemble models compared to monolithic ones, proving their theoretical validity.
- **1.2. Physics of Quantum Optimization (Italy & Korea):**
  - **Objective:** Solve the "Barren Plateau" phenomenon from a theoretical standpoint using **Gradient-Free** dynamics.
  - **Method:** Developing **Quantum Evolutionary Algorithms** and adapting the **Forward-Forward** learning paradigm.
  - **QAOA Fine-Tuning Theory:** Formulating a hybrid protocol where classical embeddings are mapped to quantum Hamiltonians, optimized via QAOA.
- **1.3 Quantum State Space Model Development (Korea, Germany, Italy):**
  - **Objective**: Develop Quantum State Space Models that can efficiently learn long range spatio-temporal dynamics.
  - **Method**: Develop **Quantum Mamba, Quantum Hydra** algorithms using Quantum Singular Value Transformation (**QSVT**) and Linear Combination of Unitaries (**LCU**).
- **1.4. Theory of Quantum Diffusion (Korea & Italy):**
  - **Objective:** Reformulate **Quantum Noise** (usually a defect) as a **Generative Resource**.
  - **Method:** Developing "Hardware-Adaptive" Diffusion Models. We will

mathematically map specific decoherence channels to the "forward diffusion" process of generative models.

Goal 2: Validation via Grand Scientific Challenges

We use two complex scientific domains as rigorous **testbeds** to validate our fundamental theories.

- **2.1. Complexity Validation: Neuroscience (Korea - SNU & Germany):**
  - **Hypothesis:** The human brain's functional connectivity (fMRI/EEG) exhibits long-range temporal correlations that classical Transformers fail to capture efficiently.
  - **Validation:** Testing **Quantum Spatio-Temporal Transformers** on fMRI data.
  - **German Contribution:** Dr. Lorenz will lead the **benchmarking of quantum medical imaging algorithms** on this dataset, specifically evaluating if the quantum models offer a statistically significant advantage in *trustworthiness* and *robustness* compared to classical AI.
- **2.2. Noise-Robustness Validation: High-Energy Physics (Korea - Yonsei & Germany):**
  - **Hypothesis:** Quantum algorithms are naturally robust to the specific stochastic noise found in particle collision data.
  - **Validation:** Using **CERN CMS data** to test quantum machine learning models across multiple HEP tasks
  - **German Contribution:** Applying the **"Quantum Randomized Smoothing"** technique to HEP data to theoretically bound the error rates of event classification, ensuring the physics results are reliable.
- **2.3. Adversarial Validation: Cybersecurity (Italy & Korea):**
  - **Hypothesis:** Quantum-generated synthetic data is harder for classical adversaries to detect.
  - **Validation:** Using Quantum Diffusion to generate synthetic network attack patterns and testing robustness.

3. Work Packages (WP) Structure

WP	Title	Lead	Tasks
WP1	**Theoretical Frameworks for Scalable QML**	**Korea**	Multi-Chip Entanglement bounds. Quantum Transformer ansatz design.
WP2	**Quantum Optimization & Fine-Tuning**	**Italy**	QAOA fine-tuning. Gradient-Free/Evolutionary

			optimization theory.
WP3	**Robustness & Benchmarking Theory**	**Germany**	Developing generalization bounds. Quantum Randomized Smoothing theory. Benchmarking protocols.
WP4	**Quantum Diffusion & Generative Physics**	**Korea/Italy**	Hardware-adaptive diffusion math. Generative protocols.
WP5	**Scientific Validation (Neuro/HEP/Security)**	**All**	Applying WP1-WP4 theories to real-world datasets. Germany leads the *rigorous evaluation* phase.

4. Key Computing Resources

- **NERSC Perlmutter:** For massive classical simulation of theoretical quantum bounds.
- **IBM Quantum (Yonsei) & IBM Quantum (Germany):** Access to real quantum hardware for validation. (Note: Fraunhofer often has IBM Quantum access via the German network).

Initial Draft

QuantERA Call 2025 Proposal Initial Draft

1. Excellence (max. 6 pages)

1.1 Targeted breakthrough, baseline of knowledge and skills

Describe the targeted breakthroughs of the project.
Describe how the science and technology contribute to the establishment of a solid baseline of knowledge and skills for the specific theme addressed.
Describe the specific objectives of the project, which should be clear, measurable, realistic and achievable within the duration of the project.

Targeted Breakthroughs

The central ambition of this project is to overcome the three fundamental barriers currently preventing Quantum Machine Learning (QML) from surpassing classical capabilities: **Scalability** (limited qubit connectivity and count), **Trainability** (the optimization trilemma of barren plateaus, measurement overhead, and classical simulability), **Temporal Expressibility** (inability to model long-range correlations), and **Reliability** (inherent NISQ fragility and lack of robustness).

Current QML research is often limited to small-scale "toy" problems on single processors, heavily reliant on heuristic ansatzes that become untrainable or unreliable at scale. Our targeted breakthrough is to shift the paradigm from "heuristic QML" to **"Physics-Aware & Reliable QML,"** establishing a rigorous theoretical and algorithmic foundation that exploits quantum resources efficiently and robustly.

We target three specific foundational breakthroughs:

1. **Foundational Scalability via Multi-Chip Ensembles:** We aim to break the monolithic chip limit by developing **Multi-Chip Ensembles** using distributed quantum computing and classical ensemble learning techniques. Rather than waiting for large-scale hardware, we will utilize distributed QML architectures to demonstrate that entanglement resources can be effectively distributed across separated quantum processing units (QPUs).
   Beyond standard parallelization, we target **Multi-Modal Data Fusion**: using separate quantum processors to encode heterogeneous data types (e.g., Chip A for sMRI, Chip B fMRI). We will demonstrate that entangled ensemble strategies can fuse these distinct quantum feature spaces to achieve a "Collective Quantum Advantage" unavailable to single-chip models.
2. **Solving the Trainability-Efficiency-Advantage Trade-off via Physics-Informed Optimization:** We target the fundamental trade-offs limiting QML optimization: **Barren Plateaus** (untrainability in deep circuits), **Measurement Overhead** (prohibitive sampling costs for both gradient and gradient-free methods), and the **Simulability Trap** (where trainable models lose quantum advantage). We will implement a hybrid **Quantum Forward-Forward (QFF)** algorithm driven by **Hybrid Quantum Genetic Algorithm (HQGA)**.
   - **Addressing Trainability & Advantage:** QFF utilizes **local goodness measurements** to bypass Barren Plateaus without restricting the circuit to classically simulable log-depth structures, thereby preserving **Quantum Advantage.**

    - **Addressing Efficiency:** By decoupling layers, QFF avoids the costly backpropagation of global error. The use of **superposition and entanglement** in the HQGA significantly speed-up the convergence of conventional classical evolutionary algorithms, balancing the exploration-exploitation trade-off to minimize measurement overhead.
3. **Next-Generation Temporal & Structural Learning:** We target the limitations of current QML in processing time-series and graph data. We will develop **Quantum State Space Models (Q-SSM)** that can effectively learn complex, high-dimensional long-range dependencies in sequential data by leveraging quantum mechanics' inherent properties. The quantum circuits exploit entanglement to create **non-local correlations across feature dimensions in a single operation**—capturing dependencies that classical models require deep stacking or attention mechanisms to achieve—while the LSTM-style gating propagates this information across arbitrarily long sequences. Unlike classical transformers that require quadratic computational complexity $O(L^2)$ where L is sequence length, our Q-SSM model has **linear complexity** $O(L)$ and access to a 2^n-dimensional Hilbert space representation. Additionally, the quantum circuits achieve this expressivity with only $O(n \times l)$ parameters versus $O(N^2)$ for equivalent classical layers (n: number of qubits; l: number of quantum circuit layers; N: hidden dimension size), making the models particularly **suitable for data-limited domains** like biomedical signal processing where capturing **subtle long-range temporal patterns** (e.g., brain state dynamics across seconds of EEG) is critical but **overfitting** is a major concern.
4. **Robustness & Reliability via Fuzzy Logic and Benchmarking Protocols:** We aim a paradigm shift from *suppressing* quantum noise to *exploiting* it as a physical generative resource. By utilizing **Fuzzy Quantum Logic** to bridge the mathematical gap between discrete qubits and continuous neural networks, we develop **Physics-Informed Fuzzy Quantum Diffusion Model** that achieves the first "Hardware-Native" AI architecture that learns the exact, drifting physical fingerprint of a specific quantum device. This enables NISQ computers to perform high-fidelity generative tasks and bespoke error mitigation without waiting for fully fault-tolerant hardware, effectively turning the intrinsic instability of quantum systems into a unique computational feature rather than a liability. To curb uncertainty from noisy models even further, we will design **benchmarking and certification protocols**, leveraging the **QUARK** (QUantum computing Application benchmaRK) framework, to ensure they meet **industrial-grade robustness** standards against adversarial perturbations and NISQ noise. QUARK enables this by providing a standardized, vendor-neutral platform to rigorously quantify critical performance metrics—such as **noise resilience**, **trainability**, and **expressibility**—under realistic hardware conditions. This systematic validation allows us to definitively prove that our Fuzzy Quantum Diffusion architecture successfully delivers a genuine quantum advantage over classical baselines.

Baseline of Knowledge and Skills

This project advances the state-of-the-art by establishing a new baseline for **reliable, scalable, and trainable QML**.

- **From Monolithic to Distributed Scalability:** Currently, QML is restricted to small, unimodal tasks. We establish a baseline for **Distributed Multi-Modal Quantum Ensembles**, providing the protocols to fuse heterogeneous data (e.g., sMRI+fMRI) across networked QPUs via classical aggregation strategies.
- **From Gradient-Dependent to Gradient-Free/Hybrid Learning:** The community currently struggles with the convergence of Variational Quantum Algorithms (VQAs). By establishing the efficacy of **QFF combined with Hybrid Quantum Genetic**

Algorithms, we provide a new baseline for training deep quantum networks, proving that learnability can be maintained where classical gradients fail.
- **From Static Snapshots to Long-Sequence Spatio-Temporal Learning:** Current QML approaches struggle to process dynamic, sequential data efficiently. By establishing **Q-SSM** as standard tools, we provide a new baseline for **learning spatio-temporal data of long sequences**, enabling the analysis of high-dimensional biological signals with linear computational complexity.
- **From Fragile to Certified Robustness:** By combining **Fuzzy Logic** strategies with the **benchmarking and certification protocols**, we establish a methodology for **Reliable QML**. This contributes to the baseline skill set needed to certify quantum algorithms for safety-critical applications.

The consortium brings together a unique, synergistic baseline of skills:

- **QML Architectures & Scalability (SNU - Cha):** Deep expertise in **Multi-Chip Ensembles**, **Spatio-temporal QML algorithms, Quantum State Space Models (Q-SSM),** and computational neuroscience.
- **Optimization & Logic (Univ. Naples - Acampora):** World-class expertise in Computational Intelligence, Evolutionary Algorithms, and Fuzzy Logic.
- **Reliability & Certification (Fraunhofer IKS - Lorenz):** Industrial-grade, **application-oriented QML** expertise focused on safety, reliability and data efficiency aspects.
- **Domain Validation (Yonsei Univ - Yoo):** Experimental High Energy Physics (HEP) expertise, providing the massive, high-dimensional datasets required to rigorously stress-test the quantum advantage.

Specific Objectives

The project is structured around four clear, measurable, realistic, and achievable objectives:

- **Objective 1: Demonstrate Scalable Distributed QML (The Scalability Objective)**
    - **Goal:** Develop **Multi-Chip Ensembles** using input partition and ensemble learning protocols.
    - **Measure:** Successfully train a QML model (e.g., Quantum Transformer) fusing sMRI and fMRI features distributed across at least **2 simulated QPUs** (and physical hardware if available).
    - **Target:** Demonstrate **Scalability** by processing a multi-modal dataset that exceeds the qubit capacity of a single available chip (simulating a **2x capacity increase** via distributed processing) while achieving **>90% classification accuracy** on neuroimaging benchmarks.
    - **Timeline:** Protocols defined by Month 12; Full validation by Month 30.
- **Objective 2: Solve the Trainability-Efficiency-Advantage Trade-off (The Trainability Objective)**
    - **Goal:** Develop the **Quantum Forward-Forward (QFF)** algorithm hybridized with **Hybrid Quantum Genetic Algorithms**.
    - **Measure:** Compare convergence rates, measurement overhead, and classical simulability against standard Backpropagation and pure Evolutionary Algorithms.
    - **Target:** Achieve **convergence** in deep circuits (>10 layers) where Backpropagation fails (Barren Plateaus), with a **20% reduction** in measurement overhead compared to parameter-shift rules, while maintaining non-classicality.
    - **Timeline:** Algorithm design by Month 12; Benchmarking by Month 24.
- **Objective 3: Advance Next-Gen Temporal & Structural Learning (The Temporal Expressibility Objective)**

    - **Goal:** Develop **Q-SSM** for high dimensional spatio-temporal data.
    - **Measure:** Compare **Q-SSM** prediction accuracy on long sequence EEG/fMRI data against classical transformers/SSMs and standard Quantum RNNs.
    - **Target:** Demonstrate **superior long-range dependency capture** (measured by memory capacity metrics) and linear scaling with sequence length.
    - **Timeline:** Theoretical framework by Month 18; Implementation by Month 30.
- **Objective 4: Tackling Uncertainty and Unreliability of Noisy Quantum Devices (The Reliability Objective)**
    - **Goal:** Create a **Fuzzy Quantum Diffusion Model** enhanced by Fuzzy Logic controllers and reduce the model's uncertainty by **certifying its robustness** with **QUARK** framework.
    - **Measure:** Generate high-fidelity synthetic data (measured by Fréchet Inception Distance - FID) under simulated NISQ noise and assess robustness using reliability metrics (e.g., Lipschitz continuity).
    - **Target:** Achieve superior generative performance and certified robustness compared to classical transformers/diffusion models and standard quantum generative models (e.g., Quantum GAN) under noise levels characteristic of current hardware.
    - **Timeline:** Theoretical framework by Month 18; Implementation by Month 36.
- **Objective 5: Validate Foundational Advantage via Domain Science (The Validation Objective)**
    - **Goal:** Use high-complexity domain data to certify the foundational breakthroughs (O1-O4).
    - **Measure:**
        - **High Energy Physics (HEP):** Apply **Multi-Chip Ensembles** to CMS calorimeter-image classification (Quantum Vision Transformer), feature-based background estimation (Q-ABCDisCo), and waveform denoising (Temporal Convolutional Network-VQC) to assess scalability and quantum advantage across multiple HEP modalities.
        - **Neuroscience:** Apply **Quantum Transformers** and **Q-SSM** to EEG/fMRI data, and **QFF-Evolutionary VQE/QAOA** for brain connectivity analysis.
        - **Cybersecurity:** Apply **Robust Fuzzy-Quantum Diffusion Models** to intrusion detection logs and certify robustness against cyber- and privacy attacks via comprehensive **benchmarking protocols**.
    - **Target:** Demonstrate quantum advantage in either accuracy, expressibility, or training efficiency compared to classical SOTA on these specific datasets.
    - **Timeline:** Data preparation by Month 6; Final Domain Validation by Month 36.

1.2 Novelty, level of ambition and foundational character

Describe the advance your proposal would provide beyond the state-of-the-art, and to what extent the proposed work is ambitious, novel and of a foundational nature. Your answer could refer to the ground-breaking nature of the objectives, concepts involved, issues and problems to be addressed, and approaches and methods to be used.

Advance Beyond the State-of-the-Art

Current Quantum Machine Learning (QML) is hitting four "hard walls": the **Hardware Wall** (single chips are too small), the **Optimization Wall** (Barren Plateaus make deep circuits untrainable), **Temporal Expressibility Wall** (long range temporal modeling is inefficient), and the **Reliability Wall** (models collapse under noise and lack industrial certification). This project proposes a foundational shift to dismantle these walls, moving beyond the current state-of-the-art (SOTA) as follows:

1. **From Monolithic to Distributed Multi-Modal QML Scalability:**
   - **State-of-the-Art:** Current QML is constrained to single QPUs and simple datasets of single modality. Scaling requires larger fault-tolerant devices, which are years away. "Distributed" approaches often rely on expensive circuit cutting that incurs heavy sampling overhead.
   - **Our Novelty (Multi-Chip Ensembles):** We introduce a **Distributed Multi-Modal Ensemble** framework, which aggregates independent quantum models trained on heterogeneous data types (e.g., sMRI on Chip A, fMRI on Chip B). Instead of cutting circuits, our **Multi-Chip Ensembles** framework utilizes input partition and classical ensemble learning techniques to knit together multiple smaller QPUs. We specifically optimize the "inter-chip entanglement cost" to train high-expressibility models (such as Quantum Transformers) across fragmented hardware without requiring a single large fault-tolerant device.
   - **Foundational Advance:** We move beyond simple parallelization to **Multi-Modal Data Fusion** in Hilbert space. We establish the resource theory for **"Collective Quantum Advantage,"** demonstrating that entangled ensemble strategies can fuse distinct quantum feature spaces to solve problems larger than any single available chip.
2. **From Global Backpropagation to Local and Evolutionary Learning:**
   - **State-of-the-Art:** Optimization in QML faces a trilemma. Gradient-based methods suffer from **barren plateaus** and **high measurement costs**. Gradient-free methods suffer from **slow convergence**. Strategies that "fix" trainability (like restricting circuit depth) often render the model **classically simulable**, destroying quantum advantage.
   - **Our Novelty (Quantum Forward-Forward & Hybrid Quantum Genetic Algorithms):** We introduces a novel symbiotic framework that integrates the architectural decoupling of the **Quantum Forward-Forward** algorithm with the quantum-native evolutionary mechanics of the **Hybrid Quantum Genetic Algorithm (HQGA)**. Unlike state-of-the-art methods that are forced to choose between the prohibitive measurement overhead of global backpropagation or the "dequantization" required to avoid barren plateaus, our approach foundationally redefines QML training by decomposing deep circuits into **locally optimized layers** solved via **entangled crossover** and **quantum elitism**. This represents a paradigm shift that allows for the training of deep, highly entangled quantum models on NISQ hardware—architectures that are currently deemed untrainable—by mathematically circumventing the vanishing gradient problem while preserving the non-classical computational advantage that gradient-free methods typically sacrifice.
   - **Foundational Advance:** We demonstrate that this hybrid approach navigates the "Classical Simulability Trap." By using HQGA's global search, we can train deep, expressive circuits that remain **mathematically non-simulable** by classical computers, preserving the core quantum resource.
3. **From Static Snapshots to Next Generation Spatio-Temporal Learning:**
   - **State-of-the-Art:** Current QML algorithms are largely static. Existing Quantum RNNs struggle with high dimensional spatio-temporal data with long sequences. While classical transformers can effectively learn complex, long range dependencies, they incur a high computational cost that scales quadratically $O(L^2)$ with sequence length L.
   - **Our Novelty (Q-SSM):** We develop **Quantum State Space Models (Q-SSM)** architecture that integrates variational quantum circuits with recurrent gating mechanisms for sequential data, addressing a critical gap in existing QML—prior quantum sequence models either lacked temporal memory (collapsing sequences

via pooling) or attempted fully quantum recurrence that breaks gradient flow. Our hybrid design uniquely combines three-branch quantum superposition for expressive feature extraction with classical LSTM-style gating for selective memory, enabling **end-to-end differentiable training while preserving quantum expressivity**.
    - **Foundational Advance:** We set a new paradigm for spatio-temporal QML algorithms that exploits quantum entanglement for capturing **correlations across high-dimensional feature spaces** that are provably hard for classical models to represent efficiently. By demonstrating that quantum circuits can serve as drop-in replacements for classical feature extractors within proven recurrent architectures, we provide a practical template for quantum-classical hybrid design that bridges near-term quantum hardware capabilities with real-world sequence modeling tasks.
4. **From Noise-induced Uncertainty to Robustness:**
    - **State-of-the-Art:** In current NISQ devices, noise is treated as a binary error to be corrected via resource-heavy error correction codes. Generative QML models (e.g., Quantum GANs) typically fail on NISQ devices because hardware noise distorts probability distributions. Benchmarking and certifying QML methods is becoming increasingly important for deployment for real-world applications. First steps towards benchmarking have already been undertaken with the QUARK benchmarking framework. However, the industrial standards for certifying robustness against cyberattacks for QML are currently lacking.
    - **Our Novelty (Fuzzy-Quantum Diffusion & Robustness Certification):** We introduce **Fuzzy Quantum Logic** to handle hardware noise not as an "error" to be corrected, but as a "degree of uncertainty" within the learning process. Crucially, we move beyond simple performance metrics by developing **benchmarking and robustness protocols** to rigorously certify the reliability and robustness of these models against adversarial perturbations and noise, defining a new industrial standard for Trustworthy QML.
    - **Foundational Advance:** We establish the first **"QML Reliability Standard."** By defining formal metrics (e.g., Lipschitz continuity, noise stability) based on the QUARK framework, we transform QML from an experimental art into a **certified engineering discipline** capable of safety-critical deployment.

Level of Ambition and Foundational Character

This proposal is highly ambitious as it does not merely apply existing algorithms to new data; it rewrites the **foundational protocols** of how quantum models are built, trained, and verified.

- **Foundational Scalability:**
  We address the foundational question: "How much entanglement is required to link distinct quantum processors?" By developing **Multi-Chip Ensemble** protocols, we are establishing a resource theory for distributed quantum learning. This is ground-breaking because it decouples QML progress from the slow roadmap of monolithic hardware scaling, enabling "Virtual Large-Scale QML" on today's fragmented NISQ machines.
- **Ambition in Complexity (The "Big Science" Validation):**
  We are not testing on toy datasets (e.g., MNIST). We are ambitiously targeting High Energy Physics (LHC data) and Neuroscience (EEG/fMRI).
    - **HEP:** CMS detector data—including calorimeter images, high-level physics features, and waveform signals—contain complex, high-dimensional structures that classical algorithms often fail to disentangle. We aim to prove that our **Multi-Chip Ensembles** can capture non-local correlations in particle jets that classical deep learning misses.
    - **Neuroscience:** Brain signals are non-stationary and exhibit complex graph

structures. We aim to demonstrate that **Quantum State Space Models (Q-SSM)** can decode long-range temporal patterns with higher fidelity than classical SOTA. Furthermore, we will apply **QFF-HQGA VQE/QAOA** to solve combinatorial brain connectivity problems, identifying functional graph structures that classical heuristics fail to optimize.

- **Foundational Trust (The "QUARK Reliability" Standard):**
  We move beyond "accuracy" to "**certified robustness**." We are establishing a foundational methodology for benchmarking the safety and reliability of QML algorithms against adversarial attacks in **Cybersecurity** contexts. This establishes the first rigorous "Quality Assurance" framework for foundational QML resources, a critical step for the QPR topic and beyond.

Issues and Problems Addressed

Critical Issue	**Proposed Foundational Solution**
Connectivity & Scale Limits: Single chips lack the qubit count and connectivity for high-dimensional data.	**Multi-Modal Multi-Chip Ensembles:** Distributed protocols that aggregate parallel quantum agents to process heterogeneous data (sMRI/fMRI) without requiring a monolithic fault-tolerant device.
The Trainability Trilemma: Gradient-based methods suffer from Barren Plateaus and high measurement costs; gradient-free methods suffer from slow convergence.	**Hybrid Physics-Informed Optimization:** A synergy of **Quantum Forward-Forward (QFF)** (local updates to bypass Barren Plateaus) and **Hybrid Quantum Genetic Algorithms** (eliminates measurement overhead and classical simulability), balancing efficiency with trainability.
Inefficient Temporal Modeling: Classical transformers are computationally expensive, scaling quadratically to sequence length. Quantum RNNs struggle to learn high dimensional, long dependencies.	**Quantum State Space Models (Q-SSM): Three-branch quantum superposition** for rich feature extraction and LSTM-style gating for **selective memory management**, which enables efficiently and effectively learning long sequence data.
Lack of Trust & Robustness: NISQ models are fragile to noise and lack industrial safety certification.	**QUARK-Certified Fuzzy Robustness:** Integrating **Fuzzy Logic** to model noise as uncertainty, rigorously benchmarked against the **QUARK reliability standards** to certify robustness against adversarial attacks.

In summary, this project is foundational because it creates the **"Operating System" for Reliable and Scalable QML**: a suite of protocols (Distributed Ensembles, Evolutionary Optimization, QUARK Reliability) that allows quantum software to run effectively on imperfect hardware, validated by the most complex scientific data available.

1.3 Concept and methodology

Describe and explain the overall concept and research approach underpinning the project. Describe the main ideas, models or assumptions involved. Identify any interdisciplinary considerations and, where relevant, use of stakeholder knowledge.
Describe any national or international research and innovation activities which will be linked with the project, especially where the outputs from these will feed into the project.

Describe the methodology and explain its relevance to the objectives.
Describe the appropriateness of the methodology to narrow down multiple options and to address high scientific and technological risks.

Overall Concept: The "Physics-Aware" QML Stack

The overarching concept of this project is to transition Quantum Machine Learning (QML) from its current heuristic, hardware-agnostic state to a **"Physics-Aware" QML Stack**. We operate on the core assumption that Near-Term Intermediate Scale Quantum (NISQ) devices will remain fragmented (limited connectivity) and noisy for the foreseeable future. Therefore, instead of waiting for fault tolerant hardware, we fundamentally redesign the *software and algorithmic layer* to exploit quantum resources under these constraints.

Our research approach is built on a **Feedback Loop between Foundational Theory and Domain Science**, addressing five specific objectives:

1. **Scalability:** The **Multi-modal Multi-Chip Ensembles** framework addresses the scalability constraints of NISQ hardware by partitioning high-dimensional multi-modal data across independent quantum processors, fundamentally operating on the premise that restricting global entanglement mitigates barren plateaus and enhances noise resilience through the classical aggregation of independent error profiles. This approach **assigns distinct data modalities** (e.g., sMRI, fMRI) **to specific chips** equipped with **orthogonal circuit architectures** to maximize feature distinctiveness without requiring lossy dimension reduction. By employing a "**Selective Entanglement**" mechanism, the model strategically introduces quantum connections only between chips processing globally dependent features, effectively optimizing the trade-off between necessary model expressibility and the robust trainability provided by isolated quantum subsystems.
2. **Optimization:** We introduce a novel **Hybrid Quantum Forward-Forward** framework that fundamentally restructures QML training by replacing global backpropagation with a layer-wise, gradient-free optimization strategy. The core concept leverages the **Quantum Forward-Forward (QFF)** method to decompose deep quantum circuits into isolated, shallow layers, each optimizing a local "goodness" objective derived from separate positive (real) and negative (noise) data passes. Within each decoupled layer, we bypass classical optimizers by implementing the **Hybrid Quantum Genetic Algorithm (HQGA)**, which encodes circuit parameters into **quantum chromosomes**—superpositions representing multiple solution states simultaneously. By utilizing **entangled crossover** to stochastically correlate successful traits across the population and **quantum elitism** to reinforcement-learn optimal rotation angles, this approach allows for the training of complex, highly entangled models that are mathematically immune to the global barren plateau

problem while minimizing measurement overhead through local, efficient convergence.

3. **Temporal Expressibility:** Our **Quantum State Space Model (Q-SSM)** framework is built on a central hypothesis: **quantum circuits** can extract richer features from sequential data than classical networks by exploiting **entanglement and superposition**, while **classical gating mechanisms** remain optimal for **temporal memory management**. Our approach decomposes sequence modeling into two distinct functions: (1) feature extraction, where variational quantum circuits process input chunks, producing measurements that capture correlations across a 2^n -dimensional Hilbert space, and (2) temporal integration, where classical LSTM-style gates (forget, input, output) selectively accumulate information across chunks to model long-range dependencies. Three quantum branches are combined via trainable complex coefficients to enable interference-like effects, with an additional layer for bidirectionality through forward, backward, and global processing paths. This hybrid design assumes that the **expressivity advantage of quantum circuits lies in feature representation** rather than recurrence, and that near-term quantum hardware (or classical simulation) can practically execute shallow circuits while classical components handle the sequential state propagation that would otherwise require fault-tolerant quantum memory.
4. **Reliability:** We replace the synthetic Gaussian noise of classical diffusion models with the authentic, physical noise of quantum hardware. The core research approach assumes that hardware imperfections—such as decoherence and crosstalk—constitute a unique, learnable noise distribution that can be deterministically reversed. To bridge the incompatibility between discrete qubit outputs and continuous neural networks, the model utilizes **Fuzzy Quantum Logic (specifically POVMs)** to treat measurements as continuous "degrees of truth" rather than binary outcomes. By encoding data into quantum states, degrading them via natural hardware relaxation, and training a classical **Hardware-Attention U-Net** to invert this process, the system achieves a "hardware-native" architecture that exploits specific physical noise profiles for high-fidelity data generation and error mitigation. We certify their robustness using the **QUARK framework**.
5. **Validation:** We use **High-Complexity Domain Science** (HEP, Neuroscience, Cybersecurity) not just as applications, but as rigorous testbeds to certify the foundational advantages developed in Objectives 1-4.
   [High Energy Physics]
   We validate the foundational advantages of **Multi-Chip Ensemble QML**, specifically targeting the computational and precision bottlenecks inherent in processing massive particle collider data. The approach integrates hybrid quantum-classical models—including **Temporal Convolutional Network-VQC, Quantum Neural Networks (Q-ABCDisCo), and Quantum Vision Transformers** —with Multi-Chip Ensembles to demonstrate that quantum feature maps can capture complex particle correlations and solve ill-posed inverse problems with superior expressivity and parameter efficiency compared to classical deep learning. By successfully applying these algorithms to tasks like Higgs signal separation, background estimation, and waveform denoising , the project proves that Multi-Chip Ensemble QML offers a scalable solution for "Big Data" science, validating its ability to achieve high-precision results with significantly reduced computational resources and training data requirements.
   [Neuroscience]
   To decode the complex dynamics of the brain, our research utilizes **Quantum Transformers** and **Quantum State Space Models** for spatio-temporal analysis, employing Quantum Singular Value Transformation (QSVT) to reduce the computational cost of modeling long-range neural dependencies from quadratic to logarithmic scale. Complementing this, we frame brain connectivity as a quantum

optimization problem: **QAOA** structurally defines functional network boundaries by solving for optimal parcellation, while **VQE** dynamically simulates the brain's active inference process by minimizing a Hamiltonian representing Variational Free Energy to find Bayes-optimal posterior beliefs. Together, these methods exploit the high expressivity of quantum circuits to achieve robust generalization on small, high-dimensional neuroimaging datasets with minimal parameters.

[Cybersecurity]
We integrate the **Physics-Informed Fuzzy Quantum Diffusion Model** as a high-sensitivity anomaly detector with the **QUARK framework** for rigorous security benchmarking. We utilize the diffusion model to learn the unique physical noise fingerprint of a quantum processor, effectively establishing a hardware baseline that allows us to detect stealthy security breaches—such as hardware Trojans or induced crosstalk—by monitoring for specific deviations in the model's attention weights. Complementing this, we employ the QUARK framework to orchestrate attack simulations and validate system integrity, ensuring that our defenses are empirically verified against realistic, hardware-specific vulnerabilities.

Interdisciplinary Considerations & Stakeholder Knowledge

This proposal is inherently interdisciplinary, fusing three distinct scientific cultures to create a sum greater than its parts:

- **Quantum Computation & Domain Science (SNU, Yonsei):** Provides the rigorous understanding of quantum algorithm development (**Multi-chip Ensembles, Quantum State Space Models**), ansatz design (**Multi-Chip Ensembles**), and the domain knowledge of **Computational Neuroscience** and **High-Energy Physics**.
- **Computational Intelligence (Univ. Naples):** Injects mature classical methodologies—specifically **Fuzzy Set Theory** and **Bio-inspired Evolutionary Computing**—into the quantum domain to solve trainability issues that pure quantum theory has failed to address.
- **Application-focused QML & Robustness (Fraunhofer IKS):** Bridges the gap to industry by defining **quality benchmarks** and establishing robustness certifications. This stakeholder knowledge ensures our algorithms are not just theoretically interesting but reliable enough for safety-critical applications like cybersecurity.

Link to National and International Activities

The project leverages significant national ecosystems:

- **Korea (SNU/Yonsei):** Aligns with the national K-Quantum initiative. Prof. Yoo's involvement connects the project directly to the **CERN/LHC community**, allowing us to utilize international "Big Science" infrastructure as a testbed.
- **Germany (Fraunhofer IKS):** The work links to the **Munich Quantum Valley** ecosystem, focusing on industrial usage and certification standards for quantum software.
- **Italy (Univ. Naples):** Connects to the vibrant European community on soft computing and computational intelligence, bringing these classical AI strengths into the European Quantum Flagship landscape.

Methodology and Relevance to Objectives

Our methodology is divided into four integrated workstreams, directly addressing the Specific Objectives:

Methodology 1: Multi-Modal Multi-Chip Ensembles (Addressing Scalability) We employ **Optimal Quantum Architecture** to design orthogonal circuits for distinct data modalities (e.g., sMRI and fMRI) and **Optimal Quantum/Classical Connection** to strategically place entanglement only between globally dependent feature groups. This framework operates by partitioning high-dimensional multi-modal data across independent quantum chips—processing locally correlated inputs within isolated devices while using selective quantum connections to model necessary global dependencies—and integrating the outputs via classical ensemble learning.

- *Relevance:* This directly tackles **Objective 1**. It allows us to process multi-modal, high-dimensional datasets that exceed the capacity of a single chip, effectively addressing the connectivity and size limits of monolithic processors. Moreover, it overcomes the "curse of dimensionality" without requiring lossy classical dimension reduction, mitigates barren plateaus by preventing unnecessary global entanglement, and achieves superior noise resilience by simultaneously reducing both quantum error bias and variance, thereby providing a scalable and trainable solution compatible with emerging modular quantum hardware.

Methodology 2: Hybrid Physics-Informed Optimization (Addressing Trainability) Our approach replaces the prohibitive global backpropagation of deep circuits with a layer-wise training scheme (QFF), where each layer is optimized locally using a gradient-free Evolutionary Algorithm—**Hybrid Quantum Genetic Algorithms (HQGA)**—instead of gradient descent. This method works by restructuring the deep quantum circuit into locally trained layers using FF, where each layer's parameters are optimized not by gradient descent, but by the HQGA running directly on the quantum processor. In this workflow, the FF framework defines local "goodness" objectives (separating positive from negative data) for shallow sub-circuits, and the HQGA evolves a population of "quantum chromosomes"—superpositions of parameter configurations —using "entangled crossover" and "quantum elitism" to maximize these local objectives.

- *Relevance:* This addresses **Objective 2**. It resolves the "Trainability Trilemma" by simultaneously eliminating the **measurement overhead** of the parameter-shift rule (by being gradient-free) and mitigates the **Barren Plateau** problem through two distinct mechanisms: QFF's local objectives prevent exponential vanishing of gradients, while the HQGA's global search based on quantum mechanics to navigate the optimization landscape more efficiently than classical solvers, effectively permitting the training of **highly entangled, non-simulatable** quantum circuits that would otherwise be untrainable.

Methodology 3: Three-branch Superposition and LSTM-Gating for Q-SSM (Addressing Temporal Expressibility) We develop **Quantum State Space Models (Q-SSM)** by combining variational quantum circuits for feature extraction with classical LSTM-style gating for temporal memory management. The models process sequential data through chunked recurrence: each chunk is passed through three parallel quantum circuits whose measurement outputs are combined via trainable complex coefficients ($\alpha, \beta, \gamma \in \mathbb{C}$) inspired by quantum superposition—this "**measurement-based superposition**" enables interference-like effects while maintaining gradient flow. The resulting quantum features are then processed by classical forget, input, and output gates that selectively retain or discard information across chunks, with a final layer for bidirectional processing (forward, backward, and global branches).

- *Relevance:* This tackles **Objective 3**. This hybrid architecture is significant for three reasons: **(1) Expressivity:** the quantum circuits access a 2^n-dimensional Hilbert space with only O(n) parameters, providing a more expressive feature representation than equivalently-parameterized classical networks; **(2) Temporal modeling:** the

LSTM gating solves the critical limitation of prior quantum sequence models (which collapsed temporal information via pooling) by enabling selective memory; and **(3) Scalability:** the quantum circuit cost Q is constant with respect to sequence length L, so the overall O(L) complexity matches classical SSMs while potentially capturing non-classical correlations through entanglement that classical models cannot represent.

Methodology 4: Fuzzy-Quantum Diffusion & QUARK Certification (Addressing Reliability) The proposed Quantum Diffusion Model fundamentally redefines generative AI by **replacing the synthetic Gaussian noise of classical models with the authentic, physical noise inherent in quantum hardware**. In this hybrid architecture, a quantum processor serves as the "forward process," degrading clean data via natural error channels—such as amplitude damping, decoherence, or crosstalk—effectively turning hardware imperfections into a unique, physics-based feature. A classical deep neural network (e.g., a U-Net) is then trained to reverse this specific physical process, resulting in a model that learns the intricate "fingerprint" of the device, which can be used for generative tasks or bespoke error mitigation.
Fuzzy Quantum Logic acts as the essential mathematical bridge that makes this hybrid training possible. By utilizing Positive Operator-Valued Measures—the operational tool of fuzzy logic—we replace sharp, binary measurements (0 or 1) with continuous, "unsharp" values. This prevents the massive information loss that usually occurs during state collapse, allowing the classical network to train on "soft" probabilistic data that accurately reflects the texture of the quantum noise. This integration transforms the system from a discrete, binary model into a continuous diffusion process, enabling the neural network to interpret and correct quantum errors as fuzzy degrees of truth rather than simple bit-flips.

We will also implement the **QUARK** framework as a rigorous "Certification Layer" for our Fuzzy-Quantum Diffusion model. The implementation involves a three-step benchmarking protocol: (1) **Adversarial Perturbation:** We will subject the QML models (e.g., the Intrusion Detection System) to quantum-specific adversarial attacks. This involves applying unitary perturbations $U(\varepsilon)$ to the input quantum state $|\psi\rangle$ to simulate worst-case scenario shifts in the data manifold. (2) **Lipschitz Continuity Analysis:** Using QUARK's mathematical tools, we will compute the **local Lipschitz constant** of the quantum circuit. This metric provides a formal upper bound on how much the model's output can deviate given a bounded input perturbation. A lower Lipschitz constant mathematically certifies a higher degree of stability. (3) **Noise Stability Profiling:** We will inject calibrated NISQ noise models (depolarizing, dephasing) into the simulation and measure the degradation rate of the model's accuracy.

- *Relevance:* This addresses **Objective 4**. Standard diffusion fails when hardware noise mimics the diffusion noise; the hybrid architecture of the **Fuzzy-Quantum Diffusion** replaces the synthetic Gaussian noise of traditional AI with authentic physical noise, allowing for the creation of hardware-aware error mitigation tools and generative models that are robust to the inherent instability of NISQ devices. Moreover, by quantifying the **Lipschitz constant**, we convert the abstract concept of "robustness" into a precise, measurable engineering metric. This allows us to mathematically prove that the Fuzzy-Quantum integration successfully dampens the impact of input uncertainty, providing the "**Certified Robustness**" required for industrial trust.

Methodology 5: Domain-Specific Validation (Addressing Validation) We will rigorously preprocess the domain data: **CMS calorimeter images, high-level physics features, and detector waveform signals** (High Energy Physics), **fMRI/EEG time-series** (Neuroscience), and **Network Traffic logs** (Cybersecurity). **(1) High Energy Physics:** We use a combination of multi-chip ensembles and QML algorithms (e.g., quantum convolutional neural network, quantum support vector machine) to overcome the computational bottlenecks of high-energy physics data analysis. Our method employs a hybrid quantum-classical framework that strategically integrates variational quantum circuits into established deep learning architectures—specifically replacing classical classifiers in the ABCDisCo background estimation model, output heads in Temporal Convolutional Networks for denoising, and linear projection layers in Vision Transformers (ViT). By utilizing specialized techniques such as Distance Correlation penalties to enforce variable independence, self-supervised "Noise2Noise" strategies to train without clean labels, and quantum-enhanced attention mechanisms for resolving merged particles, these models leverage the high dimensional expressivity and parameter efficiency of quantum states. This approach is directly relevant to the project's objectives as it addresses critical High Energy Physics bottlenecks—including data scarcity, the unavailability of ground truth for faint signals, and the computational demands of high-luminosity experiments—thereby validating QML as a robust, resource-efficient solution capable of outperforming classical baselines in complex particle analysis tasks. **(2) Neuroscience:** To conduct spatio-temporal analysis of fMRI and EEG signals, we utilize **Quantum Transformers** and **Quantum State Space Models (Q-SSM)**, which employ Quantum Singular Value Transformation to reduce the computational complexity of attention mechanisms from quadratic to logarithmic scale, enabling the efficient modeling of long-term dependencies in neural dynamics. For brain connectivity research, we apply **QAOA** to solve the combinatorial problem of optimal network parcellation and structural mapping, while utilizing **VQE** to simulate the brain's functional inference processes by minimizing a Hamiltonian that represents variational free energy. **(3) Cybersecurity:** Our research utilizes the **Physics-Informed Fuzzy Quantum Diffusion Model** as a diagnostic sentinel and the **QUARK framework** as a rigorous benchmarking environment to fortify QML against hardware-level and adversarial threats. By deploying the Quantum Diffusion model, we establish a precise baseline of quantum processor behavior, allowing us to detect subtle anomalies—such as those introduced by hardware Trojans or induced crosstalk—through deviations in attention weights that signal potential integrity breaches. Concurrently, we leverage the QUARK framework's modular architecture to simulate specific attack scenarios (like noise injection) and standardize the evaluation of security metrics, ensuring that defenses are validated against realistic, hardware-aware noise profiles.

- *Relevance:* This methodology tackles **Objective 5**. It proves that the foundational breakthroughs translate into tangible advantages in accuracy, efficiency, and robustness for real-world scientific problems. Specifically, it addresses the critical limitations of classical machine learning regarding calculation volume and precision in the face of massive LHC datasets, providing a scalable, noise-resilient pathway to discover new physical phenomena and enhance standard model verification in the era of NISQ computing; it leverages the high expressivity of quantum circuits to achieve superior generalization on small, scarce neuroimaging datasets with fewer parameters than classical models, thereby overcoming the critical bottlenecks of overfitting and memory inefficiency in precision psychiatry; it moves beyond theoretical models to address the unique physical vulnerabilities of NISQ devices, providing a reproducible and hardware-specific methodology for certifying the security and reliability of quantum processors in critical applications.

Appropriateness of Methodology to Address Risks

Our methodology is specifically designed to narrow down options and mitigate high scientific risks:

1. **Risk: Barren Plateaus make training impossible.**
    - *Mitigation:* We do not rely solely on one optimizer. By introducing **Evolutionary Algorithms**, we bypass gradient-dependence entirely. If QFF proves unstable, the Memetic Algorithm serves as a robust fallback global

search method.

2. **Risk: Hardware noise destroys the Distributed Model's accuracy.**
   - *Mitigation:* **Ensemble methods** are inherently robust to noise. Since the Multi-Chip approach relies on aggregating independent models, the stochastic errors of individual chips tend to average out, acting as a form of algorithmic error mitigation without the cost of error correction codes.
3. **Risk: Hardware noise destroys generative fidelity.**
   - *Mitigation:* Instead of expensive error correction, we use **Fuzzy Logic** to "soften" the decision boundaries, making the diffusion model tolerant to the inherent uncertainty of NISQ devices.
4. **Risk: "Quantum Advantage" is elusive in real data.**
   - *Mitigation:* We diversify the validation across three distinct domains (HEP, Neuroscience, Cybersecurity). Each domain stresses a different quantum aspect (HEP=Dimensionality, Neuroscience=Temporal Correlation, Cybersecurity=Robustness). This portfolio approach maximizes the probability of demonstrating a clear foundational advantage in at least one key area certified by the **QUARK framework**.

1.4 Interdisciplinary nature

Describe the research disciplines involved and the range of added value from interdisciplinarity, including measures for exchange, cross-fertilisation and synergy.

Research Disciplines Involved

This proposal is built upon a radical convergence of distinct scientific cultures. We move beyond standard multidisciplinary collaboration—where groups work in parallel—to true **interdisciplinarity**, where methods from one field are mathematically integrated into the core protocols of another.

1. **Quantum Information & QML Architectures:** This discipline provides the fundamental "operating system" for the project—entanglement, superposition, and Hamiltonian dynamics—and the theoretical framework for the **Multi-Chip Ensembles** architecture.
2. **Computational Intelligence & Soft Computing:** This discipline injects mature classical methodologies—specifically **Fuzzy Logic** and **Evolutionary/Memetic Algorithms**—into the quantum stack. This provides the mathematical tools to handle uncertainty and non-convex optimization, areas where pure quantum theory currently struggles.
3. **Quantum Software Engineering & Reliability:** This discipline bridges the gap between theory and industry. It provides the **QUARK Framework** (Quantum Computing for Artificial Intelligence and Reliable Knowledge), which serves as the rigorous standard for benchmarking, certifying, and ensuring the robustness of the developed algorithms.
4. **Domain Sciences (Neuroscience, High Energy Physics, Cybersecurity):** These disciplines provide the complex, high-dimensional "reality" against which the quantum algorithms are tested. They serve as rigorous validation environments for foundational claims regarding temporal correlation (Neuroscience), massive scalability (HEP), and adversarial robustness (Cybersecurity).

Range of Added Value from Interdisciplinarity

The primary added value is the creation of a **"Hybrid Robustness"** that cannot be achieved by physics or computer science alone. We leverage specific domain characteristics to solve foundational quantum challenges:

- **Neuroscience x Fuzzy Quantum Logic (The "Bio-Fuzzy" Synergy):**
  Biological signals (e.g., EEG/fMRI) are inherently non-stationary and noisy, often confounding standard quantum algorithms. By integrating Fuzzy Set Theory with Quantum Diffusion Models, we create a unique synergy: "Fuzziness" models the biological variability and noise, while "Quantum Diffusion" captures the complex, non-local temporal correlations in brain activity. This cross-fertilization creates a new class of generative models capable of synthesizing high-fidelity biological data, a feat unattainable by classical logic or pure quantum mechanics alone.
- **Cybersecurity x Evolutionary QML (The "Adversarial" Synergy):**
  In cybersecurity, network intrusion patterns change dynamically as attackers evolve. Gradient-based Quantum Machine Learning (QML) is often brittle against such adversarial shifts. By fusing Cybersecurity principles with Evolutionary and Memetic Algorithms, we develop QML models that do not merely "learn" a static dataset but "evolve" to find robust solutions in a dynamic threat landscape. We further enhance this by applying the QUARK Framework to certify the model's robustness against adversarial perturbations, transforming "Quantum Advantage" into "Certified Reliability".
- **High Energy Physics x Distributed Computing (The "Scale" Synergy):**
  The validation of distributed quantum computing requires data with intricate, non-local entanglement structures. High Energy Physics (LHC) data provides exactly this complexity. By applying Multi-Chip Ensemble protocols to particle tracking problems, we validate the "scalability resource theory" using real-world physical interactions. We prove that parallel quantum processors, working in concert via ensemble aggregation, can reconstruct the global correlations of subatomic particles without requiring a single monolithic fault-tolerant device.

Measures for Exchange, Cross-fertilisation, and Synergy

To ensure this interdisciplinary fusion occurs effectively, we have designed specific mechanisms for knowledge transfer:

1. **Cross-Disciplinary "Methodology Swaps":**
   We will organize bi-annual workshops where domain experts teach their core tools to the theory groups. For example, experts in Computational Intelligence will train the physicists on Fuzzy Logic Controller design, while QML researchers will train the domain scientists on the Multi-Chip Ensemble ansatz and Q-SSM. This ensures that "Fuzzy-Quantum" integration occurs at the code level, not just the conceptual level.
2. **Joint "Challenge" Sprints:**
   Research activities will be organized around "Challenge Datasets" provided by the domain experts.
   - **The Neuro-Challenge:** The theory group must apply **Fuzzy-Diffusion** to synthesize realistic EEG/fMRI data provided by the Neuroscience experts.
   - **The Cyber-Challenge:** The optimization group must use Evolutionary QML to detect anomalies in Network Traffic logs provided by the Cybersecurity experts, validated against QUARK robustness metrics.
     These sprints force the theoretical algorithms to adapt to the rigorous constraints of real-world domain data.
3. **Unified "Physics-Aware" Software Repository:**
   All developed algorithms (e.g., Fuzzy-Diffusion, Multi-Chip Ensembles) will be integrated into a shared, modular software repository. This enforces technical cross-fertilization, as the "Physics" modules (e.g., Hamiltonians) must interface correctly with the "Logic" modules (e.g., Fuzzy Controllers) and the QUARK

benchmarking tools, ensuring the final deliverable is a cohesive library rather than isolated scripts.

1.5 Gender dimension and open science practices

Describe how gender dimension in research content is addressed.
Describe how open science practices including sharing and management of research outputs are implemented.

Gender Dimension in Research Content

While the foundational quantum physics and abstract optimization algorithms (Goal 1) are gender-neutral by nature, the **Neuroscience application (Goal 2.1)** involves the processing of human biological data, where the gender dimension is scientifically critical.

- **Algorithmic Bias Mitigation in Neuroscience:** Research indicates that EEG/fMRI signals and brain-computer interface (BCI) performance can exhibit sex-specific variabilities due to anatomical and physiological differences. A QML model trained predominantly on male subjects may fail to generalize to female subjects, creating a biased algorithmic standard.
- **Implementation:** In the **Fuzzy-Quantum Diffusion** workstream, we will implement a "Gender-Stratified Validation" protocol. We will ensure that the training datasets for EEG/fMRI synthesis are balanced (50/50 male/female). Furthermore, we will treat "biological sex" as a distinct fuzzy variable within the Fuzzy Logic Controller, allowing the model to explicitly learn and adapt to sex-specific signal features rather than treating them as noise. This ensures the developed **Robust QML** tools are universally applicable and free from biological bias.

For the remaining research lines (HEP, Cybersecurity, and Foundational QML), the content is gender-neutral. However, we proactively address the gender dimension in the **research process** by ensuring gender-balanced project management and recruitment, aligning with the **QuantERA Gender Equality Statement**. The consortium includes female leadership (Prof. Jeanette Lorenz, Fraunhofer IKS) in a key Principal Investigator role, serving as a role model for early-career researchers.

Open Science Practices

We strictly adhere to the **Horizon Europe Open Science policy**, adopting a "As Open As Possible, As Closed As Necessary" approach to maximize the impact and reproducibility of our foundational breakthroughs.

1. Open Access to Publications

- **Policy:** We commit to **100% Open Access** for all peer-reviewed scientific publications resulting from this project.
- **Implementation:** We will adopt the "Green Road" by depositing the final peer-reviewed manuscripts or preprints in open repositories (e.g., **arXiv**, **Zenodo**) immediately upon acceptance, ensuring compliance with national funder mandates (e.g., NRF Korea, DFG). Where budget allows, we will opt for "Gold Open Access" in high-impact journals.

2. Research Data Management (FAIR Principles)

We will implement a Data Management Plan (DMP) within the first 3 months, adhering to the FAIR principles (Findable, Accessible, Interoperable, Reusable).

- **Findable & Accessible:** All non-sensitive simulation data, trained QML model weights, and benchmark results (e.g., from the HEP and Optimization tasks) will be deposited in **Zenodo** with a unique Digital Object Identifier (DOI).
- **Interoperable:** Data will be stored in standard, open formats (e.g., HDF5 for large datasets, JSON for model hyperparameters) rather than proprietary formats.
- **Reusable:** All datasets will be released under **Creative Commons licenses (CC BY)** to encourage reuse by the global quantum community.

3. Open Source Software

The "Physics-Aware QML" algorithms (QFF, Fuzzy-Diffusion) are the core output of this project.

- **Repository:** We will establish a public **GitHub repository** for the project.
- **Licensing:** All code will be released under a permissive open-source license (e.g., **MIT** or **Apache 2.0**) to foster industrial adoption and academic collaboration.
- **Documentation:** We will provide Jupyter notebook tutorials demonstrating how to apply our distributed and evolutionary QML protocols to standard datasets, lowering the barrier to entry for other researchers.

4. Management of Sensitive Data

For Cybersecurity (Goal 2.3) and Neuroscience (Goal 2.1), data privacy is paramount.

- **Anonymization:** Any human EEG/fMRI data will be strictly pseudonymized before processing.
- **Security:** Cybersecurity datasets containing potentially sensitive network vulnerability patterns will be shared only within the consortium via secure, encrypted channels, following the "As Closed As Necessary" principle to prevent misuse.

2. Impact (max. 3 pages; Weight: 25%)

2.1 Expected impacts

Be specific, and provide only information that applies to the proposal and its objectives. Wherever possible, use quantified indicators and targets.
Describe how the project will contribute to the expected impacts (see ‘Research Targeted in the Call’ of the Call Announcement).
Describe the importance of the technological outcome with regard to its transformational impact on technology and/or society.

Contribution to Expected Impacts

This project directly addresses the "Quantum Phenomena and Resources" (QPR) call objective to *"explore novel quantum phenomena, concepts, resources... and address major challenges that prevent broad applications."* We contribute to this by establishing a **Resource Theory for Scalable and Reliable QML**, impacting the field as follows:

1. Deepening the Understanding of Quantum Resources (Scalability via Ensembles)

- **Impact:** We will establish **"Parallel Quantum Processing"** as a quantifiable resource. By validating the **Multi-Chip Ensemble** architecture, we will determine the precise scaling laws between *ensemble size* (number of parallel chips) and *model expressibility*. This moves the field away from the dependency on single, large-volume entanglement towards **"Collective Quantum Advantage"** derived from aggregating diverse, smaller quantum models.
- **Target:** Demonstrate that a **Multi-Chip Ensemble** (using architectures like Quantum Transformers) can achieve **>90%** of the accuracy of a monolithic ideal circuit while reducing the qubit requirement per chip by **50%**. This proves that quantum advantage is accessible on networked NISQ devices without requiring full inter-chip connectivity or complex teleportation protocols.

2. Enhancing Robustness in the Presence of Decoherence

- **Impact:** We will fundamentally transform how noise is handled in generative QML. By shifting from resource-heavy "Error Correction" to **"Fuzzy Uncertainty Management,"** we enable quantum generative models to function on current noisy hardware. Crucially, we will use the **QUARK framework** to rigorously quantify this robustness.
- **Target:** Achieve a **2x improvement in Fréchet Inception Distance (FID)** scores for quantum-generated data (e.g., EEG/fMRI, particle jets) compared to standard Quantum GANs under realistic NISQ noise levels (10^{-3} gate error rates). Furthermore, we aim to achieve **certified robustness** (measured by QUARK metrics like Lipschitz continuity) against input perturbations.

3. Identifying New Opportunities and Applications (The Validation Feedback Loop)

- **Impact:** We will provide the first rigorous proof-of-concept that Foundational QML can solve "Big Science" problems that are intractable for classical methods due to data complexity.
- **Target:**
  - **High Energy Physics:** Demonstrate a **>5% improvement** in signal-to-background ratio for LHC particle tracking compared to classical Deep Learning, using Multi-Chip Ensembles to handle the high dimensionality of jet data.
  - **Neuroscience:** Demonstrate that **Fuzzy-Quantum** models can capture **long-range temporal correlations** in EEG/fMRI data that classical RNNs/Transformers miss, offering new biomarkers for neurological conditions.

Transformational Impact on Technology and Society

The technological outcome of this project—a **"Physics-Aware" QML Stack**—will have a transformational impact beyond the academic quantum community:

1. Technological Sovereignty & Standardization (The "Trust" Impact)

- **Context:** Currently, there is no standard for certifying that a quantum algorithm is "safe" or "robust."
- **Our Contribution:** Led by **Fraunhofer IKS**, we will leverage the **QUARK framework** to define the first **"Quantum Software Reliability Standard"** for QML. This includes specific metrics for robustness against adversarial attacks (validated via the Cybersecurity use case).

- **Society:** This is a prerequisite for deploying quantum AI in safety-critical infrastructure (energy grids, autonomous vehicles, banking). By establishing these certification protocols, we pave the way for the European and Global industry to adopt quantum solutions with confidence.

2. Democratization of Quantum Access (The "Scale" Impact)

- **Context:** "Quantum Advantage" is currently restricted to elite labs with the largest chips (100+ qubits).
- **Our Contribution:** The **Multi-Chip Ensemble** protocol allows researchers to pool multiple smaller, cheaper quantum chips (e.g., 20-qubit devices) to solve larger problems via classical aggregation.
- **Society:** This democratizes access to high-performance quantum computing. Universities and smaller companies with modest hardware can collaborate to run "Virtual Large-Scale" experiments, accelerating the global pace of quantum innovation.

3. Advanced Data Privacy in Healthcare (The "Fuzzy-Quantum" Impact)

- **Context:** Sharing medical data (like EEG/fMRI) is difficult due to privacy regulations (GDPR).
- **Our Contribution:** Our **Fuzzy-Quantum Diffusion** model can generate high-fidelity *synthetic* medical data that preserves the statistical properties of the original patients without revealing their identities.
- **Society:** This unlocks massive datasets for medical research, allowing hospitals to share "Quantum-Synthetic Data" for drug discovery or disease modeling without compromising patient privacy.

2.2 Dissemination, exploitation of results, communication

Provide a plan for disseminating and exploiting the project results beyond the project itself (scientifically and/or economically).
Results include any data produced in the framework of the project. If applicable, describe how data curation and distribution can be ensured beyond the project duration.

When applicable, describe how subsequent translation of research results into innovations and development of new products and/or services are facilitated

Describe the proposed communication measures for promoting the project and its findings during the period of the project. Measures should be with clear objectives. They should be tailored to the needs of different target audiences, including groups beyond the project's own community.
Where relevant, include measures for public/societal engagement on issues related to the project.

Dissemination of Results (Scientific & Open Science)

We adopt a "Physics-to-Product" dissemination strategy, ensuring that our foundational breakthroughs are rigorous enough for high-impact physics journals yet accessible enough for software engineering adoption.

1. Scientific Publications (100% Open Access):`

- **Target:** We plan for **12+ high-impact peer-reviewed publications** in leading venues.
    - *Nature Physics / Physical Review Letters:* For the foundational **Multi-Chip Ensembles** and **Quantum Time-Series Transformer** results.
    - *IEEE Transactions on Evolutionary Computation:* For the **Memetic QML** and **Quantum Forward-Forward** algorithms.
    - *Quantum Science and Technology:* For the **QUARK-certified reliability benchmarks** and domain validations (HEP, Neuro).
- **Mechanism:** All preprints will be uploaded to **arXiv** (quantum-ph) immediately. We will cover Gold Open Access fees where possible to ensure immediate accessibility.

2. Open Source "Physics-Aware QML" Library:

- **Deliverable:** The core algorithms—**Multi-Chip Ensembles** (Scalability), **QFF-Evo** (Optimization), and **Fuzzy-Diffusion** (Robustness)—will be released as a modular Python library. Crucially, this library will include a **QUARK-connector** module, allowing users to automatically benchmark their models against the robustness metrics defined by Fraunhofer IKS.
- **Curation:** The code will be hosted on **GitHub** with a permissive license (Apache 2.0).
- **Documentation:** We will provide comprehensive Jupyter notebooks demonstrating how to replicate our results on the specific "Challenge Datasets" (LHC , EEG/fMRI, Cyber-logs), lowering the barrier for other researchers to verify our claims.

3. Data Curation & Distribution:

- **Repositories:** The **Zenodo** platform will be used to host the non-sensitive "Challenge Datasets" (e.g., the anonymized EEG/fMRI benchmarks and specific LHC subsets). Each dataset will be assigned a DOI.
- **Long-term Preservation:** Metadata will follow the **FAIR** (Findable, Accessible, Interoperable, Reusable) principles. The data management plan (DMP) ensures these assets remain available for at least 10 years.

Exploitation of Results (Innovation & Standardization)

Our exploitation strategy focuses on transforming "Foundational Resources" into "Reliable Tools" for the quantum ecosystem.

1. Standardization of Quantum Software Reliability (Led by Fraunhofer IKS):

- **Innovation:** We will integrate the project's **Fuzzy-Quantum** robustness metrics into the industrial-grade **QUARK Framework**.
- **Facilitation:** Fraunhofer IKS will leverage its position in the **Munich Quantum Valley** and industrial networks to propose the **QUARK framework** as a preliminary industrial standard for certifying safety-critical QML (e.g., for automotive or finance sectors). This facilitates the translation of our research into a marketable "Certification Service."

2. Scalable "Virtual" Quantum Computing (Led by SNU/Yonsei):

- **Innovation:** The **Multi-Chip Ensemble** protocol enables users with small, fragmented chips to solve larger problems by aggregating parallel quantum models (e.g., Quantum Transformers).
- **Exploitation:** We will explore licensing this "Ensemble Strategy" to cloud quantum providers. It serves as an algorithmic optimization layer that can run on top of existing

NISQ hardware, instantly upgrading the capability of current machines without hardware changes.

3. Domain-Specific AI Tools (Led by Naples/SNU):

- **Innovation:** The **Neuro-Quantum Generative Model** addresses the scarcity of medical data using Fuzzy-Diffusion.
- **Exploitation:** This tool can be exploited by hospitals or MedTech startups to generate privacy-preserving synthetic patient data (EEG/fMRI), solving a major bottleneck in medical AI development while complying with GDPR.

Communication Measures (Public & Community Engagement)

We will promote the project to diverse audiences with tailored messages.

1. Academic & Developer Community (The "Users"):

- **Measure:** Organize a **"Quantum Robustness Hackathon"** (Year 3).
- **Objective:** Invite students and developers to try and "break" our Robust QML models using **QUARK's adversarial attack toolkits** or use our **Multi-Chip protocols** to solve a distributed learning challenge.
- **Target:** 50+ participants, fostering a community of early adopters for our software tools.

2. General Public (The "Society"):

- **Measure:** "Quantum for Big Science" Webinar Series.
- **Objective:** Explain *why* we need quantum computers for things like the LHC (searching for new particles) or Brain Science, demystifying the "hype" and highlighting the physics of entanglement.
- **Target:** Reach 500+ viewers via YouTube/University channels. We will specifically highlight the role of **Fuzzy Logic** as a bridge between human reasoning and quantum mechanics, making the concepts more relatable.

3. Policymakers & Industry (The "Funders"):

- **Measure:** Policy Brief on "Standardizing Quantum AI."
- **Objective:** Produced by Fraunhofer IKS, this brief will explain the necessity of **Robustness Certification (QUARK)** before QML is deployed in critical infrastructure, advocating for continued funding in reliability research (QPR topic).

3. Implementation (Weight: 25%)

3.1 Work Plan (max. 2 pages)

Provide a brief presentation of the overall structure of the work plan.
Clearly define the intermediate targets.
Provide a timing of the different work packages and their components (Gantt chart or similar).
Provide a graphical representation of the work packages components showing how they inter-relate (Pert chart or similar).

Overall Structure of the Work Plan

The project is organized into **six interconnected Work Packages (WPs)** designed to create a feedback loop between theoretical development and empirical validation.

- **WP1: Management & Coordination (Lead: SNU)** ensures scientific alignment, risk management, and administrative compliance across the consortium.
- **WP2: Scalable Distributed Quantum Architectures (Lead: SNU)** focuses on **Goal 1.1**. It develops the **Multi-Chip Ensemble** protocols to enable scalable QML on fragmented hardware by aggregating parallel quantum models.
- **WP3: Physics-Informed Optimization & Generative Models (Lead: Univ. Naples)** focuses on **Goals 1.2 & 1.3**. It develops the **Quantum Forward-Forward (QFF)** algorithm, **Memetic/Evolutionary strategies**, and **Fuzzy-Quantum Diffusion** models to solve trainability and noise challenges.
- **WP4: High-Complexity Domain Validation (HEP & Neuroscience) (Lead: Yonsei)** focuses on **Goals 2.1 & 2.2**. It prepares the "Big Science" datasets (LHC, EEG/fMRI) and benchmarks the algorithms from WP2/3 against classical SOTA to prove quantum advantage.
- **WP5: Reliability, Safety & Certification (Lead: Fraunhofer IKS)** focuses on **Goal 2.3** and the **Reliability Standard**. It benchmarks the robustness of the developed models against adversarial attacks (Safety & Cybersecurity) and defines certification metrics using the **QUARK framework**.
- **WP6: Dissemination, Exploitation & Communication (Lead: Fraunhofer IKS)** ensures Open Science compliance, code release, and industrial engagement.

Intermediate Targets

We have defined four critical Intermediate Targets (ITs) to monitor progress:

- **IT-1 (Month 12): Theoretical Foundations Established.**
    - Mathematical formulation of the **Multi-Chip Ensemble** ansatz complete.
    - Fuzzy Logic Controllers for Quantum Diffusion defined.
    - Domain datasets (LHC, EEG, Network Logs) preprocessed and feature-mapped to Hilbert space.
- **IT-2 (Month 24): Algorithmic Validation (Simulation).**
    - QFF-Evolutionary algorithm demonstrates convergence on >10 qubit simulations where Backpropagation fails (Barren Plateau proof).
    - Fuzzy-Diffusion generates synthetic EEG data with statistical fidelity comparable to classical GANs.
- **IT-3 (Month 30): Cross-Domain Integration.**
    - **Multi-Chip Ensemble** protocol successfully reconstructs correlations in LHC particle jets by aggregating outputs from 2+ parallel chips.
    - Cybersecurity models pass the preliminary robustness certification against gradient-based attacks using **QUARK metrics**.
- **IT-4 (Month 36): Final "Physics-Aware" Library.**
    - Release of the integrated open-source library.
    - Publication of final benchmarks demonstrating advantage in scalability (WP2) and robustness (WP5).

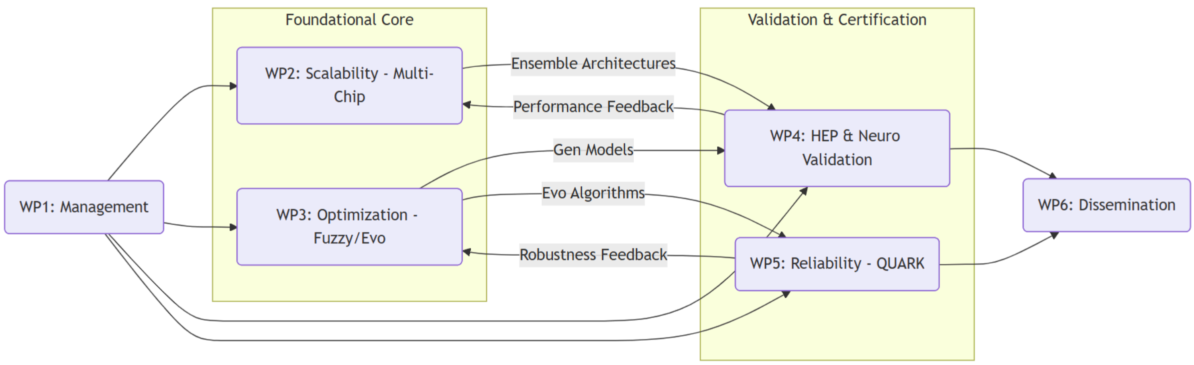
Timing of Work Packages

Work Package	M1-6	M7-12	M13-18	M19-24	M25-30	M31-36
WP1: Management	**====**	**====**	**====**	**====**	**====**	**====**
WP2: Scalable Arch (Multi-Chip)	**Definition of Protocols/ Architectures**	**Algorithm Development**	**Simulation-based Proof of Concept**	**Optimization & Refinement**	**Integration into Unified Library**	**Reporting (Deliverables/Periodic Reports)**
WP3: Opt & Gen Models (Fuzzy)	**Definition of Protocols/ Architectures**	**Algorithm Development**	**Simulation-based Proof of Concept**	**Optimization & Refinement**	**Integration into Unified Library**	**Reporting (Deliverables/Periodic Reports)**
WP4: Domain Validation (HEP/Neuro)	**Data Collection & Preprocessing**	**Quantum Feature Mapping**	**Initial Testing & Debugging**	**Benchmarking against Classical SOTA**	**Final Domain Validation**	**Reporting (Deliverables/Periodic Reports)**
WP5: Reliability (QUARK) & Cyber	**Requirements**	**Metric Definition (Robustness/Fuzziness)**	**Initial Testing & Debugging**	**Adversarial Attack Testing**	**Reliability Certification**	**Reporting (Deliverables/Periodic Reports)**

WP6: Dissemination	**Dissemination Planning**	**---**	**Scientific Publication**	**---**	**Hackathon Event**	**Final Conference Presentation**

The workflow follows a clear dependency path where **Foundational Theory (WP2/3)** feeds into **Domain Validation (WP4/5)**, with continuous feedback loops for refinement.

1. **Input:** Domain Requirements (from **WP4/WP5**) → define the constraints for **WP2** (Scalability needs for LHC) and **WP3** (Robustness needs for Neuro/Cyber).
2. **Development: WP2** (Multi-Chip Ensembles) and **WP3** (Fuzzy/Evo) develop the core algorithms in parallel.
3. **Integration:** The algorithms are integrated into a unified framework.
4. **Validation:**
   - **WP4** takes the **WP2** Multi-Chip engine to process LHC data.
   - **WP4** takes the **WP3** Fuzzy-Diffusion model to synthesize EEG/fMRI data.
   - **WP5** takes the **WP3** Evolutionary models and subjects them to Adversarial Attacks (Cybersecurity) to certify reliability using the **QUARK framework**.
5. **Output:** Validated results flow into **WP6** for dissemination.



3.2 Work Packages

(max. 1 page per WP)

For the description of each work package, please use the template provided. Use as many templates as needed.

WP 1	**Project Management & Coordination**						Start Month: 1	End month : 36
Contribution of project partners								
Partner number[1]	**1**	2	3	4	5	6	7	8
Total effort per partner (person*months)								

[1] ***Bold*** *the partner number of the work package leader*

Aim of the WP
To ensure efficient administrative and scientific coordination, timely reporting to national funding agencies and the QuantERA secretariat, and effective risk/data management.

Tasks	
T1.1	**Consortium Coordination & Monitoring (M1–M36: SNU; All)**[2] Day-to-day management of the project; organization of bi-annual consortium meetings (virtual/physical); monitoring of milestones and KPIs.
T1.2	**Risk & Innovation Management (M1–M36: SNU; Fraunhofer)** **Task title (start month – end month: responsible partner; involved partners)** Continuous monitoring of scientific and technical risks; execution of mitigation plans; management of Intellectual Property Rights (IPR).
T1.3	**Data Management & Ethics (M1–M36: SNU; Fraunhofer)** Implementation of the Data Management Plan (DMP) ensuring FAIR principles; oversight of ethical compliance regarding EEG/fMRI data (Neuroscience) and sensitive network logs (Cybersecurity).

Deliverable	**Month of delivery**	**Title of deliverable**
D1.1	M3	Data Management Plan
D1.2	M18	Mid-term Progress Report
D1.3	M36	Final Project Report

WP 2	**Scalable Distributed Quantum Architectures**					Start Month: 1	End Month : 36
Contribution of project partners								
Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								
Aim of the WP To develop theoretical protocols for "Virtual Large-Scale" quantum computing by implementing Multi-Chip Ensembles, enabling scalable QML on fragmented NISQ hardware via parallel model aggregation.								
Tasks								
T2.1	**Multi-Chip Protocol Definition (M1–M12: SNU; Naples, Fraunhofer)**							

[2] *For instance: T1.1 Development of something (M3-M6; responsible: 3; involved: 1, 4)*

	Mathematical formulation of ensemble learning strategies for distributed QPUs; development of aggregation logic (voting/averaging) to combine outputs from independent quantum models.
T2.2	**Multi-chip Architecture Optimization (M7–M24: SNU; Naples, Fraunhofer)** Design of scalable QML architectures (Quantum Transformers, HQTCN) optimized for ensemble deployment, targeting improved generalization and data-efficiency; with a focus on minimization of qubits-per-chip requirements.
T2.3	**Integration & Testing (M13–M36: SNU; Naples, Fraunhofer)** Implementation of the Multi-Chip protocols into a unified software module; verification of scalability, trainability, performance and fidelity retention across simulated distributed environments.

Deliverable	**Month of delivery**	**Title of deliverable**
D2.1	M12	Report on Multi-Chip QUARK Protocols
D2.2	M24	Software Module for Distributed Circuit Cutting

WP 3	**Physics-Informed Optimization & Generative Models**						Start month: 1	End month : 36
Contribution of project partners								
Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To solve the "Trainability" and "Noise" bottlenecks by developing the Quantum Forward-Forward algorithm (QFF) driven by Memetic Evolution, and Fuzzy-controlled Quantum Diffusion models.

Tasks	
T3.1	**Evolutionary QFF Development (M1–M18: Naples; SNU)** Development of Memetic Algorithms (MA) to drive Quantum Forward-Forward learning; benchmarking convergence against Backpropagation in Barren Plateau landscapes.
T3.2	**Fuzzy Logic Controller Design (M7–M24: Naples; SNU)** Design of Fuzzy Logic Controllers (FLC) to model NISQ noise as uncertainty; integration of FLC into the reverse diffusion process of generative models.

T3.3	**Robust Model Implementation (M19–M36: Naples; SNU, Fraunhofer)** Full implementation of the Fuzzy-Quantum Diffusion model with built-in noise resilience.

Deliverable	Month of delivery	Title of deliverable
D3.1	M18	Algorithm Definition: QFF-Evolutionary & Fuzzy-Diffusion
D3.2	M30	Benchmarking Report: Trainability & Noise Resilience

WP 4	**High-Complexity Domain Validation (HEP & Neuro)**						Start month	End month
Contribution of project partners								
Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To validate the foundational breakthroughs (WP2/WP3) using massive, complex datasets from High Energy Physics (LHC) and Neuroscience (EEG, fMRI), proving quantum advantage in scalability and temporal modeling.

Tasks

T4.1	**Domain Data Preprocessing (M1–M6: Yonsei; SNU, Fraunhofer)** Preparation of LHC particle jet data (Yonsei) and EEG/fMRI time-series (SNU/Naples); implementation of quantum feature maps (Fraunhofer).
T4.2	**HEP Scalability Validation (M13–M30: Yonsei; Naples, Fraunhofer)** Application of Multi-Chip Ensembles (from WP2) to LHC jet classification; benchmarking accuracy and generalization vs. single-chip and classical SOTA baselines.
T4.3	**Neuro-Generative Validation (M13–M30: SNU; Naples, Fraunhofer)** Application of Fuzzy-Diffusion (from WP3) to synthesize EEG/fMRI data; evaluation of statistical fidelity (FID score) and privacy preservation.

Deliverable	Month of delivery	Title of deliverable
D4.1	M6	Curated "Challenge Datasets" (LHC & EEG/fMRI)
D4.2	M36	Validation Report: Roadmap to Quantum Advantage in HEP & Neuro

WP 5	Reliability, Safety & Certification	Start month: 1	End month : 36

Contribution of project partners

Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To define the first "QML Reliability Standard" and benchmark and certify the robustness of the developed QML models against cybersecurity threads and privacy attacks.

Tasks

T5.1	**Reliability Metrics and Safeguard techniques (M1–M12: Fraunhofer; All)** Definition of formal metrics for QML robustness (e.g., Lipschitz continuity, noise stability, privacy budget, data leakage)
T5.2	**Cybersecurity Adversarial Testing (M13–M30: Fraunhofer; Naples)** Application of Evolutionary QML to Network Intrusion Detection; stress-testing the models with gradient-based adversarial attacks to measure robustness.
T5.3	**Certification Framework (M25–M36: Fraunhofer)** Synthesis of results into a formal QUARK Certification protocol for industrial QML applications.

Deliverable	**Month of delivery**	**Title of deliverable**
D5.1	M18	Robustness Report:
D5.2	M36	QML Reliability Standard, e.g. certification protocols

WP 6	Dissemination, Exploitation & Communication	Start month: 1	End month : 36

Contribution of project partners

Partner number	1	2	3	4	5	6	7	8
Total effort per partner (person*months)								

Aim of the WP

To maximize impact through Open Science publications, release of the "Physics-Aware QML" library, and engagement with the industrial and public sectors.

Tasks	
T6.1	**Scientific Dissemination & Open Science (M1–M36: All)** Publication of results in Open Access journals; management of the GitHub repository and Zenodo data archives.
T6.2	**Exploitation & Standardization (M13–M36: Fraunhofer; All)** Promotion of QUARK Reliability Standards to industrial bodies (e.g., Munich Quantum Valley); exploration of licensing models for the Multi-Chip middleware.
T6.3	**Communication & Events (M1–M36: Fraunhofer; SNU)** Organization of the "Quantum Robustness Hackathon" (M30); public webinars on "Quantum for Big Science"; website maintenance.

Deliverable	**Month of delivery**	**Title of deliverable**
D6.1	M36	Open Source "Physics-Aware QML" Library Release
D6.2	M36	Final Plan for Exploitation and Standardization

Work package overview (total effort per WP and partner in person.months)

Use as many lines and columns as needed.

Partner	WP1	WP2	WP3	WP4	WP5	WP6	Total
1							
2							
3							
4							
5							
6							
Total							

3.3 Management structure, milestones, risk assessment

(max. 2 pages)

Describe the organisational structure and the decision-making including a list of milestones (template provided). A milestone is a major and visible achievement. It should be SMART: Specific, Measurable, Attainable, Relevant, Time-bound.
Explain why the organisational structure and decision-making mechanisms are appropriate to the complexity and scale of the project.

Organisational structure and decision-making

The project management structure is designed to ensure efficient coordination between the Asian (Korea) and European (Germany, Italy) partners, facilitating the rigorous feedback loop between foundational theory and domain validation.

- **Project Coordinator (PC):** Prof. Jiook Cha (SNU, Partner 1) is the single point of contact for the QuantERA Secretariat. The PC is responsible for overall progress monitoring, submission of deliverables, and conflict resolution.
- **General Assembly (GA):** The ultimate decision-making body, composed of one representative from each partner (Cha, Yoo, Acampora, Lorenz). The GA meets bi-annually to review progress against Milestones. Decisions are made by consensus; if consensus fails, a majority vote (one vote per partner) applies.
- **Scientific Steering Committee (SSC):** Composed of the Work Package Leaders. The SSC meets quarterly to ensure technical alignment (e.g., ensuring WP2's Multi-Chip Ensemble protocols meet WP4's HEP data requirements).
- **Innovation & Ethics Board:** Led by **Fraunhofer IKS (Partner 4)**, this board oversees IPR management, open science compliance via the QUARK framework standards, and ethical handling of Neuroscience/Cybersecurity data.

This structure is appropriate because it is **lean** (minimizing administrative overhead for a 4-partner consortium) yet **robust** (ensuring all scientific disciplines—Physics, CS, Engineering—have equal voting power in the GA).

List of milestones

Use as many lines as needed but try to limit the number of milestones.

Milestone	Delivery month	WP involved	Title
M1	M12	WP2, WP5	**Foundational Architectures Defined:** Multi-Chip Ensemble protocol specs and QUARK Reliability Metrics defined.
M2	M18	WP3	**Algorithmic Proof-of-Concept:** QFF-Evolutionary algorithm demonstrates convergence in simulation (Barren Plateau avoidance).
M3	M30	WP4, WP5	**Cross-Domain Validation:** Successful application of Multi-Chip Ensembles to LHC data and Fuzzy-Diffusion to EEG data.

M4	M36	WP6	**Final Release:** "Physics-Aware QML" Open Source Library released with QUARK Certification Standard.

Describe any critical risks, relating to project implementation, that the stated project's objectives may not be achieved. Detail any risk mitigation measures. Please provide a table with critical risks identified and mitigating actions (template provided).

Critical risks for implementation

Use as many lines as needed.

Description of risk (indicate level of likelihood: Low/Medium/high)	**WP(s) involved**	**Proposed risk-mitigation measures**
Scientific: Barren Plateaus persist despite QFF/Evolutionary methods. (Medium)	WP3, WP4	**Mitigation:** If the Forward-Forward approach proves unstable, we will fall back to a "Hybrid Memetic" strategy using standard VQE but optimized via Prof. Acampora's proven Evolutionary Algorithms, which do not rely on gradients.
Technical: Ensemble Aggregation fails to achieve accuracy targets due to weak individual learners. (Medium)	WP2, WP4	**Mitigation:** We will implement advanced "Weighted Voting" mechanisms where the weight of each chip's output is determined by its **QUARK reliability score**. Additionally, we will use the **Quantum Transformer** ansatz, which is designed to maximize expressibility even on small chips.
Technical: Fuzzy-Diffusion fails to capture temporal correlations in EEG. (Low)	WP3, WP4	**Mitigation:** We will pivot from pure Diffusion Models to **Quantum GANs** enriched with Fuzzy Logic Controllers. GANs are notoriously unstable, but the Fuzzy Controller is designed to stabilize learning dynamics, providing a viable alternative generative architecture.
Operational: Delays in accessing sensitive data (LHC/Cyber logs). (Low)	WP4, WP5	**Mitigation:** All partners have direct access (Yoo is CMS member; Lorenz is Fraunhofer Head). However, if delays occur, we have identified open-source proxy datasets (e.g., CERN Open Data Portal, KDD Cup 99) to proceed with algorithmic benchmarking without delay.
Management: IPR disputes regarding industrial exploitation. (Low)	WP1, WP6	**Mitigation:** A comprehensive Consortium Agreement (CA) based on the DESCA model will be signed before the project start, clearly defining

		background/foreground IP and licensing terms for the open-source library.

3.4 Consortium as a whole

(max. 1 page)

The individual members are described in section 3.5, there is no need to repeat that information there.
Describe the consortium. How will it match the project's objectives and bring together the necessary expertise? How do the members complement one another?
In what way does each of them contribute to the project? Show that each has a valid role and adequate resources in the project to fulfil that role.
If applicable, describe the industrial/commercial involvement in the project and explain why this is consistent with and will help to achieve the specific measures which are proposed for exploitation of the results of the project.

Matching Objectives and Expertise

The consortium is designed as a trans-continental feedback loop that bridges the gap between Foundational Quantum Theory (Goals 1.1–1.3) and Rigorous Domain Validation (Goals 2.1–2.3). No single partner possesses the full spectrum of skills required to deliver the "Physics-Aware QML Stack," necessitating this specific collaboration:

- **The "Architects" (SNU):** To achieve **Objective 1 (Scalability)**, we require deep expertise in ansatz design. SNU provides the **Multi-Chip Ensemble** protocols and **Quantum Transformer** architectures needed to distribute learning across fragmented hardware.
- **The "Optimizers" (Univ. Naples):** To achieve **Objective 2 (Trainability)** and **Objective 3 (Robustness)**, we need to solve the Barren Plateau and Noise problems. Univ. Naples injects **Fuzzy Logic** and **Evolutionary/Memetic Algorithms**—classical AI methodologies absent in standard physics labs—to stabilize the quantum models.
- **The "Validators" (Yonsei):** To achieve **Objective 4 (Domain Advantage)**, we need massive, high-dimensional data. Yonsei, through the CMS Collaboration, provides the **LHC particle jet data** that acts as the ultimate stress test for SNU's scalability protocols.
- **The "Certifiers" (Fraunhofer IKS):** To ensure these breakthroughs translate to reality, Fraunhofer IKS provides the **QUARK Framework**. They act as the consortium's "Quality Assurance" department, using QUARK to benchmark the reliability and robustness of all developed algorithms, ensuring they meet industrial standards.

Complementarity and Contribution

The partners complement each other through a "Scale-Logic-Trust" triangle:

- **SNU (Scale)** provides the *structure* of the quantum models. Without SNU, the consortium would lack the scalable architectures (Multi-Chip) to run on NISQ devices.
- **Naples (Logic)** provides the *training engine*. Without Naples, SNU's deep architectures would be untrainable (Barren Plateaus) or fragile to noise. Their **Fuzzy Logic Controllers** provide the "soft" resilience layer.
- **Fraunhofer IKS (Trust)** provides the *standard*. Without Fraunhofer, the results would remain academic proofs-of-concept. The **QUARK framework** transforms the consortium's outputs into certified, reliable software assets ready for safety-critical deployment.
- **Yonsei (Physical Reality)** grounds the project. Their experimental HEP expertise prevents the project from becoming purely theoretical, forcing the algorithms to handle the noise and dimensionality of real subatomic physics.

Industrial/Commercial Involvement

While the core consortium focuses on foundational QPR research, Fraunhofer IKS serves as the bridge to industry. As a leading German RTO (Research and Technology Organisation), Fraunhofer's involvement ensures the path to exploitation is built-in from day one.

- **Consistency with Exploitation Measures:** The proposed **"Quantum Software Reliability Standard"** (WP5) is directly consistent with Fraunhofer's mandate within the **Munich Quantum Valley**. By using the **QUARK framework** to certify the consortium's algorithms, Fraunhofer validates the commercial viability of "Trustworthy QML" services. This allows the consortium to offer not just algorithms, but "Certified Algorithms" to industrial stakeholders (e.g., automotive, finance) who demand reliability guarantees, maximizing the economic impact of the project.

3.5 Description of the consortium

(max. 1 page per partner)

Describe expertise and role in the project for each partner (templates provided). The information provided here will be used to judge the operational capacity. Use the following templates for the coordinator, the other partners requesting funding, and partners not requesting funding if any. If the project relies on input to be provided by a third party, append a letter of commitment at the end of the proposal.

Jiook Cha Project Coordinator	**Seoul National University / Department of Psychology, Department of Brain and Cognitive Sciences, Interdisciplinary Program in Artificial Intelligence**

Expertise:

Seoul National University (SNU) is South Korea's premier research university. Prof. Jiook Cha's lab specializes in **Quantum Machine Learning (QML) architectures** and their application to **Computational Neuroscience**. The group possesses unique expertise in:

1. **Multi-Chip Ensembles:** Developing scalable approaches to train QML models across multiple chips, applied to image data, EEG/fMRI, and Reinforcement Learning.
2. **Quantum Time-Series Analysis:** Pioneering quantum algorithms for temporal data, including Quantum Transformers for fMRI analysis.
3. **QML for Neuroscience:** Bridging quantum computing with brain science through specialized neural network architectures.

Principal Investigator: Prof. Jiook Cha (Male)

Prof. Cha is a leading expert in applied QML architectures. His research focuses on creating scalable quantum algorithms for high-dimensional biological data.

Relevant Achievements:

- Developing **multi-chip ensembles** approaches and applying them to image data, EEG data, and reinforcement learning environments.
- Developing **quantum time-series transformers** and applying them to fMRI analysis.
- Developing **hybrid quantum temporal convolutional neural network (HQTCN)** algorithms and applying them to EEG analysis.

Role in project: Coordinator & QML Architecture Lead.

SNU leads WP1 (Management) and drives WP2 (Scalable Architectures) by implementing the Multi-Chip Ensemble protocols. Prof. Cha also leads the theoretical development of the Neuro-Generative validation in WP4, utilizing his expertise in Quantum Transformers and HQTCNs to process EEG/fMRI data.

Use this template if this partner is requesting funding

Hwidong Yoo	**Yonsei University / Department of Physics**

Expertise:
Yonsei University is a leading private research university in Korea with a strong legacy in High Energy Physics (HEP). Prof. Hwidong Yoo is a key member of the **CMS Collaboration at CERN**, providing the consortium with direct access to massive-scale "Big Science" data.

1. **High Energy Physics:** Deep expertise in LHC particle jet analysis, tracking, and event classification.
2. **Big Data Analysis:** Infrastructure for processing petabyte-scale datasets from particle collisions.
3. **Quantum for HEP:** Experience in benchmarking quantum algorithms against classical Deep Learning (CNNs/RNNs) for particle physics tasks.

Principal Investigator: Prof. Hwidong Yoo (Male)

Prof. Yoo is an expert in experimental high-energy physics and the application of advanced computing to particle detection. He leads the validation efforts, bridging the gap between abstract quantum algorithms and physical reality.

Relevant Achievements:

- Leadership roles in the CMS Collaboration (CERN) for particle tracking analysis.
- Development of classical Deep Learning benchmarks for LHC jet tagging.
- Pioneering work in applying Quantum SVMs and QNNs to HEP event classification.

Role in project: Domain Validation Lead (HEP).

Yonsei leads **WP4 (High-Complexity Domain Validation)**. They are responsible for curating the LHC datasets, implementing the quantum feature maps for high-dimensional particle data, and rigorously benchmarking the Scalable QML protocols (WP2) against classical SOTA methods.

Jeanette Miriam Lorenz	**Fraunhofer Institute for Cognitive Systems IKS / Department of Quantum Computing**

Expertise:

Fraunhofer IKS is Germany's leading applied research institute for **Safe and Reliable Software Engineering and Artificial Intelligence**. PD Dr. habil. Jeanette Miriam Lorenz's group specializes in **Reliable Quantum Computing for Applications,** focusing on a broad range of topics encapsulating QML**,** quantum optimization, quantum simulations and physics-inspired methods.

1. **Application-driven QML:** Broad expertise in key QML performance aspects, including data-efficiency, impact of feature maps' design, noise stability and generalization.

2. **Quantum Robustness and Safety:** Extensive hands-on experience in QNN certification and verification against noise impacts and cyberattacks, which are critical for the project's cybersecurity validation and "Quality of Service" standards
3. **Benchmarking:** Fraunhofer IKS is one of the European's leaders in application-driven benchmarking of QCs, coordinates the European QC Benchmarking Coordination Committee, and engages in co-developping the QUARK framework.

Principal Investigator: PD Dr. habil. Jeanette Miriam Lorenz (Female)

PD Dr. Lorenz heads the department 'Quantum Computing' at Fraunhofer IKS and is associate professor at LMU Munich. She also leads the consortium focussing on applications of quantum computing in the Munich Quantum Valley. The mission of her research is to enable the use of quantum computing in safety-critical areas such as aerospace, mobility, logistics and medtech.

Relevant Achievements:

- Establishes the guidelines to perform application-centric benchmarking of QC within Europe, e.g. via the EQCBC and the QUARK-framework
- Extensive work on security and safety of QML, e.g. for the BSI (Federal Ministry of Information Security
- Bridging theoretical work with practical aspects of 'getting quantum computing to work'

Role in project: Reliability & Certification Lead

Fraunhofer leads **WP5 (Reliability, Cybersecurity & Certification)** and **WP6 (Dissemination)**. They apply the **QUARK framework** to benchmark the consortium's algorithms, define the **Quantum Reliability Standards**, and conduct the rigorous **Cybersecurity stress-tests** (Adversarial Attacks) to certify the consortium's algorithms.

Giovanni Acampora	**University of Naples Federico II / Department of Physics "Ettore Pancini"**
Expertise:	

The University of Naples Federico II is one of the oldest universities in the world. Prof. Giovanni Acampora’s group is globally recognized for **Computational Intelligence** and **Fuzzy Logic**. They bring the critical "classical intelligence" layer to the quantum stack. 1. **Fuzzy Logic Systems:** World-class expertise in Type-2 Fuzzy Sets and Fuzzy Logic Controllers, essential for the project's noise-resilience goals. 2. **Evolutionary Computation:** Deep experience in Memetic Algorithms and Bio-inspired optimization to solve non-convex problems. 3. **Quantum Computational Intelligence:** Pioneering the fusion of Fuzzy Logic with Quantum Computing (Quantum Fuzzy Logic). **Principal Investigator: Prof. Giovanni Acampora (Male)** Prof. Acampora is a Senior Member of IEEE and a pioneer in Quantum Computational Intelligence. His work focuses on hybridizing soft computing with quantum mechanics to solve trainability issues. **Relevant Achievements:** ● Definition of **Quantum Fuzzy Logic** frameworks for uncertainty management. ● Development of **Evolutionary strategies** for optimizing Variational Quantum Circuits. ● Application of **Memetic Algorithms** to prevent Barren Plateaus in QML.
Role in project: Optimization & Intelligence Lead Naples leads **WP3 (Physics-Informed Optimization)**. They provide the **Evolutionary/Memetic algorithms** to drive the QFF training and design the **Fuzzy Logic Controllers** for the Robust Quantum Diffusion models. They also support WP5 in adversarial evolutionary testing.

3.6 Consortium agreement principles (partner’s rights and duties, IPR management, data management, open access, GDPR)

(max. ½ page)

Consortium Agreement & Decision Making

The consortium will sign a Consortium Agreement (CA) based on the DESCA (Horizon Europe) model prior to the project start. The CA will define the rights and duties of partners, specifying that SNU (Coordinator) manages the administrative interface, while the General Assembly (one vote per partner) decides on major scientific pivots.

Intellectual Property Rights (IPR) Management

- **Background IP:** Partners retain ownership of their pre-existing know-how. Specifically, **Fraunhofer IKS** retains all rights to the core **QUARK framework**, and **Univ. Naples** retains rights to their pre-existing Fuzzy Logic libraries.
- **Foreground IP:** The new "Physics-Aware QML" algorithms (e.g., Multi-Chip Ensemble protocols, Fuzzy-Diffusion architectures) developed jointly will be jointly owned. We adopt an **"Open Source by Default"** strategy for software outputs, releasing the unified library under an **Apache 2.0 license** to maximize adoption.
- **Commercial Exploitation:** Fraunhofer IKS leads the exploitation strategy. If a patentable invention arises (e.g., a specific industrial control system based on Robust QML), the CA outlines fair access rights and royalty-sharing mechanisms for the inventors.

Data Management & Open Access

- **FAIR Data:** We adhere to the **FAIR principles** (Findable, Accessible, Interoperable, Reusable). All non-sensitive "Challenge Datasets" (LHC jets, synthetic EEG) will be deposited in **Zenodo** with DOIs.
- **Open Access:** The consortium commits to **100% Open Access** for peer-reviewed publications (Green or Gold), ensuring immediate availability on **arXiv**.

GDPR & Ethical Compliance

- **Neuroscience Data:** The processing of EEG/fMRI data (WP4) involves human subjects. **SNU** will ensure all data is **pseudonymized** at the source before sharing within the consortium, in strict compliance with **GDPR** and Korean Personal Information Protection Act (PIPA).
- **Cybersecurity Data:** Network logs used in WP5 will be screened to ensure no personally identifiable information (PII) or critical infrastructure vulnerabilities are exposed.

3.7 Significant facilities and large equipment available to the consortium to perform the project

(max. ½ page)

The consortium brings together world-class infrastructure that enables "Big Science" validation without the need for external hardware procurement.

1. High-Performance Computing & Simulation (SNU & Naples)

- **SNU:** Access to extensive GPU clusters required to simulate the **Multi-Chip Ensemble** protocols. Since we are simulating distributed quantum learning (validating >20 qubits across multiple virtual chips), high-memory classical simulation resources are critical.
- **Univ. Naples:** Dedicated computational clusters optimized for **Evolutionary Algorithms**, which require massive parallel processing capabilities to evolve the populations for the Memetic QFF optimization tasks.

2. "Big Science" Data Infrastructure (Yonsei)

- **Yonsei (CMS Collaboration):** Through Prof. Yoo's leadership in the CMS Collaboration, the consortium has authorized access to the **CERN LHC computing**

grid and petabyte-scale storage systems. This provides the massive **High Energy Physics datasets** (particle jets) required for WP4, which are otherwise inaccessible to standard research groups.

3. Reliability Benchmarking & Quantum Hardware Access (Fraunhofer IKS)

- **The QUARK Framework:** Fraunhofer provides the proprietary **QUARK benchmarking suite**, a significant software facility that automates the assessment of reliability metrics (Lipschitz continuity, robustness). This pre-existing infrastructure accelerates WP5 significantly.
- **Munich Quantum Valley (MQV):** As a key member of MQV, Fraunhofer IKS provides access to a diverse ecosystem of quantum hardware backends (Superconducting, Ion Trap) for validating the **Multi-Chip** protocols on physical NISQ devices where applicable.

3.8 Link with ongoing projects

(max. ½ page)

For each partner indicate (if applicable) the ongoing projects linked to the proposal topic, and their funding sources.

3.9 Financial plan

(max. 1 page)

The resources to be committed for each project partner have to be described in the Electronic Submission System by the coordinator. These resources include: Personnel, Consumables, Equipment, Travel, Subcontracting, Provisions, Licensing fees, other. Justify them here. Both the justification and the information in the system will be communicated to the Evaluation Panel.

4. Ethical issues (max. ½ page)

Describe any foreseeable ethical issue that may arise during the course of the research project. Describe all mitigation strategies employed to reduce ethical risk, and justify the research methodology with respect to ethical issues.

5. References (max. 30 references)

Provide references of articles and publicly available documents directly supporting the proposal.

QuantERA Call 2025

Call 2025
for Transnational Research Proposals

supporting the topics of

Quantum Phenomena and Resources

and

Applied Quantum Science

Launch of the Call:	4 September 2025
Submission deadline:	5 December 2025, 17.00 CET
Documents and procedures:	**https://www.quantera.eu**
Call information:	Maurice Tia, Ph.D. +33 (0)1 72 73 06 90 Maurice.Tia@anr.fr
Call budget:	ca. € 53 M

Co-funded by
the European Union

History of changes

Date	Changes to the document
10/10/2025	The maximum funding per project for Italian partners (MUR) has been corrected in the funding overview table. The sentence “All project-related costs (of Estonian partner) must be incurred no later than 31.08.2029” has been added in the Estonian national annex (ETAG).

QUANTERA

KEY FACTS & FIGURES

Topics	**Quantum Phenomena and Resources** **Applied Quantum Science**
Indicative Call budget	Approx. € 53 M
Project consortium	The project consortia must be composed of at least 3 partners eligible to receive funding from three or more different countries participating in the Call Standard consortium size: 3 to 6 partners
Participating countries	Austria, Belgium, Bulgaria, Croatia, Czechia, Estonia, Finland, France, Germany, Hungary, Ireland, Israel, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Romania, Slovakia, Slovenia, South Korea, Spain, Sweden, Switzerland, Türkiye, United Kingdom
Partner search tool	Applicants are offered a Partner search tool that can be used to look for partners or to look for projects: https://www2.ncn.gov.pl/partners/quantera2025/
Project duration	24-month or 36-month period
Evaluation	Proposals are evaluated based on the criteria of *Excellence*, *Impact*, and *Quality and efficiency of the implementation*
Eligibility for funding	The project consortium as a whole must be eligible with respect to transnational rules. Each partner must fulfil the conditions of the national/regional funding organisation, as described in the Annex 1.

CALL CALENDAR

5 December 2025 **17.00 CET**	**Deadline for the proposal submission**
May 2026	Notification of accepted proposals
June-September 2026	Funded project start

QUANTERA

Table of Contents

List of Abbreviations

Abbreviation	Full name
AQS	Applied Quantum Science
Call	QuantERA Call 2025 for proposals
CO	Coordination Office
CoI	Conflict of Interest
CS	Call Secretariat
CSC	Call Steering Committee
EC	European Commission
EP	Evaluation Panel
ESS	Electronic Submission System
EU	European Union
PI	**Principal Investigator** – Partner in the project consortium, the leader of the national/regional team
PC	**Project Coordinator** – Coordinator of the project consortium and the leader of the transnational team
QPR	Quantum Phenomena and Resources
QT	Quantum Technologies
RFO	Research Funding Organisation
SAB	**Strategic Advisory Board** - Advisory body for QuantERA consisting of prominent scientists and industry representatives of Quantum Technologies (QT)

1. SCOPE AND OBJECTIVES

About QuantERA

The QuantERA programme[1], supported by the European Commission, forms a network of Research Funding Organisations across Europe and beyond dedicated to advancing high-quality research and innovation in the field of Quantum Technologies (QT). The programme brings together researchers, funding agencies, and policymakers to make sure that cutting-edge quantum projects get the support they need.

The funding organisations of QuantERA jointly finance multilateral projects with the potential to initiate or foster new lines of QT. The support is intended to advance multidisciplinary science and drive cutting-edge engineering.

Information about the Call

The QuantERA Call 2025 is a joint transnational Call for research proposals in QT cofunded by organisations gathered under QuantERA III Research & Innovation Action, with additional EC contribution.

Through this Call, the QuantERA III consortium aims at stimulating and intensifying research and innovation in the field of QT, contributing to the Strategic Research and Industry Agenda of the European Quantum Technology community[2].

Targeted Research

Projects funded in QuantERA III should contribute to the development of European research and innovation in QT. The transformative research funded within QuantERA III should explore collaborative advanced interdisciplinary science and/or cutting-edge engineering with the potential to initiate or foster new lines of QT and help Europe grasp leadership early on in promising future technology areas.

Projects are expected to start at 1-4 Technology Readiness Level[3] (TRL) and achieve TRL in the range 1-6 commensurate with the Call topic.

Call Topics

The submitted proposals are expected to be aligned with one of the two QuantERA Call 2025 topics:

Quantum Phenomena and Resources (QPR)

where the goal is to lay the foundations for the QT of the future. The focus is on basic quantum science and fundamental physics, and the projects should explore novel quantum phenomena, concepts, resources, protocols, algorithms, and/or address major challenges that prevent broad applications of some quantum technologies;

[1] See: https://quantera.eu/
[2] See Strategic Research and Industry Agenda of the European Quantum Technology community at https://qt.eu/media/pdf/Strategic-Reseach-and-Industry-Agenda-2030.pdf
[3] For definition of TRL check: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/wp-call/2023-2024/wp-13-general-annexes_horizon-2023-2024_en.pdf

Co-funded by the European Union

Applied Quantum Science (AQS)

where the goal is to take known quantum effects and established concepts from quantum science, translate them into technological applications and develop new products. These could be novel devices that are based on known quantum effects and that will serve a novel application in QT, or devices and systems that translate known quantum applications into products and industrial applications.

Note: If the tasks of (a) partner(s) are not well in line with the consortium's chosen call topic (and closer to the other topic), the said partner(s) are strongly encouraged to contact their respective RFOs to check their eligibility. Each funding organisation participating in the Call decides to allocate its budget to the QPR topic, or the AQS topic, or both. The budget table is available in the section Funding: Overview.

Research areas

Funded projects in both topics are expected to address one or more of the following areas:

Quantum communication
Methods/tools/materials/strategies to deal with the issues of distance, reliability, efficiency, robustness and security in quantum communication; novel protocols for multipartite quantum communication and quantum cryptography; quantum memory and quantum repeater concepts.
Novel photonic sources for quantum information and quantum communication, coherent transduction of quantum states between different physical systems; integrated quantum photonics; quantum communication embedded in optical telecommunications systems; other communication protocols with functionality enhanced by quantum effects. Methods for quantum communications in space, between satellites and Earth.

Quantum simulation
Platforms and materials for quantum simulation; development of new measurement and control techniques and of strategies for the verification of quantum simulations.
Application of quantum simulations to condensed matter, chemistry, thermodynamics, biology, high-energy physics, quantum field theories, quantum gravity, cosmology and other fields.

Quantum computation
Development of noisy intermediate-scale quantum platforms; devices to realise multiqubit algorithms; demonstration and optimisation of error correction codes; progress towards fault-tolerance; interfaces between quantum computers and communication systems.
Development of novel quantum algorithms and software stacks; demonstration of quantum speed-up; new architectures and programming paradigms for quantum computation, including hybrid approaches.

Quantum sensing and metrology
Use of quantum properties for time and frequency standards (including precise frequency distribution), light-based calibration and measurement, gravimetry, magnetometry, accelerometry, and other applications. Development of detection schemes that are optimised with respect to extracting relevant information from physical systems; novel solutions for quantum imaging and ranging. Implementation of micro- and nano- quantum sensors, for instance for quantum limited sensitivity in the measurement of magnetic fields at the nanoscale. Extension of the reach of quantum sensing and metrology to other fields of science including e.g. the prospects of offering new medical diagnostic tools.

General quantum science
Novel sources of non-classical states and methods to engineer such states. Development of

device-independent quantum information processing. Methods for the reconstruction and

estimation of complex quantum states or channels and certification of their properties. Development of resource theory for quantum information. Study of topological systems for quantum information purposes. Understanding and control of open quantum systems and quantum measurement processes; development of methods to confine dynamics in controllable decoherence-free subspaces. Study of quantum energetics and thermodynamic processes at the quantum scale.
Novel ideas and applications in quantum science and technologies, based on e.g., superposition, interference, and entanglement, as means to achieve new or radically enhanced functionalities.

Expected impacts

Funded projects are expected to significantly advance the state-of-the-art of quantum sciences and technologies[4] by achieving one or more of the following targets:

- Develop a deeper fundamental and practical understanding of systems and protocols/algorithms for manipulating and exploiting quantum information,
- Enhance the robustness and scalability of quantum information technologies in the presence of environmental decoherence, hence facilitating their real-world deployment,
- Develop reliable technologies for the different components of quantum architectures,
- Identify new opportunities and applications fostered through quantum technologies, and the possible ways to transfer these technologies from laboratories to industries,
- Enhance interdisciplinarity in crossing traditional boundaries between disciplines in order to enlarge the community involved in tackling these new challenges,
- Move towards a gender diverse and inclusive quantum community, in particular targeting PhD students and early-career researchers,
- Spread excellence throughout Europe by involving partners from the Widening Countries,
- Build leading innovation capacity across Europe by involvement of key actors that can make a difference in the future, for example excellent young researchers, ambitious high-tech SMEs or first-time participants.

[4] QuantERA projects shall not duplicate research funded as part of the projects of the previous QuantERA Calls (see funded projects: Call 2017, Call 2019, Call 2021, Call 2023) and the projects of the EC Quantum Technologies Flagship Call (see here)

2. FUNDING OVERVIEW

Country	Funding Organisation	QPR budget (M €)	AQS budget (M €)	Expected no of projects	Limits on amount which may be requested (M €)	Contact prior to submission	Additional forms required	Industrial partner eligible for funding	Contact
Austria	**FFG**	**-**	**2.00**	4-6	0.5 per project	Recommended	Yes	Yes	Fabienne.Nikowitz@ffg.at
Austria	**FWF**	**1.5**	**-**	3-5	-	Recommended	Yes	No	Sabine.Ertl@fwf.ac.at
Belgium	**FNRS**	**0.30**	**-**	1	0.30 per project	Recommended	Yes	No	international@frs-fnrs.be
Belgium	**FWO**	**0.7**	**-**	2-3	0.35 per project (overhead included)	Recommended	Yes	No	europe@fwo.be
Bulgaria	**BNSF**	**0.46**		2-3	0.15 per project (overhead included)	Mandatory	Yes	No	Aleksandrova@mon.bg
Croatia	**HRZZ**	**0.20**		1	0.20	Recommended	Yes	Yes	international@hrzz.hr Milan@hrzz.hr
Czechia	**MEYS**	**0.50**	**-**[5]	-	-	Recommended	Yes	Yes	Michal.Vavra@msmt.gov.cz
Czechia	**TA CR**	**-**	**1.00**	3	-	Recommended	Yes	Yes	Magdalena.Pillasagua@tacr.cz
Estonia	**ETAG**	**0.30**			0.15 (partner) or 0.30 (coordinator)	Recommended	No	Yes	Margit.Suuroja@etag.ee
Finland	**AKA**	**1.00**		2-3	Up to 0.45 per partner	Recommended	No	See national annex	Katrine.Mahlamaki@aka.fi
France	**ANR**	**2.50**			Up to 0.45 per project and up to 0.30 per partner	Recommended	No	Yes	Maurice.Tia@agencerecherche.fr
Germany	**DFG**	**4.00**		-	-	-	Yes	No	Andreas.Deschner@dfg.de Michael.Moessle@dfg.de
Germany	**BMFTR**		**4.50**	-	-	Mandatory (see annex 1)	No	Yes	Krug@vdi.de
Hungary	**NKFIH**	**0.45**		1-2	-	Not required	No	Yes	ncp@nkfih.gov.hu
Ireland	**TÉ – RI**	**0.75**		1-2	Up to 0.33 for project partner Up to 0.405 per project coordinator	Yes, notification required	No	No	Maria.nash@researchireland.ie EU-Cofund@researchireland.ie
Israel	**IIA**	**0.75**		3-4	According to national regulations Please check eligibility rules for Academia	National application to be submitted mandatorily	Yes	Yes	Dan@iserd.org.il Batzion.F@innovationisrael.org.il
Italy	**MUR**	**0.5**		5	Up to 0.10 per project	National application to be submitted mandatorily	Yes	Yes	Aldo.Covello@mur.gov.it Sonia.Angiolin@est.mur.gov.it

[5] In exceptional circumstances, the MEYS can finance also the projects from the AQS call.

Country	Funding Organisation	QPR budget (M €)	AQS budget (M €)	Expected no of projects	Limits on amount which may be requested (M €)	Contact prior to submission	Additional forms required	Industrial partner eligible for funding	Contact
Latvia	LZP	0.60		2-3	Up to 0.10 per year per partner	Recommended	No	Yes	Maija.Bundule@lzp.gov.lv
Lithuania	LMT	0.3		-	Up to 0.15 per project (for project partner) or up to 0.20 EUR per project (for project coordinator)	Recommended	No	Yes	Saulius.Marcinkonis@lmt.lt
Luxembourg	FNR	0.35		1-2	-	Recommended	Yes	No	Christiane.Kaell@fnr.lu Helena.Burg@fnr.lu
Malta	MEYR	0.05		1	-	Mandatory	Yes	Yes, but only basic research, not pursuing direct commercial goals	David.D.Muscat@gov.mt
Netherlands	QDNL	2.00		-	0.35 per project	Recommended	Yes	No	Julia.Oesterling@quantumdelta.nl
Norway	RCN	0.434		-	-	Recommended	No	Yes	twa@rcn.no
Poland	NCBR	-	1.50	-	-	Recommended	No	Yes	Krystyna.Maciejko@ncbr.gov.pl
Poland	NCN	1.50	-	5-6	-	National application to be submitted mandatorily	Yes	Yes	quantera@ncn.gov.pl alicja.dylag@ncn.gov.pl
Romania	UEFISCDI	0.50		2-3	• 0.25 in case a Romanian institution is the coordinator (together with other Romanian partner(s) – if it is the case) • 0.20 for one/all Romanian partner(s) participating in a proposal.	Recommended	No	Yes	Nicoleta.Dumitrache@uefiscdi.ro
Slovakia	SAS	0.12		1	0.12	Recommended	No	No	Mnovak@up.upsav.sk Panisova@up.upsav.sk
Slovenia	MVZI	0.90		3	max. 0.3 per project	Recommended	No	Yes	Anamarija.Meglic@gov.si
South Korea	NRF	1.6		2	-	Mandatory	Yes	Yes	rock@nrf.re.kr
Spain	AEI	1.00		4-6	0.175 per project with one AEI applicant 0.275 per project if AEI applicant is coordinator 0.325 per project if two AEI applicants with one as coordinator	National application to be submitted mandatorily	Yes	No	era-ict@aei.gob.es
Sweden	VR	1.00	-	-	See National Annex	Recommended	Yes	No	Tomas.Andersson@vr.se

Country	Funding Organisation	QPR budget (M €)	AQS budget (M €)	Expected no of projects	Limits on amount which may be requested (M €)	Contact prior to submission	Additional forms required	Industrial partner eligible for funding	Contact
Switzerland	**SNSF**	**2.00**		5-8	For a 3-years project: CHF 0.4 Mio. / partner For a 2-years project: CHF 0.266 667 Mio. / partner	Recommended	Yes	Yes, but only basic research, not pursuing direct commercial goals	quantera@snf.ch
Türkiye	**TÜBİTAK**	**0.20**		2-3	See National Annex	Recommended	Yes	Yes	Kivilcim.Koseoglu@tubitak.gov.tr
United Kingdom	**UKRI EPSRC**		**5.0**	-	Up to £0.40M per partner, up to £0.75M per project	Recommended	No	No	QuantumTechnologies@epsrc.ukri.org

1. The budget is indicative and represents the committed funding available for all the proposals expected to be funded in this Call.
2. Partners are strongly discouraged from requesting greater budgets than are necessary for the activities being proposed.
3. Please check the RFOs requirements in Annex 1.

3. APPLICATION GUIDE

Project Consortia: Formation Principles

When setting up project consortia, applicants are strongly encouraged to take into account the following aspects:

Gender balance

To promote equal opportunity and gender balance, QuantERA encourages the participation of consortia with a fair representation of female researchers as PCs, PIs and in the research team.

Geographical diversity

To spread research excellence throughout Europe, QuantERA projects are encouraged to include partners from the Widening countries[6] participating in the Call: Bulgaria, Croatia, Czechia, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Romania, Slovakia, Slovenia, Türkiye and Outermost territories of France (defined in Art. 349 TFEU).

Academic age balance

To ensure heterogeneity of the project teams allowing for inter-generation transfer of knowledge, skills, etc. QuantERA encourages involvement of researchers at various stages of their research careers, including post-doctoral and PhD students as participants in the project.

Involving industry actors

To leverage innovation capacity across Europe and connect with industry, QuantERA projects are encouraged to involve industry actors (mostly for the AQS topic) that can contribute to development of QT domain in the future, e.g. ambitious high-tech SMEs, etc.

Project Costs

The typical total project cost, including own contributions across all partners, is approximately 2,500,000.00 EUR. Proposals requesting other amounts are also welcome and will be considered. The average requested funding per project in previous QuantERA calls ranged between 750,000.00 EUR and 1,500,000.00 EUR, which has proven appropriate for effectively addressing the objectives of the QuantERA calls.

Funding per entity

The grant for one legal entity being a partner organisation in one or more project consortia in the Call cannot exceed 1,000,000.00 EUR[7].

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[6] See the list of Wideninig countries in Horizon Europe at https://rea.ec.europa.eu/horizon-europe-widening-who-should-apply_en

[7] This maximum amount, however, may be higher if the objective of the action would be impossible or overly difficult to achieve (and duly substantiated).

Partner Search Tool

In order to facilitate the process of forming research consortia, we offer applicants a Partner Search Tool available here: https://www2.ncn.gov.pl/partners/quantera2025/.This tool can be used by projects looking for partners and partners looking for projects.

Eligibility

Consortium structure

- Each project must comprise **at least three eligible partners**, based in institutions eligible to Research Funding Organisations (RFOs) participating in the QuantERA Call **from at least three countries**.
- Each consortium partner is led by a **Principal Investigator (PI)**, who liaises with the respective RFO. The project is coordinated by one of the PIs, the Project Coordinator, who acts as the single point of contact towards the QuantERA Call Secretariat. One PI may not coordinate more than one proposal.
- The consortium **budget must be balanced**:
  - at most 60% of the total requested funding may be requested by partners from one country,
  - at most 40% of the total requested funding may be requested by a single partner.
- The consortium should maintain a clear focus, that is, the proposed research must have a clearly defined goal.
- Each partner can choose only one funding organisation when submitting a proposal.

Note that partners who are not requesting funding may be part of a consortium if they are able to financially secure their participation. However, countries subject to sanction(s) by the European Union authorities are excluded from this Call. At the time of publication, these countries include the following: Belarus, Russia. Ukrainian territories out of control of the Ukrainian government are also concerned.

Additionally, Beneficiaries implementing FSTP schemes also have to respect sanction regimes (also called ‘EU restrictive measures’ (EURM)) against persons, entities or certain economic sectors, including notably Russia and Belarus and also restrictions to participation of entities registered outside Israel’s 1967 borders (‘occupied Palestinian territories’) or Russian occupied territories of Ukraine[8].

The list of sanctioned countries and territories[9] might evolve, and application measures will be taken accordingly.

Project duration

In general, QuantERA projects have a duration of 36 months. Consortia may also propose 24 months, but each member of the consortium is strongly advised to read the national rules of their RFOs (Annex 1), to be aware of the durations authorised by respective RFO.

[8] See: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/other/guidance_fstp-good-practices_en.pdf
[9] See: https://data.europa.eu/apps/eusanctionstracker/

Eligibility of Partners

- The consortium as a whole must be eligible with respect to transnational rules.
- Each consortium partner must be eligible with national rules as stated in Research Funding Organisation Information specified in Annex 1.
- If an eligibility issue is detected at the transnational or national level, the proposal's Coordinator will be contacted and asked to provide the solution to the Call Secretariat within one week.
- Submitted proposals cannot request funding to cover any work, that is part of a project benefitting from any other EU grant.
- In order not to jeopardise the whole consortium, partners should enquire about the eligibility at the national level regarding:
  - institution (university, academic institution, industry, end user, standard organisation etc.),
  - partners (permanent staff, position secured for the duration of the project, etc.),
  - eligible costs and budget balance,
  - any other conditions stipulated in the Annex 1 or in the applicable national regulations.

Submission

Prior to the submission proposers are asked to check if:

- The research is in line with the scope of the Call,
- The consortium meets all transnational eligibility criteria,
- Each partner meets the national eligibility criteria.

It is the responsibility of the partners to ensure their eligibility (see the *Eligibility* section).

The Call follows a **one-stage submission and evaluation procedure**, with each proposal comprising a scientific document (a proposal form typically between 25 and 35 pages) and financial document (a table listing the detailed requested funding and lists of participants).

The Project Coordinator (PC) prepares a joint proposal for the whole consortium, using the templates available on the QuantERA website: http://www.quantera.eu/. The two forms (proposal form and financial form) composing the proposal are submitted **via the Electronic Submission System (ESS)** for QPR and for AQS.

The ESS will require administrative and financial information to be filled in by the Project Coordinator (PC) when creating the submission on the platform. Once the email addresses of the other partners are filled in by the PC, they receive an automatic email from the platform inviting them to connect to the submission platform and adjust the administrative and financial information already filled in by the PC.

It is crucial that each person involved in a proposal (Project Coordinator, Principal Investigator, permanent staff, non-permanent staff such as postdocs and PhD students) has their first and last name specified in the financial template as well as in the table appearing on the first page of the proposal form.

Partners whose funding organisation requires submitting forms alongside the joint proposal submission must do so (see requirements of respective Research Funding Organisation in Annex 1).

We recommend submitting the proposal several days before the deadline in case that unforeseen issues occur. Submitted proposals can be modified until the Call deadline.

The Project Coordinator and all partners must be prepared to diligently answer email queries after the submission.

Note: The Electronic Submission System will be open within several weeks after the Call launch. Its opening will be announced on the QuantERA website: http://www.quantera.eu/. Detailed step-by-step user instructions will be made available at the same time.

4. EVALUATION AND SELECTION

Evaluation

The evaluation relies on international Evaluation Panels of experts (EP), one for the Call topic of Quantum Phenomena and Resources (QPR) and one for Applied Quantum Science (AQS).

By default, each evaluation criterion is scored 0, 1, 2, 3, 4 or 5.

The scores indicate the following with respect to the criterion under examination (the definitions of scores are taken from Horizon Europe Evaluation Form[10]):

0. The proposal fails to address the criterion or cannot be assessed due to missing or incomplete information (unless the result of an 'obvious clerical error').
1. Poor. The criterion is inadequately addressed, or there are serious inherent weaknesses.
2. Fair. The proposal broadly addresses the criterion, but there are significant weaknesses.
3. Good. The proposal addresses the criterion well, but a number of shortcomings are present.
4. Very good. The proposal addresses the criterion very well, but a small number of shortcomings are present.
5. Excellent. The proposal successfully addresses all relevant aspects of the criterion. Any shortcomings are minor.

The eligible proposals recommended by the EPs are those proposals that reach at least a score of 3 in each of the three criteria used for the evaluation and detailed below. While the "Quality" threshold is the same for each criterion and each topic, the weights of the other two criteria differ between the QPR and the AQS topics (see below).

Excellence (weight in QPR: 50%; weight in AQS: 25%)

- Clarity and novelty of long-term vision, and ambition and concreteness of the targeted breakthroughs towards that vision.
- Novelty, non-incrementality and plausibility of the proposed research for achieving the targeted breakthrough and its foundational character.
- Soundness of the research methodology and its suitability to address high scientific and technological risks in the context of potential benefits.

[10] See: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/temp-form/ef/ef_he-ria-ia_en.pdf

- Appropriate consideration of the gender dimension in research content and the quality of open science practices.
- Range and added value from interdisciplinarity, including measures for exchange, cross-fertilisation and synergy.

Impact (weight in QPR: 25%; weight in AQS: 50%)

Extent to which the outputs of the project contribute at the European or international level to:

- The expected impacts (see p.8).
- Positive transformative effects to the economy, environment and/or society.

Quality of the proposed measures to:

- Exploit and disseminate the project results (incl. management of intellectual property rights), and to manage research data where relevant.
- Communicate the project activities to different target audiences, especially considering the need to inspire STEM students and early-career researchers.
- Facilitate subsequent translation of research results into innovations and development of new products and/or services when applicable.

Quality and efficiency of the implementation (weight in QPR: 25%; weight in AQS: 25%)

- Quality, coherence and effectiveness of the work plan, including extent to which the resources assigned to work packages are in line with their objectives and deliverables.
- Appropriateness of the management structures and procedures, including risk mitigation measures and innovation management.
- Complementarity of the participants and extent to which the consortium as a whole brings together the necessary expertise.
- Appropriateness of the allocation of tasks, ensuring that all participants have a clearly defined role and adequate resources in the project to fulfil that role.

The Evaluation Panels will meet to discuss the proposals and each one of them will establish a ranking list of the proposals. The evaluation of each proposal will be summarised in a consensus report, which will be made available to the applicants at the end of the evaluation and selection process.

Selection

Based on the two ranking lists (AQS, QPR) provided by the Evaluation Panels, a final list of projects selected for funding will be approved by the QuantERA CSC. The CSC will strive to fund as many excellent collaborative projects as possible. If at a given rank in the list not all ex aequo proposals can be selected, the criteria listed below will be applied in the following order:

1. The output of the Call, i.e. the overall funding, should be maximised.
2. The success rates (i.e. the overall number of funded projects relative to the overall number of proposals) of both topics should be as close as possible to each other.
3. If possible, each Research Funding Organisation should fund at least one project.
4. Projects involving partners from the Widening countries and territories should be prioritised.
5. The distribution of genders among the PIs and co-PIs considering all funded proposals should be as balanced as possible. Note that this criterion should not be understood as a distribution of genders per funded proposal. As an example, a proposal with more

female PIs and co-PIs than male PIs or co-PIs, will be prioritised if it results in a better overall balanced distribution of genders considering all the funded proposals.

5. MANAGEMENT OF FUNDED PROJECTS

Establishing the Project Consortium

The administrative and financial management of projects is overseen by the respective Research Funding Organisations, according to the respective national/regional rules and guidelines.

In order to ensure that the collaborative research can be conducted as planned, all partners of a consortium should strive to **start their part of the project at approximately the same date**.

The project must follow the *European Charter for Researchers* and the *Code of Conduct for the Recruitment of Researchers*[11]. The project should bear in mind gender balance, especially at the PhD level and early-career opportunities resulting from the funded projects, and promote equal opportunities between women and men at all levels in the implementation of the research activities.

A consortium agreement managing inter alia the ownership and access to key knowledge (including IPR, data management, GDPR etc.) has to be signed by all partners and sent to QuantERA Coordination Office no later than three months after the start of a project. Some funding organisations require that the consortium agreement be signed before the contract can be finalised or have other requirements regarding the consortium agreement (see Annex 1). Please see DESCA Model Consortium Agreement as an example: http://www.desca-agreement.eu.

Reporting and Publications

Consortia must present the status of their project at **three follow-up events** organised by QuantERA. The related costs are eligible and must be secured by the consortia as part of the project's budget.

Each Coordinator of a funded project has to submit to QuantERA **a mid-term report and a final report**.

PIs will have reporting duties to their respective national Research Funding Organisations regarding administrative and financial matters. This is specified in the individual contracts with respective national/regional funders.

Any publication resulting from this Call must **acknowledge QuantERA,** the European Commission and the relevant national funders involved according to the national requirements. The acknowledgement to be used is: *This work has been supported by QuantERA [insert project acronym], by [insert the funding organisation and the grant number for that funding organisation], [insert the next funding organisation and the grant number for that funding organisation] and co-funded by the European Commission ".*

[11] See: https://euraxess.ec.europa.eu/hrexcellenceaward/european-charter-researchers

Open Access Publishing / Open Research Data

Project consortia are encouraged to adhere to the Horizon Europe regulation on Open Science[12] for all output of scholarly research as long as it is in line with the national/regional requirements of the partners in the project.

The project consortia should agree upon and lay out publishing strategies and data management plans in their consortium **agreements at the beginning of the project and should always aim for Open Access research data and publications, as well as Open Science best practices**. Journal articles should be archived in either institutional or subject-specific repositories. Applicants should check the national information about open access. Research artefacts (data, software, etc.) should be shared on repositories fulfilling the FAIR data principles[13].

Compliance with EU obligations

Due to the obligations imposed on the QuantERA Consortium by the European Commission, the funded project partners are subject to the obligations defined in the standard provisions of Horizon Europe grants:

- EU regulations under the following Model Grant Agreement (MGA)[14] Articles: 12 (conflict of interest), 13 (confidentiality and security), 14 (ethics), 17.2 (visibility), 18 (specific rules for carrying out action), 19 (information) and 20 (record-keeping),
- The exercise of rights by the bodies listed in Article 25 of MGA (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.).

Additionally, all funded projects need to follow anti-discrimination rules and principles commonly shared by the EU (notably concerning racism, incitement to violence, discrimination based on religion, gender or sexual orientation).

The specific obligations of the project consortia partners will, where applicable, be laid down in individual contracts with the respective national or regional funders.

6. MANAGEMENT OF THE CALL

The Call is administered by the Call Secretariat (CS) and the Call Steering Committee (CSC).

The Call Secretariat (CS) is hosted by French Research Agency (ANR, FR), State Research Agency (AEI, ES) and supported by National Science Centre (NCN, PL). The CS assists the EPs and the CSC with the implementation of the evaluation and selection process.

The Call Steering Committee (CSC) is the highest decision-making body of the QuantERA Call 2025 for proposals, being composed of representatives of QuantERA III Research Funding Organisations (RFOs) participating in the Call. The CSC steers all activities of the Call. In particular the CSC nominates the members of each Evaluation Panel (EP) and establishes the final list of projects recommended for funding to the RFOs, based on the

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[12] For more information see: https://rea.ec.europa.eu/open-science_en

[13] For more information see: https://www.go-fair.org/fair-principles/

[14] The Horizon Europe Model Grant Agreement (MAG) available here: https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/common/agr-contr/general-mga_horizon-euratom_en.pdf

ranking lists provided by the EPs. All decisions concerning the Call implementation are taken by the CSC (e.g. management of confidentiality and conflicts of interest).

The Evaluation Panels (EP) are the assessment bodies. The two Evaluation Panels, one for the Quantum Phenomena and Resources (QPR) topic, and one for the Applied Quantum Science (AQS) topic, are the evaluation bodies of the QuantERA Call 2025, being composed of scientific and industry experts in QT. The EP evaluate each eligible proposal. Each panel establishes a ranking list of proposals which serves as a basis for the CSC funding recommendation.

With the aim of providing as much diversity as possible to the scientific evaluation, gender balance will be actively promoted in the panel formation.

Annex 1: Requirements of Research Funding Organisations

Austria FFG

Participating organisation

Austrian Research Promotion Agency

Contact

Fabienne Nikowitz
Programme Leader
Telephone number: +43 5 7755 5081
Email: fabienne.nikowitz@ffg.at

Funding criteria and regulations

Funding commitment: 2,000,000.00 EUR

Funding regulations

Maximum funding per proposal: 500,000.00 EUR

For Austrian proposers, the same regulations hold as for cooperation projects, see the application guidelines and regulations available at the FFG website (available mid-September 2025): https://www.ffg.at/quantentechnologien/quantera2025

Eligibility of a partner as a beneficiary institution:

Consortium requirement: at least one Austrian industry partner.

Eligible for funding are Austrian applicants from:

- Companies;
- Universities and research institutes/organisations - only in cooperation with at least one Austrian company.

Eligible TRL range: 2-7

All Austrian applicants are encouraged to contact FFG prior to submission.

Submission of the proposal at the national level

In addition to the QuantERA application, Austrian applicants are requested to submit a national application to FFG, through the eCall available at the following link: https://ecall.ffg.at/Cockpit/Cockpit.aspx

Mandatory are:

- QuantERA proposal application form and financial form as requested by QuantERA;
- Costs and requested funding.

No application will be considered after the final submission date.

Additional requirements

Further information and requirements can be found at the FFG website (available mid-September 2025): https://www.ffg.at/quantentechnologien/quantera2025

Participating organisation

FWF Austrian Science Fund

Contact

Sabine Ertl
Department Head - Department of Natural Sciences and Technology
Scientific Project Officer
Telephone number: +43 676 83487 8403
Email: sabine.ertl@fwf.ac.at

Natascha Dimovic
Operative project officer
Department of Natural Sciences and Technology
Telephone number: +43 676 83487 8402
Email: natascha.dimovic@fwf.ac.at

Funding criteria and regulations

Funding commitment

Total FWF funding commitment: 1,500,000.00 EUR

This commitment is solely to the Call topic "Quantum Phenomena and Resources"

Funding regulations

The FWF is a funding agency that funds basic sciences only. Hence, only proposals in the research Call topic "Quantum Phenomena and Resources" will be accepted.

Eligibility of a partner as a beneficiary institution

The same eligibility criteria apply as for regular Principal Investigator Projects. These criteria need to be fulfilled by the applicants.

Please find the application guidelines on the FWF Webpage: Principal Investigator Projects - FWF

Austrian applicants are encouraged to contact FWF prior to submission.

Submission of the proposal at the national level

The submission criteria follow the same as Principal Investigator Projects - FWF. This includes eligible costs as well.

In addition to the application submitted to the Call Secretariat, administrative and financial data as well as academic abstracts (in accordance with the FWF guidelines for Principal Investigator Projects) must also be submitted to the FWF using the online portal elane. The applicants must choose **"KIN – International Multilaterale Initiativen".**

The submission deadline, **December 9th 2025, 14:00 CET** must be strictly adhered to. Submissions after the deadline will not be considered.

The FWF does not finance infrastructure or basic equipment at research institutions. Overheads may not be requested.

Austrian applicants are encouraged to contact FWF prior to submission.

Additional requirements

The FWF only funds proposals in the Call topic "Quantum Phenomena and Resources".

The FWF generally does not have any minimum or maximum limits but, given the limited nature of the financial commitment of the FWF to this Call (see above), we do expect Austrian participants not to exceed the average range of an FWF stand-alone Project (typically 300,000.00 EUR to 450,000.00 EUR).

Please be aware that the FWF eligibility criteria apply. See Principal Investigator Projects - FWF.

In case one or more partners would be found not eligible this would lead to dismissal of the proposal.

Participating organisation

Fund for Scientific Research – FNRS

Contact

Florence Quist
Senior Scientific Officer
Telephone number: +32 2 504 9351
Email: Florence.quist@frs-fnrs.be

Funding criteria and regulations

Funding commitment

300,000.00 EUR / project

Funding regulations

PINT-MULTI Regulations (FR / EN)

Project duration is set at max. 36 months. If the project involves the recruitment of a PhD student, the project duration of the F.R.S.-FNRS sub-project could be up to four years. Since this is a co-financed call, this extra year should not be included in the proposal see article III.6 of the PINT-MULTI Regulations (FR / EN).

Eligibility of a partner as a beneficiary institution

All eligibility criteria can be found in the PINT-MULTI Regulations (FR / EN)

Submission of the proposal at the national level

Applicants to F.R.S.-FNRS funding must provide basic administrative data by submitting an administrative application on e-space within 5 working days after the general deadline of the QuantERA Call to be eligible. Please select the “PINT-MULTI” funding instrument when creating the administrative application. Proposals invited to the second stage will be able to complete the pre-proposal form and provide information for the full proposal upon validation by the F.R.S.-FNRS.

Additional requirements

N/A

Participating organisation

The Research Foundation – Flanders (FWO)

Contact

Toon Monbaliu (FO)
Advisor European Programmes
Telephone number: +32 (0)2 550 15 70
Email: europe@fwo.be

Kristien Peeters (SBO)
Science Policy Advisor
Telephone number: +32 (0)2 550 15 95
Email: europe@fwo.be

Funding criteria and regulations

Funding commitment

Total FWO funding commitment: 700,000.00 EUR

Maximum funding per project/consortium: 350,000.00 EUR (overhead included)

Anticipated number of fundable projects with FWO partners: 2-3

Funding regulations

In essence, the FWO only intends to fund projects within the Quantum Phenomena and Resources (QPR) topic, focusing on basic quantum science and fundamental physics. If duly justified that the tasks performed by a researcher from Flanders (FWO) within an Applied Quantum Science (AQS) project/consortium fit the FWO funding channels that are integrated within this transnational Call (and as such include basic research), this might also be accepted.

Eligibility of a partner as a beneficiary institution

The FWO integrates two of its national/regional funding channels within this multilateral framework. The choice of the FWO funding channel depends on the type of project the researchers from Flanders wish to undertake (i.e. more fundamental in nature (FO) vs. strong emphasis on valorisation potential (SBO)).

The scope and the eligibility of institutions and its researchers can be verified in the relevant chosen funding channel documents and regulations, which can be consulted on the FWO website:

- FWO Research Projects (FO)
- Strategic Basic Research (SBO)

Or by contacting the FWO contact points mentioned above.

Additional eligibility requirements for this framework

- Participation in this Call does not interfere with the ‘national’ FWO calls and is consequently not taken into account for calculating the max. available number of new

applications and running projects combined. **However, researchers can only participate within 2 different international consortia in this Call as partner, and only once as coordinator.**

- Projects aiming at the development of a spin-off company are not eligible in this context. This potential 'route' should not be mentioned in the FWO pre-registration form, as such.
- The project duration is limited to 36 months, which implies the funding has to be budgeted and spent accordingly. An automatic prolongation and using positive (financial) balances after the end date is not applicable in this framework. As such article 28 of the FWO Research Projects and article 14 of the Strategic Basic Research (SBO) regulations do not apply here.
- The PI, for each of the participating institutions applying for FWO funds, must **hold an appointment that fully covers the duration of the research project.**
- Linked to the above, when it comes to the FWO research project regulations (FO): article 10, §7 is not applicable in this Call. I.e. supervisors (-spokespersons), or coordinators/consortium partners in this context, who are granted **emeritus status** during the calendar year of submission of the project application or during the duration of the project execution, are **not eligible.**

Eligibility of costs

The FWO foresees a budget of 700,000.00 EUR in total, which guarantees at least two projects. The respective funding channel regulations apply (see links to national rules above: 'Eligibility of a partner as a beneficiary institution' section), unless specified otherwise, and both are **capped at max. 350,000.00 EUR per project/consortium** (incl. overhead, for which the calculation method diverges per funding channel - see explanation below).

For the **overhead calculation**, even though the overhead rates differ, the fundamental (FO) and strategic research projects (SBO) use the same approach. More specifically, a structural overhead rate should be applied on the project costs, with an overhead rate of **6% for 'FO projects'**, and a **17% overhead rate for 'SBO projects'**. Some practical examples:

- *FO: the sum of all costs (personnel, consumables, travel, subcontracting, etc.) amounts to 200.000 EUR, then the overhead will amount to 12,000.00 EUR (6% of 200,000.00 EUR) and the total requested cost is 212,000.00 EUR. This **total requested cost may never exceed the max. available amount of 350,000.00 EUR.***
- *SBO: the sum of all costs (personnel, consumables, travel, subcontracting, etc.) amounts to 200,000.00 EUR, then the overhead will amount to 34,000.00 EUR (17% of 200,000.00 EUR) and the total requested cost is 234,000.00 EUR. This **total requested cost may never exceed the max. available amount of 350,000.00 EUR.***

The overhead cost should be mentioned in the Quantera budget tables, but it should not be inserted in the national/regional FWO pre-registration form on the FWO E-loket (see below under 'Submission of the proposal at the national level').

Submission of the proposal at the national level

A **mandatory administrative application** via the personal FWO e-loket of the PI from Flanders is an essential part of the application process.

- For fundamental research projects (FO) select the application type: 'Research projects – European programme fundamental research'.

- <u>For strategic basis research projects (SBO)</u> select the application type: 'Research projects – European programme strategic basic research'.

In case the consortium includes more than one partner requesting funding from FWO, a <u>single</u> <u>online form</u> should be submitted by the main PI (promotor(-spokesperson)) itself, containing all relevant information for the different FWO applicants.

The deadline to submit this administrative application to the FWO is identical to the deadline of the joint transnational Call (pre-proposal stage). To ensure the eligibility of the proposal, it is recommended to consult the FWO administration at least one week in advance. <u>Failure to comply with these requirements can lead to ineligibility.</u>

Again, it is strongly advised to approach the FWO contact points mentioned above, in order not to jeopardize any research projects/consortia.

Participating organisation

Bulgarian National Science Fund - BNSF

Contact

Milena Aleksandrova
Expert
Telephone number: +359 884 171 363
Email: aleksandrova@mon.bg

Funding criteria and regulations

Funding commitment

Total BNSF funding commitment: 460,162.00 EUR

Maximum funding per project/consortium: 153,387.00 EUR (overhead included)

Anticipated number of fundable projects with BNSF partners: 2-3

Funding regulations

Type of research funded - Basic research, TRL 1-4

The budget contains costs directly related to the implementation of the project (direct costs) and indirect (overhead) costs. The direct costs must be used to carry out the project and be separately identifiable.

Eligible costs are specified in "National requirements and eligibility conditions" of Bulgarian National Science Fund available at:

Link (in

Bulgarian) Link

(in English)

Eligibility of a partner as a beneficiary institution

1) Accredited universities as defined in Art.85 para.1, p. 7 of the Higher Education Act,

2) Research organizations as defined in Art. 47, para 1 of the Higher Education Act:
http://lll.mon.bg/uploaded_files/zkn_visseto_obr_01.03.2016_EN.pdf

Submission of the proposal at the national level

Applicants have to submit an application form for national eligibility when submitting the porposals. The formulier should be filled and elecronically submitted in both Bulgarian and in English at: https://nims.egov.bg/login#/

BNSF requests the Bulgarian applicants to send the following documentation:

1. Proposal (in PDF format, sealed on the date of Call deadline),
2. Financial Plan for the Bulgarian research group, extracted from the - Financial Form submitted by the project consortium,
3. Work plan for the Bulgarian research group.

Additional requirements

The funded Bulgarian applicants will have to submit annual Narrative and Financial reports to BNSF, along with a Declaration on the prevention of double financing.

Eligible costs are:

- Personnel costs (employment of postdoctoral researchers),
- Equipment purchases and equipment maintenance costs,
- Consumables,
- Travel costs,
- Costs for open access publishing,
- Indirect costs - maximum 7% of total funds requested,
- Audit - maximum 1% of total funds requested.

Croatia HRZZ

Participating organisation

Croatian Science Foundation

Contact

Dejan Košćak
Senior advisor, Department for International Cooperation
Telephone number: +385 1 2356 607
Email: dejan@hrzz.hr

Funding criteria and regulations

Funding commitment

Max. funding per project: 200.000,00 EUR (1 project can be funded)

Proposed projects may last from 24 to 36 months.

Principal Investigators are not allowed to apply for funding in more than one proposal within this Call.

Funding regulations

Croatian applicants (Principal Investigators) are advised to consult the Guidelines for Applicants to Calls of the Croatian Science Foundation prior to submitting a proposal, in order to verify compliance with national funding terms and conditions.

The End Beneficiary, i.e. Principal Investigator may hold the role of Principal Investigator (PI) and/or team member on a maximum two HRZZ projects: either as the PI of one project and team member or co-PI on another, or as a team member or co-PI on two projects. Therefore, at the time of application, an eligible PI may hold only one active role on HRZZ projects. Active roles do not include the role of PI or collaborator on HRZZ projects scheduled to conclude by May 31, 2026. PIs of Croatian research groups on projects in bilateral programmes, Multilateral Academic Projects, and other ERA-NET projects are considered Co-PIs for the purpose of this Call.
Applicant can participate in only one project consortium in one ERA-NET Cofund Call.

Eligibility of a partner as a beneficiary institution

Eligible applicants are:

- higher education institutions
- scientific institutes
- other research-oriented legal entities that employ at least five PhD holders who have been employed at the institution for at least one year

Submission of the proposal at the national level

The HRZZ requests the Croatian applicants to send the following documentation (in English), not later than 3 working days after the submission of the proposal:

4. Proposal (in PDF format, sealed on the date of Call deadline),

5. Financial Plan for the Croatian research group (the part to be financed by the HRZZ), extracted from the - Financial Form submitted by the project consortium,
6. Work plan for the Croatian research group,
7. Institutional Support Letter - a written commitment from the Croatian applicant's institution accepting the proposed research and committing to its administration (signed and certified by the authorized representative of the institution),
8. Signed Letter of Commitment for participation in the project for all associates not employed at the applicant's institution.

The electronic version of the requested documentation shall be submitted via email to the following address: international@hrzz.hr.

Additional requirements

The funded Croatian applicants will have to submit annual Narrative and Financial reports to HRZZ, along with a Declaration on VAT status and Declaration on the prevention of double financing for the year in question.

Eligible costs are:

- Research costs,
- Personnel costs (employment of postdoctoral researchers),
- Equipment purchases and equipment maintenance costs,
- Dissemination, training, and cooperation costs,
- Costs for open access publishing,
- Indirect costs - maximum 15% of total funds requested, only if they are directly connected with project activities and they cannot be placed into any of the categories of eligible costs.

Participating organisation

Ministry of Education, Youth and Sports

Contact

Michal Vávra
Policy Officer
Telephone number: +420 773 793 439
Email: michal.vavra@msmt.gov.cz

Funding criteria and regulations

Funding commitment

The overall funding commitment of the Ministry of Education, Youth and Sports of Czechia is 500,000.00 EUR for the Call Quantum Phenomena and Resources (QPR). Under exceptional circumstances, MEYS can finance also Czech participants in the AQS Call – e.g., when Czech partner is ineligible for funding by TA CR due to the fact that the basis of its activities in the project proposal is defined as fundamental research. In such a case, the applicant is obliged to contact the MEYS beforehand and consult the situation.

Funding regulations

Eligible costs for a Czech participant involved in a project consortium are defined *by the Act No. 130/2002 Coll. on Support of Research, Experimental Development and Innovation from Public Funds and on Amendment to Some Related Acts*. The maximum indirect costs set for the present Call are 25% (flat rate) of direct costs without the sub-contracting.

The aid intensity for activities carried out by a research organisation might be at the level of 100% provided that the research organisation complies entirely with requirements stipulated by the Article 2.1.1 "Public funding of non-economic activities" of the *Framework for State Aid for Research and Development and Innovation (2014/C 198/03)* and proves it by means of the *Statutory Declaration*.

Should the above-stated criteria not be fulfilled by the Czech participant, funding rates will be adjusted appropriately by the Ministry of Education, Youth and Sports and will reach the level of 100% for fundamental / basic research activities, 50% for applied research activities and 25% for experimental development activities.

Each Czech participant in a project consortium is also requested to specify the costs related to the envisaged R&D activities in detail by using the Eligible Costs Specification template available on websites of the Ministry of Education, Youth and Sports: http://www.msmt.cz/vyzkum-a-vyvoj2/quantera.

Eligibility of a partner as a beneficiary institution

The participants from Czechia in the projects´ consortia must meet the criteria of *the research and knowledge-dissemination organization* (hereinafter referred to as "research organisation") in accordance with the *Framework for State Aid for Research and Development and Innovation (2014/C 198/03)*. These might be public universities, public research institutes and/or other entities classified as research organizations.

It is obligatory that the Czech participants involved in the projects´ consortia prove compliance with the eligibility criteria and fulfilment of the conditions set by *§ 18 of the Act No. 130/2002 Coll. on Support of Research, Experimental Development and Innovation from Public Funds and on Amendment to Some Related Acts* by means of a *Statutory Declaration*. The required procedure is described, and Statutory Declaration form will be available on website of the Ministry of Education, Youth and Sports: http://www.msmt.cz/vyzkum-a-vyvoj-2/quantera.

Submission of the proposal at the national level

The Full Project Proposal form and all the other requested documentation (Statutory Declaration and Eligible Costs Specification) shall be sent by each Czech participant in a project consortium to the Ministry of Education, Youth and Sports no later than **5th December 2025** via means specified at the website of the Ministry of Education, Youth and Sports: http://www.msmt.cz/vyzkum-a-vyvoj2/quantera.

In the case that project with Czech participants is recommended for funding based on the international evaluation outcomes and is approved by the Call Steering Committee, the Ministry of Education, Youth and Sports may ask the successful Czech applicants to submit additional documents in order to issue the Grant Agreement needed for providing the national financial support according to the rules stipulated by the Ministry of Education, Youth and Sports

Additional requirements

N/A

Participating organisation

TA CR – Technology Agency of the Czech Republic

Contact

Magdalena Pillasagua Ptakova
Coordinator for International Cooperation
Telephone number: +420 775 871 321
Email: magdalena.pillasagua@tacr.cz

Funding criteria and regulations

Funding commitment

Funding available: 1,000,000.00 EUR

Maximum permissible aid intensity for the Czech part of the project is 80% of the total eligible costs, based on the type of entity.

Czech applicants will be financed from the Programme to Support Applied Research and Innovation SIGMA (Sub-objective 4 – International collaboration).

Funding regulations

Type of research funded
Applied research, i.e. industrial research and experimental development.

Eligible costs:

- Personnel costs (including scholarships),
- Subcontracting costs (max. 20% of total eligible costs throughout the whole project period),
- Other direct costs (write-offs, protection of intellectual property, operating expenses, travel costs, consumables),
- Indirect costs (overheads) – full cost/flat rate 25% (indirect costs in the respective year are calculated as 25% of the sum of the personnel costs and other direct costs in the same year).

Eligibility of a partner as a beneficiary institution

Eligible applicants:
1. Enterprises (according to Annex 1 of the Regulation),
2. Enterprises who act as natural persons according to Annex 1 of the Regulation engaged in an economic activity pursuant to Act no. 455/1991 coll. on Trades (Trade Act),
3. Research organizations (according to Article 2 paragraph 83 of the Regulation). TA CR excludes the disbursement of individual aid to an enterprise:

- Against which a recovery order has been issued which is unpaid,
- Meeting the definition of an “undertaking in difficulty”,
- Which has not met the obligation to publish the financial statements for the years 2022, 2023, 2024 in the respective register - the so-called "Veřejný rejstřík”
- Which has not disclosed its ownership structure in the so-called “Evidence skutečných majitelů”.

Submission of the proposal at the national level

Mandatory forms to be submitted:

- A Sworn statement of the applicant.
- Completed “TACR Application Form” Excel file (submitted by the main Czech applicant only)*
- If the applicant plans to achieve “Nmet” type of result, the mandatory form for Nmet result needs to be attached*
- If the applicant plans to achieve the “Patent” type of result, patent search must be substantiated*
- Sworn statement of the composition of the consortium (only if Czech enterprise is part of the project consortium; submitted by the main Czech applicant only).

All documents proving the eligibility of the Czech partner stated above shall be submitted via the TACR data box (TACR data box ID: afth9xp).

Submission deadline: 5 December 2025, 17.00 CET
All mandatory documents to be found on TA CR website.

**Applicants who will not submit this mandatory form (if relevant) via databox before the deadline will be considered as not eligible for TA CR funding.*

Additional requirements

Supported results

Projects that achieve at least one of the following types of results can be supported in this Call. The type of the result has to be clearly described in the project proposal:

- P - patent
- G - technically realised results - prototype, functional sample
- Z - pilot plant, proven technology
- R - software
- F - results with legal protection - utility model, industrial design
- N - Certified methodologies and practices, treatment, conservation methods, procedures and specialised maps with professional expert content
- O – Other

Results not to be recognized only in combination with at least one other result listed above, i.e. not as a single result of a given project:

- H - results reflected in non-legislative directives and regulations binding within the competence of the respective provider and results reflected in the approved strategic and conceptual documents of the state or public administration.

Project start and end

Please note that following the national legislation, Czech applicants shall start within 120 days from the funding decision being communicated by the Call Management (60-day period to make a contract + 60-day period to start the project).

Guide for Czech applicants and all mandatory forms are available on TA CR website (in Czech).

Participating organisation

Estonian Research Council

Contact

Margit Suuroja
Senior Adviser
Telephone number +372 731 7360
Email: margit.suuroja@etag.ee

Funding criteria and regulations

Funding commitment

Total: 300,000.00 EUR

The maximum amount of requested funding per project is 150,000.00 EUR for a project partner and maximum 300,000.00 EUR for a project coordinator for a total period of up to three years.

Funding regulations

The budget contains costs directly related to the implementation of the project (direct costs) and indirect (overhead) costs. The direct costs must be used to carry out the project and be separately identifiable.

Direct costs:

Personnel costs are monthly salaries (along with all state taxes, contributions, and compensations arising from law) of the project participants, calculated according to their commitment and in proportion to their total workload at their Host Institution.

Other direct costs are:

- Travel costs that may cover expenses for transport, accommodation, daily allowances and travel Insurance. If the project is funded from the European Regional Development Fund (Mobilitas 3.0) resources, travel costs are eligible only for travels abroad,
- Consumables and minor equipment directly and fully related to the project,
- Publication and dissemination of project results,
- Organising meetings, seminars or conferences (e.g., room rent, catering, equipment rental and related costs),
- Fees for participating in scientific forums, conferences and other events directly and fully related to the project,
- Patent costs,
- All other costs that are identifiable as clearly required for carrying out the project (e.g., translation, copy editing, webpage hosting, postage costs etc.) and are directly and fully related to the project.

Indirect costs (overhead) are costs that cannot be identified as specific costs directly linked to the performance of the action and/or should cover the general expenses of the Host Institution related to the management of the grant. Office consumables and costs for equipment and services intended for general use (e.g., phone bills, copy service, printer) should be

covered from the indirect costs. Indirect costs are 15% of the personnel costs of the Estonian sub-project.

Subcontracting costs are direct costs. Subcontracting costs should cover only additional or complementary research related tasks (e.g., analyses, conducting surveys, building a prototype, etc.) performed by third parties. Subcontracting costs should not be included in the overhead calculation. The activities and budget should be described in the proposal. Core project tasks should not be subcontracted. Subcontracting costs may not exceed 15% of the total costs of the Estonian sub-project.

Double funding of activities is not acceptable (i.e. the same cost item may not be reimbursed twice).

If several Estonian institutions participate in one proposal, the sum of their requested budgets may not exceed the maximum contribution of the respective national Funding Organisation indicated in the Call documents.

State Aid

If the Host Institution is an undertaking, EU Regulations on State aid and de minimis aid must be taken into account when requesting funding. An undertaking is any entity, be it a natural or a legal person, engaged in an economic activity, regardless of its legal status and the way in which it is financed.

Eligibility of a partner as a beneficiary institution

The Host Institution may be any legal entity that is registered and located in Estonia and has an Estonian bank account. Restrictions to eligible institutions and applicants may however be stated in the general Call conditions which also apply to Estonian participants.

The Principal Investigator (PI) is a researcher who acts as the Estonian team leader in the project proposal. He/she will be responsible for how the grant is used and how the Estonian part in the project is executed. The PI must be clearly identified in the proposal.

The Principal Investigator:

1. Must have an updated public profile in the Estonian Research Information System (ETIS) by the submission deadline (in case of two-stage application, by the preproposal deadline).

2. Must hold a doctoral degree or an equivalent qualification. The degree must be awarded by the submission deadline (in case of two-stage application, by the preproposal deadline) at the latest.

3. Must have published at least three articles that comply with the requirements of Clause 1.1 of the ETIS classification of publications, or at least five articles that comply with the requirements of Clauses 1.1, 1.2, 2.1 or 3.1, within the last five calendar years prior to the proposal submission deadline (in case of two-stage application, prior to the preproposal deadline).1 Patents are equalled with publications specified under Clause 1.1. A monograph (ETIS Clause 2.1) is equalled with three publications specified in Clause 1.1 if the number of authors is three or fewer. If the applicant has been on maternity, paternity, or parental leave, in compulsory military service, or if there have been other exceptional circumstances (e.g., serious illness), they can request the publication period requirement to be extended by the relevant period of time.

If the Principal Investigator has received the PhD degree outside Estonia, its correspondence to an Estonian doctoral degree must be recognised by either the Estonian ENIC-NARIC Centre or the Host Institution in accordance with the Regulation of the Government of the Republic of

April 6, 2006, No. 89 "Evaluation and academic recognition of documents proving foreign education and the name of the qualification awarded in the foreign education system terms and conditions of use". The Funding Organisation may ask for a relevant Evaluation Report2.

If several Estonian institutions participate in a proposal, all institutions must have a Principal Investigator who meets the national eligibility requirements.

Submission of the proposal at the national level

No

Additional requirements

Following the restrictions laid down in Article 7 of the Regulation of the European Parliament and of the Council No 2021/1058 of 24 June 2021 on the European Regional Development Fund and on the Cohesion Fund research and other activities related to fossil fuels and their use, as well as other activities not eligible as per Article 7 of the Regulation, cannot be funded from the European Regional Development Fund (Mobilitas 3.0) resources.

All project-related costs (of Estonian partner) must be incurred no later than 31.08.2029.

Participating organisation

Research Council of Finland

Contact

Katrine Mahlamäki
Science Adviser
Telephone number: +358 29 533 5062
Email: katrine.mahlamaki@aka.fi

Funding criteria and regulations

Funding commitment

Total: 1,000,000.00 EUR

The Research Council of Finland plans to grant up to 450,000.00 EUR per partner (150,000.00 EUR per year).

Funding regulations

In general, the conditions and restrictions on Academy Projects apply to the funding. The Research Council applies the full cost model in its funding, and the Research Council's funding contribution for a project can come to no more than 70% of the total project costs. Funding can be granted to research teams for the purposes of hiring scientific staff, for the acquisition of equipment and supplies, and for other expenses arising, for instance, from researcher mobility and networking activities. The costs of making publications open access are included in the overheads of the sites of research and thus form part of the costs of the basic facilities they provide. The PI's salary costs must not be significant in relation to the project's total costs. For example, a three-year research project must not include more than 4.5 months of the PI's effective working hours. This is equivalent to about 1.5 months a year.

Research Council funding cannot be used for economic activity. Economic activity is defined as all activity where goods or services are offered on an open market regardless of whether profits are pursued or generated. When an organisation is also engaged in economic activities, separate accounts must be kept of the funding and costs of and the revenue generated by such activities.

The Research Council of Finland advises that funding recipients as soon as possible after the funding decision has been made agree in writing between all project partners on the rights of ownership and use of the project's research results. At the launch of the project, the partners should also agree on the rights of ownership and use of the materials and data to be used by the project.

Eligibility of a partner as a beneficiary institution

Finnish applicants are recommended to contact the Academy of Finland before proposal submission for the purpose of checking national funding terms and conditions (e.g. national eligibility criteria).

In addition to their doctorate, the PI must have other significant scientific merits. Usually, they are a professor or docent-level researcher. In addition, the applicant must have a close

connection with Finland to support the implementation of a multi-year project. A PI requesting funding from the Academy of Finland may not participate in more than two proposals.

Submission of the proposal at the national level

Finnish partners of projects that have been selected for funding will be invited to submit national application forms through the Research Council's online system. The application must include an annex outlining the research security risk assessment and mitigation plan.

Additional requirements

Open Access to Publications and Research Data Policy

We require that the scientific publications on the results of research projects funded by the Research Council of Finland are open access, and that the research data produced by the projects are made widely available. The degrees of data openness may justifiably vary, ranging from fully open to strictly confidential. More information is provided here: https://www.aka.fi/en/research-funding/responsible-science/open-science/

Participating organisation

French National Research Agency

Contact

Maurice Tia
Scientific Project Officer
Telephone number: +33 1 72 73 06 90
Email: Maurice.TIA@agencerecherche.fr

Funding criteria and regulations

Funding commitment

The total ANR funding commitment is 2,500,000.00 EUR, the maximum funding per project is 450,000.00 EUR and 300,000.00 EUR per partner.

The document presenting the modalities of participation of the French applicants will be available in September 2025 at https://anr.fr/quantera-call-2025.

Please note that companies with economic difficulties cannot receive ANR subventions.

A DMP (Data management plan) must be transmitted to ANR and updated during the course of the project. Please refer to the ANR funding regulations for more details: https://anr.fr/fr/rf/

If applicable, Declarations of Due Diligence for the financed projects (Nagoya Protocol) must also be transmitted to ANR in due time.

Participating organisation

German Research Foundation

Contact

Andreas Deschner
Programme Director
Telephone number: +49 228 885 2959
Email: andreas.deschner@dfg.de

Funding criteria and regulations

Funding commitment

4,000,000.00 EUR for the proposals of the topic QPR. No funds are allocated for the topic AQS.

Funding regulations

- Funding is implemented as a Research Grant (*Sachbeihilfe*). Information about eligible cost categories (related to specific *Programme Modules*) and other specifics of the programme are available on the DFG website.
- The standard module that fits most needs is the *Basic Module* (form 52.01).
- The DFG grants 22% of the direct costs as overhead (*Programmpauschale*). Please choose 22% of the direct costs as the overhead for the submission to QuantERA.

Eligibility of a partner as a beneficiary institution

- The general DFG rules and conditions as defined in the current DFG *Guidelines Research Grants Programme* apply.
- The guidelines regarding the *Duty to Cooperate* do **not** apply.

Participation of commercial enterprises

If commercial companies are involved in the project, the cooperation agreement must meet the DFG requirements concerning exploitation rights granted to the cooperation. The model cooperation agreement provided by the DFG or the DESCA model cooperation agreement can be used for this purpose.
When using the DESCA model cooperation agreement, the following must be observed with regard to the granting of rights of use to the commercial enterprises involved in the project regarding the work results achieved by the DFG grant recipients in the funded project: If these rights of use, granted by the DFG recipients, are to be utilised by the commercial enterprises for the purpose of exploiting the work results achieved by the commercial enterprises in the project, *fair and reasonable conditions* must be agreed to for the rights of use (see Section
9.4.1 Option 1 of the DESCA Horizon Europe 2.0 model cooperation agreement in conjunction with Annex 5 of the Model Grant Agreement (Specific Rules on IPR)). In principle, the market value of the work results achieved by the grant recipient must be taken into account.

Submission of the proposal at the national level

- Proposals have to be submitted at the latest on 5 December 2025 (same day as submission deadline of QuantERA).
- If you do not have an elan account, you must register on the elan portal before submitting your proposal. Confirmation is usually provided within three working days. It is not possible to submit a proposal without prior registration.
- For each project, one copy of the proposal, the financial form, necessary offers for scientific equipment and accompanying documentation (see below) has to be submitted via the DFG's elan portal. All documents should be submitted using the Portable Document Format (PDF).
- A CV for each applicant requesting funds from the DFG has to be submitted. Please use the current template on DFG's site with submission instructions.
- Regardless of the number of PIs requesting funding from the DFG, the proposal has to be submitted to the DFG only once. The person uploading the documents enters the relevant data for all PIs requesting funding from the DFG.
- First steps on elan: Log-In > Proposal Submission > New Project / Draft proposal > Individual Grants Programme > "Proposal for a Research Grant" > Start online form.
- Please select the Call "QuantERA 2025 (QPR)".
- Overheads (*Programmpauschale*) will be calculated automatically in elan during submission do the DFG.
- It is compulsory to submit the project data form (to be made available before October 2025).

Equity and Diversity

The DFG strongly welcomes proposals from researchers of all genders and sexual identities, from different ethnic, cultural, religious, ideological or social backgrounds, from different career stages, types of universities and research institutions, and with disabilities or chronic illness. With regard to the subject-specific focus of this Call, the DFG encourages female researchers in particular to submit proposals.

Good Research Practice

According to a resolution of the DFG General Assembly, DFG funding may only be awarded to research institutions that have implemented the guidelines laid down in the Code of Conduct for Safeguarding Good Research Practice in their own regulations. The management of your institution is responsible for implementing the guidelines in a legally binding manner. In order to avoid delays in the disbursement of funding, please verify implementation within your institution in good time. For information regarding the implementation, please refer to the Research Integrity Portal. If you have any questions on this subject, please contact the Research Integrity team at the DFG Head Office.

Privacy Policy

We, the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation), take the protection of your personal data and its confidential treatment extremely seriously. Therefore, please refer to the DFG's Privacy Policy. If you intend to transmit personal data of third parties, please make sure to do so only if the necessary legitimation under data protection law exists. Before transmitting data of third parties to the DFG, please forward the DFG's Data Protection Notice to the individuals affected (data subjects). If there is a legitimate interest not to inform individuals beforehand (e.g. for reasons of secrecy or in case of a nomination or candidate proposal), these individuals should be informed no later than at the time of publication.

Participating organisation

Federal Ministry of Research, Technology and Space

Contact

VDI Technologiezentrum GmbH (VDI TZ) – funding agency on behalf of the BMFTR

Sebastian Krug
Procedural contact person
Telephone number: +49 211 6214 472
Email: krug@vdi.de

Dr. Bastian Hiltscher
Scientific contact person
Telephone number: +49 211 6214 441
Email: hiltscher@vdi.de

Funding criteria and regulations

Funding commitment

BMFTR budget earmarked for the funding of German participants in projects under the AQS topic: max. 4,500,000.00 EUR.

Funding regulations

Project proposals involving German participants requesting funding from the BMFTR must:

1. Clearly address and be in line with the Applied Quantum Science (AQS) topic,
2. Be in line with the BMBF Research Programme *'Quantum Systems'* and the Federal Government´s 'Quantum Technologies Conceptual Framework' which can be downloaded at https://www.quantentechnologien.de/service/publikationen.html.

Funding for German applicants under this Call will be granted based on §§ 23 and 44 BHO ('Bundeshaushaltsordnung') and the related administrative regulations and BMFTR/ BMBF guidelines. An entitlement to funding cannot be derived.

In particular, the national regulations *NABF*, *NKBF 2017* and *BNBest-mittelbarer Abruf-BMBF* apply. The documents can be found at: foerderportal.bund.de/easy/easy_index.php

Eligibility of a partner as a beneficiary institution

BMFTR funding of German participants is available:

- For universities and research institutes/organisations (funding up to 100% of the total eligible costs/expenditures; while overhead is not eligible for universities, they will receive an additional lump sum ('*Projektpauschale*') equivalent to 20% of the eligible expenditures),
- For companies (funding up to 50% of the total eligible costs; in addition, bonuses may be granted to SMEs under GBER Article 25(6)(a) ('SME bonus') and to companies under Article 25(6)(b)(i) ('bonus for effective cooperation')).

Submission of the proposal at the national level

No additional national forms need to be submitted during the QuantERA application process. However, previous registration with the funding agency (VDI TZ) is mandatory (see section *additional requirements* below).

Successful German participants must submit complete BMFTR application forms (AZA/AZK including annexes) within two months after the official notification of the project selection. Non- compliance may result in exclusion.

Additional requirements

Extending the stipulations of the Call announcement, the following requirements apply to German participants respectively to project proposals involving German participants:

- **Mandatory registration:** German applicants are required to contact and to register with the responsible BMFTR funding agency ('*Projektträger'* VDI TZ – see section *contact* above) prior to the submission of a project proposal they are involved in.
- **Compulsory industry involvement:** Due to the application orientation of the AQS topic, the involvement of at least one company (from Germany or another EEA country) in the project is compulsory.
- **Obligatory exploitation plan:** In section 2.2 proposal must contain a concrete, appropriately detailed plan (who? what? when?) for the subsequent innovation steps, the technical application and the future commercial utilisation of the results in the EEA. Please note that in this context, the term *application* does not imply the utilisation of the results for merely scientific purposes.
- **Demonstrating additional benefit:** To be eligible for BMFTR funding, project proposals must demonstrate an additional benefit due to the transnational cooperation.
- **Funding limits:** The requested BMFTR funding must amount to at least 120 k€ per German partner over the entire duration of a project. Moreover, at least 50% of the total eligible costs/expenses on the part of the German participant(s) in a project must be personnel costs/expenses.

Participating organisation

NKFIH, Hungarian National Research, Development and Innovation Office

Contact

Flora Csesznok
National Contact Point
Email: ncp@nkfih.gov.hu

Funding criteria and regulations

Funding commitment

The total indicative national funding for this Call is 450,000.00 EUR.

Funding regulations

The conditions for the call for ERA-NET COFUNDs (2024-1.2.2-ERA_NET) are applicable. For details see: https://nkfih.gov.hu/palyazoknak/nkfi-alap/palyazat-era-net-cofund-ejp-cofund-egyeb-multilateralis-programok-2024-122-era-net/palyazati-felhivas

Eligibility of a partner as a beneficiary institution

The conditions for the call for ERA-NET COFUNDs (2024-1.2.2-ERA_NET) are applicable. For details see: https://nkfih.gov.hu/palyazoknak/nkfi-alap/palyazat-era-net-cofund-ejp-cofund-egyeb-multilateralis-programok-2024-122-era-net/palyazati-felhivas

Submission of the proposal at the national level

Not required prior to the QuantERA Call submission.

Following the international selection of the projects to be funded, a proposal should be formally submitted to NKFIH.

Participating organisation

Taighde Éireann – Research Ireland

Contact

Dr Maria Nash
Senior EU affairs Executive

Maria.nash@researchireland.ie

Eu-cofund@researchireland.ie

Funding criteria and regulations

Funding commitment

Maximum budget per project:

Up to €330,000 direct costs, €405,000 including overheads, for Irish-based researchers applying as a project partner.

Up to €405,000 direct costs, €530,000 including overheads, for Irish-based researchers applying as a project coordinator.

Eligible costs:

- Salary-related costs for research personnel. Please note the Irish-based partner cannot request their own salary.
- Equipment
- Travel
- Direct running costs (materials and consumables)
- Dissemination and knowledge exchange costs
- Subcontracting costs are considered an eligible budget category however strong justification for subcontracting must be provided and pre-approved directly with Research Ireland in advance of proposal submission.
- Overheads should be calculated as 30% of the direct costs, but excluding therefrom the cost of all equipment identified in the application.

Unless otherwise stated, all rules regarding eligible and ineligible costs apply as defined within Research Ireland's Grant Budget Policy.

Funding regulations

Please refer to the Research Ireland call webpage for funding regulations and applicant eligibility in full: https://www.researchireland.ie/funding/

Eligibility of a partner as a beneficiary institution

Only an academic partner or coordinator based in an eligible Irish Host Research body may apply for Research Ireland funding. Please refer to Research Ireland's Policies and Guidance for the list of eligible Research Performing Organisations: Eligible Research Bodies Policy

The Irish-based applicant must:

- hold a PhD or [equivalent qualification] for at least 3 years by the proposal deadline. The official date is defined as the day, month and year that the degree was conferred i.e., the month and year printed on the official PhD certificate.

AND

- be a member of the academic staff of an eligible Research Body (permanent or with an active contract that covers the period of the grant)

OR

- be a contract researcher with a contract that covers the period of the grant, who is recognised by the eligible Research Body as an independent investigator and will have an independent office and research space for which he/she will be fully responsible for at least the duration of the Research Ireland grant

OR

- be an individual who will be recognised by the eligible Research Body upon receipt of the grant as an academic staff or as a contract researcher as defined above. The applicant does not necessarily need to be employed by the Research Body at the time of the application submission.

AND

- be an author on at least three international peer-reviewed articles. Only original research publications, and not review articles or other secondary research literature, are acceptable.

Submission of the proposal at the national level

Please give a brief notification of your intent to submit a proposal through email before the submission deadline. Within the notification, please include the following information:

- List of project partners
- Irish Host institution
- Total budget request to Research Ireland
- Whether you intend to apply as a coordinating or non-coordinating partner

Additional requirements

Applicants are advised that funding awarded by Research Ireland under the QuantERA Programme will be subject to, and must comply with, State aid rules and the conditions of the EU Commission General Block Exemption Regulation (GBER). Funding will be awarded to successful applicants under Article 25, in respect of aid for research and development projects. For further details please consult: Taighde Éireann-Research Ireland Research and Innovation Scheme 2021-2026

Participating organisation

Israel Innovation Authority - ISERD

Contact

Mr. Danny Seker
Senior Director: Digital, Industry and Space; Civil Security for Society
+972 52 733 4127 – Dan@iserd.org.il

Bat Zion Fisch
NCP Cluster 4: AI, Digital and Industry
+972 54 435 8285 - Batzion.F@innovationisrael.org.il

Funding criteria and regulations

Funding commitment

0.75 million
Anticipated number of fundable projects with IIA partners: 3-4

Funding regulations

According to national regulations. Please refer to the following documents:

Sub-route 5a' – Applied Research Consortia

Procedure for the implementation of international agreements related to the European Framework Programme

The QuantERA III call will appear at ISERD site in due course, please refer to ISERD site before submitting the national application.

Eligibility of a partner as a beneficiary institution

Israeli company engaged in R&D

Israeli Academia Institute –subject to the following:
1. Commitment of an Israeli company (a partner in the consortium) to fund at least 10% of the Academia's approved budget.
2. Joint tasks/working package of the specific academia and company
Eligible cost and expenses – In line with national rules and regulation as detailed in;
The national call of QuantERA

The QuantERA III call will appear at ISERD site in due course,
Please refer to ISERD site before submitting the national application.

Submission of the proposal at the national level

National application to be submitted mandatorily

Deadline for submission: 12.12.2025, at 12:00 PM (IST)

Additional requirements

Israeli applicants must submit a parallel national application to the Israel Innovation Authority (IIA) in addition to the international proposal submission on the QuantERA III website.

The national call for proposals for Israeli entities will be published on the **Israel Innovation Authority** website shortly after the official announcement on the QuantERA III partnership website.

Where to Submit

The submission of the national application in Israel is made exclusively through the online portal on the **Israel Innovation Authority (IIA)** website. Applicants must log into their personal account on the website to access the dedicated submission forms.

To submit the application in Israel, one must act according to the following:

- The "Procedure for the implementation of international agreements related to the European Framework Programme" (hereinafter: "ISERD Procedure").

The instructions of Sub-route 5a' – Applied Research Consortia, as detailed in the ISERD Procedure.

Italy MUR

Participating organisation

Ministero dell'università e della ricerca

Contact

Aldo Covello
Programme manager
Telephone number: 375 510 2431
Email: aldo.covello@mur.gov.it

Sonia Angiolin
Project officer
Email: sonia.angiolin@est.mur.gov.it

Funding criteria and regulations

Funding commitment

Total: 500,000.00 EUR

Funding regulations:

- Maximum funding per project: 100,000.00 EUR

Eligibility of a partner as a beneficiary institution:

- Eligible partners: Universities, Research institutes, Research organizations in accordance with EU Reg. n. 651/2014 of the European Commission - June 17, 2014, Enterprises, foundations and non-economic entities.

Submission of the proposal at the national level

- In addition to the project proposal, which shall be submitted at European level, the Italian participants are requested to submit a national application to MUR, through the national web platform, available at the following link: https://banditransnazionali.mur.gov.it
- **The national application must be submitted by the same deadline established by the international Call. Any participant who does not submit its national documents by this deadline will be considered not eligible for funding.**

Additional requirements

- The complete list of criteria and provisions legally valid, which must be respected by all the Italian participants, is included in the "Avviso integrativo nazionale", to be published on the dedicated web page on MUR website: http://www.ricercainternazionale.miur.it/era/european-partnership-2021-27 and in the national web platform https://banditransnazionali.mur.gov.it

Participating organisation

Latvian Council of Science

Contact

Maija Bundule
Head of Unit - International Research Programs Unit
Telephone number: +371 26514481
Email: maija.bundule@lzp.gov.lv

Funding criteria and regulations

Funding commitment

Total: 600,000.00 EUR

Funding regulations

For project coordinator or partner a maximum of 100,000.00 EUR can be requested per project year. The grant can be used to cover all project-related costs, which include as direct costs personnel costs (corresponding to the level of activity in the project), travel costs, subcontracts (up to 25% of total direct costs), other direct costs such as materials, consumables and publication costs, equipment (only depreciation costs), and etc. Indirect project costs can reach a maximum of 25% of total direct costs, excluding subcontracting costs.

Eligibility of a partner as a beneficiary institution

Following legal persons are eligible for funding:

(1) R&D institutions - research institutes, universities, higher education establishments, their institutes and research centers. R&D institution (research institute, university, higher education establishment, research center etc.) must be listed in the Registry of Research Institutions operated by the Ministry of Education and Science of the Republic of Latvia.

(2) Enterprises. Enterprise must be registered in the Registry of Enterprises of the Republic of Latvia and most of its R&D&I activities must be carried out in the Republic of Latvia, be able to submit its financial statements for the last two closed financial years and has a Latvian bank account.

Submission of the proposal at the national level

Not requested

Additional requirements

No more than two partners from Latvia may participate in the project.

A person, acting as a project leader or partner’s principal investigator can submit or take part only in one project proposal per Call.

Lithuania LMT

Participating organisation

Lietuvos mokslo taryba (Research council of Lithuania)

Contact

Saulius Marcinkonis
Adviser
Telephone number: +370 676 17256
Email: saulius.marcinkonis@gmail.com

Funding criteria and regulations

Funding commitment

Total: 300,000.00 EUR

Up to 150,000.00 EUR per project (for project partner) or up to 200,000.00 EUR per project (for project coordinator).

Funding regulations

The general funding rules of Research Council of Lithuania (LMT) apply:

The applicant is Lithuanian higher education and research institution (which is listed in the Register of Ministry of Education and Science of Republic of Lithuania),

SME eligible to apply only in collaboration with Lithuanian higher education and research institution,

The applicant who intends to act as a project leader (PL) or principal investigator (PI) has to be a scientist (researcher holding at least a Ph.D. degree).

A person, acting as a PL, PI or a core group member can participate only in one proposal per Call.

The workload of the core members of project team (PL and PI) must be ≥20 hours multiplied by the duration of the project in months.

National funding will be provided according to the General Rules of the Research Council of Lithuania for the competitive funding of the Research and Dissemination Projects approved by the Order No V-176 of the Council Chairman on 4 April, 2019.

Eligibility of a partner as a beneficiary institution

-

Submission of the proposal at the national level

Applicants are not requested to submit any national forms during pre-proposal and full proposal submission stages.

Additional requirements

Requested to submit regular financial, mid-term and final reports at the national level.

Participating organisation

Luxembourg National Research Fund

Contact

Christiane Kaell

Head of Research Programmes

Telephone number: +352 691 362 817

Email: christiane.kaell@fnr.lu

Funding criteria and regulations

Funding commitment

350,000.00 EUR

Funding regulations

FNR’s programme for international collaborations: INTER - FNR

The FNR financial regulations specify the different costs categories and expenses eligible for reimbursement by the FNR.

Eligibility of a partner as a beneficiary institution

Public institutions performing research: https://www.fnr.lu/fnr-beneficiaries/

Submission of the proposal at the national level

Additional documentation must be submitted the FNR no later than 7 working days after the QUANTERA III deadlines. Submissions to the FNR must be done FNR Online Grant Management System.

Applicants will need to provide basic information and attach the proposal (including all further relevant documents) based on the form provided by the lead agency/organising network.

Additional requirements

A principal investigator may act in one proposal only. A consortium leader from a Luxembourg- based public research institution should be a researcher with solid experience of managing collaborative research projects.

Participating organisation

Ministry of Education, Sport, Youth, Research and Innovation - MEYR

Contact

David Muscat
Snr Manager – Programme
Implementation Telephone number: +356
2598 1074 Email: david.d.muscat@gov.mt

Funding criteria and regulations

The following legal persons are eligible for funding:

- Research institutes, universities, higher education establishments, their institutes and research centres etc.
- Enterprises and companies.

Entities must be registered in Malta.

Funding commitment

Total: 50,000.00 EUR

Funding regulations

YES

Eligibility of a partner as a beneficiary institution

YES

Submission of the proposal at the national level

Mandatory

Additional requirements

N/A

Participating organisation

Quantum Delta NL – QDNL

Contact

Dr. Julia Oesterling
Senior Advisor EU programs Telephone
number: +31-6-81110955 Email:
julia.oesterling@quantumdelta.nl

Funding criteria and regulations

Funding commitment

Total: 1,500,000.00 EUR

Maximum funding per the awarded project: 350,000.00 EUR

Funding regulations

Maximum number of fundable research groups

3-5

Eligibility of project duration

3 years

Eligibility of costs, types and their caps

One Postdoctoral researcher and/or non-scientific personnel (up to a max of 3 years) and/or PhD student, material costs, traveling costs, network and consortium costs. With a maximum project duration for three years, only funding for 3 years can be granted. However, in case of a 4-year PhD employment contract, this may be indicated and included in the project budget.

For all eligible applicants standard tariffs for personal costs apply (see: https://www.nwo.nl/en/salary-tables).

A standard material budget can be requested up to a maximum of 15,000.00 EUR per year. This budget covers third party costs (related to external services such as consulting-provided by companies or freelancers, capped at a rate of 125.00 EUR per hour), costs for consumables, small instruments and aids, cost for travel and lodging, costs for buying the rights to patents necessary for the project, for the duration of the project, miscellaneous costs directly related to the project. Costs for cleanroom (access to NanoLabNL) can be claimed, additionally.

Eligibility and assessment criteria are accessible on the website of QDNL (http://quantumdelta.nl/quantera).

Eligibility of a partner as a beneficiary institution

Full, associate, and assistant professors and other researchers with a comparable position may submit an application if they have a tenured position (and therefore a paid position for an indefinite period) or a tenure track agreement at Dutch universities, or NWO institutes, or at

other research organisations, that -at least in part- are funded by the Ministry of Education, Culture, and Science.

SMEs are eligible to join a project together with one of the eligible organisations mentioned above. SMEs cannot however request funding within this call.

Individuals who, in the framework of a tenure-track position, -do not yet have a permanent appointment can also be (co-) applicant. The employing institution, thus the dean or director of the institute, has to ensure a full 4-year PhD position employment contract.

Submission of the proposal at the national level

Yes. Submission of financial and scientific reports at national level is required in accordance with the rules of QDNL.

Additional requirements

Applicants must fit one of the themes of the NAQT (National Agenda of Quantum Technology), or the CAT (Catalyst) themes of QDNL. QDNL will host an information session in September 2025 for applicants from Dutch entities to brief them of the selection criteria and how to ensure compliance with scope and increase a successful application.

Participating organisation

The Research Council of Norway

Contact

Torgeir Waaga
Senior Advisor
Telephone number: +47 48234747
Email: twa@rcn.no

Funding criteria and regulations

Funding commitment

Total: 434,000.00 EUR

The project period can only be extended for the Norwegian project partner if the project period is extended for the project as a whole. The project period must not exceed the project period of the framework of the partnership.

The National Budget of the Call is 5 M NOK

The official exchange rate can be found here:

https://www.ecb.europa.eu/stats/policy_and_exchange_rates/euro_reference_exchange_rates/html/eurofxref-graph-nok.en.html

The exchange rate at the timepoint of the Call application submission deadline will be the valid exchange rate.

Funding regulations

The aided part of the research and development project is set by the State Aid Rules. The aid is assessed to fall within, and must only be used in accordance with, the EU's Block Exemption Regulation Article 25 (Commission Regulation (EU) No. 651/2014). Undertakings that fall under the state aid regulations' definition of "undertaking in difficulty" cannot receive support from the Research Council. For more information see: https://www.forskningsradet.no/en/apply-for-funding/funding-from-the-research-council/Conditions-for-awarding-state-aid/

Depending on the volume of submitted and eligible projects, up to 25% additional funding may be allocated to the Call to fund additional projects on the ranking list.

Eligibility of a partner as a beneficiary institution

Only research organizations on the RCN's list of approved research organizations are eligible for funding.

Submission of the proposal at the national level

Project partners of funded projects will have to submit national application forms to The Research Council of Norway after notification.

Additional requirements

- The participation must follow RCN's General Terms and Conditions for R&D Projects.
- Norwegian project partners will sign a separate contract with the RCN. The currency in the Norwegian contracts will be in NOK to avoid currency risk.
- The budget for the Norwegian partners shall follow RCN cost model and RCN regulations.

Participating organisation

National Centre for Research and Development

Contact

Krystyna Maciejko
Project Coordinator
Telephone number: +48 22 39 07 489
E-mail: krystyna.maciejko@ncbr.gov.pl

Funding criteria and regulations

Funding commitment

Maximum funding for the Call 2025: 1,500,000.00 EUR

Funding regulations

Detailed information (e.g. eligibility of the costs and funding rates) can be found at:
https://www.gov.pl/web/ncbr/platforma-konkursowa

NCBR supports only applied research and experimental development.

Eligibility of a partner as a beneficiary institution

- Research organisations[15],
- Enterprises[16] (micro, small, medium or large),
- Groups of entities (scientific consortia).

For research organisations an industrial partner is not required to establish a national scientific consortium.

Submission of the proposal at the national level

After international evaluation has been completed and the ranking list established, Polish participants from consortia recommended for funding will be invited to submit the National Application Form (NAF).

All eligible entities invited to submit NAF are obliged to use the rate of exchange of the European Central Bank dated on the day of opening of the Call.

NAFs will be then examined by the interdisciplinary panel of experts for international projects for the appropriateness of funding requested. Based on recommendations of the panel, the Director of NCBR issues a funding decision.

Additional requirements

N/A

[15] As defined in Article 2 (83) of General Block Exemption Regulation (Regulation No 651/2014).
[16] As defined in Article 1 of Annex 1 to the General Block Exemption Regulation (Regulation No 651/2014).

Participating organisation

National Science Centre

Contact

Alicja Dyląg
International Cooperation Officer
Email: alicja.dylag@ncn.gov.pl

Dr Anna Koteja
Coordinator for Physical Sciences and Engineering
Email: anna.koteja@ncn.gov.pl

National Science Centre: quantera@ncn.gov.pl

Funding criteria and regulations

Funding commitment

Total: 1,500,000.00 EUR

Funding regulations

Only proposals involving basic research (experimental or theoretical endeavours undertaken to gain new knowledge of the foundations of phenomena and observable facts, without any direct commercial use) may be submitted in response to the Call for proposals.

What are the eligible costs for Polish researchers?

We strongly encourage all applicants to read information on *eligible costs* included in the Annex to NCN Council's Resolution on awarding funding for research tasks funded or co-funded under international calls launched by the National Science Centre and carried out as multilateral collaboration UNISONO.

Applicants can apply for funding for all costs relevant, necessary and directly connected to the proposed research project including:

1. **Personnel costs (salaries):**

- **Full time remuneration:** funds for full-time employment of the principal investigator or post-doc(s); Please note, if the principal investigator is also the coordinator of an international consortium submitting a joint proposal a full-time remuneration of PI can be increased.
- **Additional remuneration for members of the research team**. Please note: if the principal investigator is also the coordinator of an international consortium submitting a joint proposal a budget for additional remuneration can be increased.
- **Salaries and scholarships for students and PhD students**. Please note: if the principal investigator is also the coordinator of an international consortium submitting a joint proposal the budget for salaries and scholarships for students and PhD students can be increased.

2. **Purchase or manufacturing of research equipment, devices and software** (the cost of an individual item of equipment must not exceed PLN 500,000.00).
3. **Other costs** such as:

- Purchase of material and small equipment,

- Outsourcing (costs of services rendered by third parties),
- Business trips (travel and subsistence costs),
- Visits and consultations,
- Compensation for collective investigators,
- Other costs crucial to the research project which comply with the Annex to NCN Council's Resolution on awarding funding for research tasks funded or co-funded under international calls launched by the National Science Centre and carried out as multilateral collaboration UNISONO.

4. **Indirect costs** must not exceed a maximum of 20% of the total eligible costs and may not be increased during the course of a research project.

   Additionally, indirect costs of up to 2% of direct costs may be spent on Open Access to publications and research data. Administrative personnel costs and costs of organisation of conferences have to be covered from overheads.

EURO exchange rate: 1 EUR = 4,2858 PLN

Eligibility of a partner as a beneficiary institution

Proposals may be submitted by the entities specified in Article 27 (1) of the Act on the National Science Centre (NCN Act). For further details on eligibility of applicants, as well as terms of entry, please check the Annex to the NCN Council Resolution on funding granted within calls for proposals for international research projects (UNISONO).

Who can act as Principal Investigator?

The Principal Investigator in the Polish research team must be at least a PhD holder when submitting a proposal. The Principal Investigator must be a person employed at the host institution for the project for the entire project duration period pursuant to at least a part-time employment contract. The Principal Investigator must reside in Poland for at least 50% of the project duration period.

Submission of the proposal at the national level

Up to 7 days from the full proposal submission deadline Polish applicants must submit their national applications in the OSF submission system. The application will include a budget that should be calculated according to the Annex to NCN Council's Resolution on awarding funding for research tasks funded or co-funded under international calls launched by the National Science Centre and carried out as multilateral collaboration (UNISONO).

If one international project includes partners from two or more different Polish institutions, these institutions apply as a group of entities. Each entity within the group has a separate budget, but the limit on the remuneration applies to the group as a whole (please see: UNISONO). Please note that groups of entities have higher limits on the remuneration.

Additional requirements

Applicants are obliged to adhere to the rules included in the Annex to NCN Council's Resolution on awarding funding for research tasks funded or co-funded under international calls launched by the National Science Centre and carried out as multilateral collaboration: UNISONO.

The rules for awarding NCN scholarships are laid down in the Regulations on awarding scholarships.

- Personal Data Processing at the NCN
- Open Access Policy at the NCN
- Data Management Plan requirements

Participating organisation

Executive Agency for Higher Education, Research, Development and Innovation Funding

Contact

Nicoleta Dumitrache
Expert international cooperation
Telephone number: +40 2 1302 3886
Email: nicoleta.dumitrache@uefiscdi.ro

Funding criteria and regulations

Funding commitment

Total: 500,000.00 EUR

Funding regulations:

- 250,000.00 EUR in case a Romanian institution is the Coordinator (together with other Romanian partner(s) – if it is the case),
- 200,000.00 EUR for one/all Romanian partner(s) participating in a proposal.

Eligible costs:

a. Staff costs,

b. Logistics expenses:

- Capital expenditures (cannot exceed 25% of the funding from the public budget),
- Expenditure on stocks – supplies and inventory items,
- Expenditure on services performed by third parties cannot exceed 25% of the funding from the public budget. The subcontracted parts should not be core/substantial parts of the project work,

c. Travel expenses,

d. Overhead (indirect costs) is calculated as a percentage of direct costs: staff costs, logistics costs (excluding capital costs and cost for subcontracting) and travel expenses. Indirect costs will not exceed 25% of direct costs.

Eligibility of a partner as a beneficiary institution: Universities, national institutes, SMEs, other institutions with R&D activity.

Submission of the proposal at the national level - the proposal will not be submitted at national level.

Additional requirements:

Funding rates vary in accordance with state aid legislation.

A person, as project manager, regardless of whether the Romanian institution is a project coordinator or partner, at the level of a transnational competition, can participate in a single project proposal.
For more information: https://uefiscdi.gov.ro/parteneriate-si-misiuni-europene

Participating organisation

Slovak Academy of Sciences

Contact

Martin Novak,
Program Officer
Telephone number: +421257510119
Email: mnovak@up.upsav.sk

Zuzana Panisova
Program Officer
Telephone number: +421257510245
Email: panisova@up.upsav.sk

Funding criteria and regulations

Funding commitment:

Total: 120,000.00 EUR

Funding regulations

Personnel costs up to 15 % of all direct costs or up to 30% of all direct costs, if the Slovak team is the consortium's coordinator.

*Overhead*s up to 20% of all direct costs.

No capital expenditures are allowed.

Eligibility of a partner as a beneficiary institution

Only research institutes and/or centres of the Slovak Academy of Sciences are eligible organisations for funding by SAS (up to 100%). The main applicant must have, at the time of submission, a contract(s) with one or several of the institutes/centres equivalent to more than 50% of the full-time employment valid for the whole duration of the project. Each member of the applicant's team must also have an employment contract or a fellowship with the same or another SAS institute/centre.

Submission of the proposal at the national level

No

Additional requirements

N/A

Participating organisation

Ministry of Higher Education, Science and Innovation

Contact

Anamarija Meglič
Function: Science Policy Officer - Directorate for Science and Innovation
Telephone number: + 386 1 478 46 14,
Email: anamarija.meglic@gov.si

Funding criteria and regulations

Funding commitment

Max. 900,000.00 EUR with the possibility of additional EC funding depending on the EC contribution (in the form of top-up funding).

Funding regulations

Estimated number of research partners that can be cofunded: 3 research projects with Slovenian applicants.

Maximum co-funding or grant awarded to a partner: max. 300,000.00 EUR per project (the actual amount and dynamics of co-funding shall be defined in the national co-funding contract).

Eligibility of a partner as a beneficiary institution

Type of research eligible for co-funding on the TRL scale:
Basic or industrial research TRL 1-6 (foreseen activities of the Slovenian partner).
The type of research to be carried out by Slovenian researchers must be identified and explained in the project proposal.

Submission of the proposal at the national level
Programme title:
Programmes of International Scientific Cooperation

Geographical coverage of the programme (national/regional):
National programme

Programme website:
https://www.gov.si/zbirke/projekti-in-programi/obzorje-evropa/partnerstva/

Legal basis:

- State Administration Act (Official Gazette of the Republic of Slovenia [*Uradni list RS*], Nos 113/05 – official consolidated version, 89/07 – Constitutional Court Decision, 126/07 – ZUP-E, 48/09, 8/10 – ZUP-G, 8/12 – ZVRS-F, 21/12, 47/13, 12/14, 90/14, 51/16, 36/21, 82/21, 189/21, 153/22 and 18/23; hereinafter: ZDU-1);
- Public Finance Act (Official Gazette of the Republic of Slovenia *[Uradni list RS*], Nos 11/11 – official consolidated text, 14/13 – corr., 101/13, 55/15 – ZFisP, 96/15 – ZIPRS1617, 13/18, 195/20 – Constitutional Court Decision, 18/23 – ZDU-1O, 76/23

and 24/25, hereinafter: ZJF);

- Scientific Research and Innovation Activity Act (Official Gazette of the Republic of Slovenia [*Uradni list RS*], Nos 186/21, 40/23, 102/24 and 102/24, hereinafter: ZZrID);
- Rules on co-funding scientific research activities (Official Gazette of the Republic of Slovenia [*Uradni list RS*], No 69/23);
- Rules determining special projects of national importance and the regulatory framework for the determination of salaries for researchers (Official Gazette of the Republic of Slovenia [*Uradni list RS*], No 126/23 and 106/24);
- Act Regulating the Implementation of the Budgets of the Republic of Slovenia for 2025 and 2026, ZIPRS2526 (Official Gazette of the Republic of Slovenia [*Uradni list RS*], No 104/24 in 17/25 – ZFO-1E);
- Budget of the Republic of Slovenia for 2025 (Official Gazette of the Republic of Slovenia [Uradni list RS], No 123/23 and 104/24), Budget of the Republic of Slovenia for 2026 (Official Gazette of the Republic of Slovenia [*Uradni list RS*], No 104/24);
- Integrity and Prevention of Corruption Act (Official Gazette of the Republic of Slovenia *[Uradni list RS*], No 69/11 – official consolidated text, 158/20, 3/22 – ZDeb and 16/23-ZZPri);
- Personal Data Protection Act (Official Gazette of the Republic of Slovenia *[Uradni list RS*], No 163/22; hereinafter: ZVOP-2);
- Trade Secrets Act (Official Gazette of the Republic of Slovenia [*Uradni list RS*], No 22/19, hereinafter: ZPosS);
- Public Procurement Act (Official Gazette of the Republic of Slovenia *[Uradni list RS*], Nos 91/15, 14/18, 121/21, 10/22, 74/22 – Constitutional Court Decision, 100/22 – ZNUZSZS, 28/23 and 88/23 – ZOPNN-F, hereinafter: ZJN-3);
- Resolution on the Slovenian Scientific Research and Innovation Strategy 2030 (Official Gazette of the Republic of Slovenia *[Uradni list RS*], No 49/22);
- Decree on scientific research public funding from the budget of the Republic of Slovenia (Official Gazette of the Republic of Slovenia *[Uradni list RS*], Nos 35/22, 144/22 and 79/23);
- The criteria for determining the achievement of internationally comparable research results and the period covering internationally comparable research achievements for research project or programme leaders, effective from 22 May 2024 (available at https://www.arrs.si/sl/akti/kriteriji-vodja-maj24.asp) and Amendment of the Criteria for determining the achievement of internationally comparable research results and the period covering internationally comparable research achievements for research project or programme leaders, effective from 24 May 2024 (available at https://www.aris- rs.si/sl/akti/kriteriji-vodja-dop-maj24.asp)
- Framework for State Aid for Research and Development and Innovation (2014/C 198/01);
- Programme of Collaborative Research and Development Projects and Other Projects Subject to State Aid of the Ministry of Higher Education, Science and Innovation 2022– 2030, Minister, Version 2.0 (Consolidated Text), No. 440-4/2022/1 of 26 September 2022, as amended and supplemented (Version 2.0) on 20 July 2023, available at: https://www.gov.si/assets/ministrstva/MVZI/Znanost/Strategije-predpisi-in-drugi-dokumenti/Program-sodelovalnih-raziskovalno-razvojnih-projektov-in-drugih-projektov-ki-so-predmet-drzavnih-pomoci-MVZI-2022-2030.pdf);
- Decree on the implementation of scientific research work in accordance with the principles of open science (Official Gazette of the Republic of Slovenia *[Uradni list RS*], No 59/23 and 39/25);

- Open Science Action Plan for the implementation of Action 6.2: Open science to improve the quality, efficiency and responsiveness of research in the framework of the Resolution on the Scientific Research and Innovation Strategy of Slovenia 2030 of 31 May 2023, No 63100-1/2023/5 (available at:
- https://www.arrs.si/sl/dostop/inc/24/Akcijski_nacrt_odprta%20znanost.pdf).

Co-funding rate:

- 100% for research organisations (such as universities, public and private research institutes) whose funded activity is **non-commercial** in accordance with the provisions of the Community Framework for State Aid for Research and Development and Innovation. Wide dissemination of all research results on a non-exclusive and non- discriminatory basis is required.
- For research organisations whose funded activity is **commercial** in accordance with the provisions of the Community Framework for State Aid for Research and Development and Innovation, the provisions of the Community Framework for State Aid for Research and Development and Innovation (OJ EU C 198, 27.6.2014) and the national scheme for State aid for research and development (the Programme of collaborative research and development and other projects subject to State aid from the Ministry of Education, Science and Sport 2022-2030, as amended, published at: https: //www.gov.si/podrocja/izobrazevanje-znanost-in-sport/znanost/), the maximum co-funding rates shall be as follows:

Type of research	**Large enterprises**	**Medium-sized enterprises**	**Small enterprises**
Basic research	100%	100%	100%
Industrial research	65%	75%	80%
Experimental development	40%	50%	60%

Main eligibility criteria (e.g. types of organisations, thematic restrictions, types of costs, level of co-funding, etc.)

"Research organisations", as defined in the ZZrID, are eligible for co-funding. All participating organisations must be registered in the register of research organisations of the Public Agency for Scientific Research and Innovation of the Republic of Slovenia (Slovenian Current Research Information System, hereinafter: the SICRIS).

The eligibility of the research leader and other members of the research team: The project activities of **each** Slovenian partner must be supervised by a research leader who fulfils the conditions for a project leader as defined in paragraph one of Article 63 of the ZzrID. Regarding the definition of internationally comparable research outputs, the criteria for determining the achievement of internationally comparable research results and the period covering internationally comparable research achievements for the management of a research project or programme of 22 May 2024 are available at https://www.arrs.si/sl/akti/kriteriji-vodja- maj24.asp

All participating researchers must be registered in the SICRIS and have available research hours.

The total funding requested per project for all Slovenian partners within a consortium may not exceed EUR 300,000.00 for the entire duration of the project.

Eligible costs

The Ministry of Higher Education, Science and Innovation will finance all eligible costs of Slovenian applicants in selected transnational projects, in accordance with the Decree on scientific research public funding from the budget of the Republic of Slovenia. Eligible costs must be directly related to the research carried out and must include all of the following categories of direct costs:

- Personnel (including social, health, pension and other contributions in accordance with national legislation),
- Materials (travel and meeting expenses, consumables, dissemination and knowledge exchange costs and other costs),
- Depreciation costs.

Indirect costs are eligible for up to 25% of direct costs.

This Call is suitable for implementing Article 64 of the Scientific Research and Innovation Activity Act (Official Gazette of the Republic of Slovenia [Uradni list RS], Nos 186/21, 40/23, 102/24 and 40/25 (ZZrID)).
For more detailed explanations, please refer to the *Navodila za prijavitelje in upravičence* (Instructions for Applicants and Eligible Persons) (available at: https://www.gov.si/zbirke/projekti-in-programi/obzorje-evropa/partnerstva/

In the event of the application of the Rules determining special projects of national importance and the regulatory framework for the determination of salaries for researchers, costs will be planned in accordance with the provisions of this Annex.

Public expenditure eligibility period: from the financial year 2027, until the end of the financial year 2029.

Project expenditure eligibility period: From the start date of the transnational project, as specified in the National Documentation for the duration of the project, with the prescribed additional 30-day deadline for the payment of invoices related to the project costs. The project expenditure eligibility period starts after the entry into force of the consortium agreement between the selected consortium partners from the date on which the national co-funding contract enters into force.

The co-funding contract enters into force on the date of the fulfilment of the suspensive condition, i.e. on the date of the conclusion/signing of the consortium agreement. The consortium agreement shall be deemed to be signed when the last signatory party signs it. If the consortium agreement is not signed at the latest within one year from the date of the start of the project (as specified in the national documentation), the contract between the parties shall not enter into force and therefore no rights and obligations shall arise between them In this case, the contracting party cannot demand performance from the counterparty as agreed in the above-mentioned contract. As from the date of the entry into force of the contract, the eligible person shall be entitled to the payment of eligible costs and expenses incurred by it as a result of the implementation of the project from the date of the start of the project.

Applicants must submit the transnational application in accordance with the instructions and deadlines set out in the joint transnational Call for tender.

Submission of the project proposal at the national/regional level (method, content, deadlines, etc.)

The submission of an application at the national level is not foreseen.

All information necessary to establish the eligibility for the co-funding of Slovenian applicants must be evident from the duly submitted transnational project application.

The national contracting procedure will start once the projects are proposed for co-funding at the level of the transnational Call for tenders. The national documentation submitted, including evidence of the start date of the transnational project, will be a prerequisite for the signature of the contract at the national level (a draft contract is annexed to the *Navodila za prijavitelje in upravičence* (Instructions for Applicants and Eligible Persons), available at: https://www.gov.si/zbirke/projekti-in-programi/obzorje-evropa/partnerstva/

The submission of financial and substantive reports at the national/regional level

In accordance with the co-funding contract and the National Annex.

Additional requirements

All Slovenian applicants are recommended to contact the Slovenian national contact person.

All Slovenian applicants should refer to the information published on the Call for tenders at the following website:
https://www.gov.si/drzavni-organi/ministrstva/ministrstvo-za-visoko-solstvo-znanost-in-inovacije/javne-objave/

For more detailed explanations, please refer to the *Navodila za prijavitelje in upravičence* (Instructions for Applicants and Eligible Persons) (available at: https://www.gov.si/zbirke/projekti-in-programi/obzorje-evropa/partnerstva/).

Participating organisation

National Research Foundation of Korea

Contact

Seok-Ho Kim
Program Officer
Telephone number: +82-42-869-7831
Email: rock@nrf.re.kr

Funding criteria and regulations

Budget

Maximum funding committed: 1,600,000.00 EUR

Maximum number of fundable research groups: 2

Eligibility of project duration: 3 years

Funding regulations

National funding will be provided according to the

NATIONAL RESEARCH AND DEVELOPMENT INNOVATION ACT.

Eligibility of a partner as a beneficiary institution

Full, associate, and assistant professors and other researchers with a comparable position may submit an application if they have a tenured position (and therefore a paid position for an indefinite period) or a tenure track agreement at Korean universities, or national research institutes, or at other research organisations, that -at least in part- are funded by the Ministry of Science and ICT.

SME eligible to apply only in collaboration with Korean higher education and research institution.

Submission of the proposal at the national level

Additional documentation must be submitted the NRF no later than 7 working days after the QUANTERA III deadlines.

Additional requirements

A principal investigator may act in one proposal only. A consortium leader from a Republic of Korea-based public research institution should be a researcher with solid experience of managing collaborative research projects.

Participating organisation

Agencia Estatal de Investigación

Subdivisión de Programas Científico - Técnicos Transversales, Fortalecimiento y Excelencia

c/Torrelaguna, 58 bis – planta 6ª

20827 Madrid

+34 916037990

Contact

Administrative and technical issues:
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Representative:
Beatriz Goméz Miguel
Telephone number: +34 916 038 707
Email: beatriz.gomez@aei.gob.es

Funding Criteria and Regulations

Funding commitment

Maximum funding for the QuantERA Call 2025: 1,000,000.00 EUR.

Funding Programme

Funding Programme: The framework for this funding action is the Plan Estatal de Investigación Científica, Técnica e Innovación 2024-2027. On a national level, the Call will be managed by the Subdivisión de Programas Científico-Técnicos Transversales, Fortalecimiento y Excelencia (STRAN) of the AEI.

Instrument for funding the Spanish groups

The instrument for funding the Spanish groups is “Proyectos de Colaboración Internacional” (Projects of International Collaboration).

The applicants are strongly advised to read the call text of the PCI 2025-1 (“Resolución de 5 de marzo de 2025”) for reference and especially the PCI Requirements document.

Data Protection

By submitting a grant application to the AEI, the applicants consent to communication of the data contained in the application to other public administrations, with the aim of further processing of the data for historical, statistical or scientific purposes, within the framework of the Organic Law 3/2018, of December 5, on Personal Data Protection and Guarantee of Digital Rights.

Eligibility criteria

Eligible entities for the AEI funding are:

Non-profit research organizations (such as universities, public research institutions, technological centres and other private non-profit institutions performing RDI activities in Spain), which must comply with the requirements established by this transnational Call and with the rules on eligibility defined in the corresponding Spanish national funding instrument “Proyectos de Colaboración Internacional” (PCI) and the PCI Requirements document. The entities must have been previously beneficiaries of any of the AEI calls. They have to ensure contractual relationship with the Principal Investigator (PI) during all the implementation of the project.

Although private enterprises are not funded by the AEI, the Spanish industrial sector is welcome to participate in the transnational consortia using their own funds or obtaining funds from the CDTI or other innovation and technological development funding agencies.

Additional eligibility criteria

Principal Investigators (PIs) requesting funding to the AEI must:

- Be eligible to the corresponding PCI call (see PCI2025-1 as an example) and the PCI Requirements document
- Demonstrate experience as investigators in projects funded by the different Plan Estatal I+D+i: 2013-2016, 2017-2020, 2021-2023, 2024-2027, ERC Grants, European Framework Programmes or other relevant national or international programmes.

Incompatibilities: These must be taken into account when participating in different ERA-Nets, European Co-funded Partnerships or other international initiatives.

- PIs are not allowed to apply for funding in (i) more than one proposal of this transnational Call, (ii) in more than one proposal in the same PCI call and (iii) in PCI calls in consecutive years.
- If the same PI submits two or more proposals in this transnational Call, they will all be declared ineligible, except one, without the possibility of changing the PI.
- PIs must remain unchanged between the proposal to this transnational Call and the corresponding PCI call should the proposal be recommended for funding.
- A PI that has been granted a PCI the previous year will be declared ineligible, without the possibility of changing the PI.

Important

1. Submission of proposals at the AEI. Within one week after the QuantERA short proposal submission (until 12th of December 2025 at the latest), the Spanish PI must submit a copy of the international joint proposal, **and** the “Declaración responsable del investigador principal (templates in PDF and WORD available on AEI website). You can find below the link to the AEI application system to upload these documents: https://aplicaciones.ciencia.gob.es/proyectostransnacionales/
2. The applicants should include the PI’s full name and the full name of their institution as it is stated in the Sistema de Entidades (SISEN).

Eligible costs

- Research and innovation activities are eligible. Mere dissemination, communication or other similar activities will not be eligible
- Only Personnel costs for new temporary employment dedicated to the project are eligible. This must be clearly stated in the contract. The costs of permanent staff linked to the beneficiary entity or members of the research team will not be considered eligible costs.
- Direct costs such as current costs, disposable materials, travelling expenses, coordination costs and other costs that can be justified as necessary to carry out the proposed activities.
- Overheads (25% of all direct costs, including the subcontracting costs).

The AEI will avoid double funding (overlapping with other EU or National funding) and will not grant projects or parts of projects already funded.

Funding rates

The following funding limits (including direct + 25% indirect costs) are considered eligibility criteria. Proposals not respecting these limits could be declared ineligible.

Maximum funding per project	CD (EUR)	CI (25%) (EUR)	TOTAL (EUR)
One AEI applicant	140.000	35.000	175.000
One AEI applicant - coordinator	220.000	55.000	275.000
Two AEI applicants- one coordinator	260.000	65.000	325.000

Additional 30,000 EUR (direct costs) can be granted for the entire proposal if the work plan includes substantial experimental tasks.

IMPORTANT

- Only ONE applicant asking funding for AEI is allowed per proposal as partner,
- Two AEI applicants are allowed in the same proposal if one of them is acting as coordinator,
- These amounts refer to 3 years projects. In case of shorter projects, the amount will be adjusted accordingly.

Other Funding Criteria

- Centres formed by different Spanish legal entities will be considered as a unique entity, and thus the maximum funding should not exceed the limits per proposal established above (for example mixed centres).
- Two centres or institutions belonging to the Consejo Superior de Investigaciones Científicas (CSIC) will be treated as two separate partners one from another when one of them is acting as Coordinator of the proposal and their tasks and identity in the project are sufficiently separated and justified.

The final funding will take into account the transnational evaluation of the collaborative proposal, the scientific quality of the Spanish group, the added value of the international collaboration, the participation of the industrial sector, and the financial resources available.

Mandatory acknowledgement

Any publication or dissemination activity resulting from the granted projects must acknowledge funding by the Agencia Estatal de Investigación: “Project (reference nº XX) funded by the Agencia Estatal de Investigación through the PCI (year) call (or its equivalent)”.

Open Access and Open Data policy

Applicants should comply with Open Access/Open Data specified in the respective PCI call or equivalent. The results of the funded research actions, including both the results disseminated through scientific publications and the data generated in the research, must be available in open access, with the exceptions indicated in the PCI call or equivalent.

Participating organisation

Swedish Research Council

Contact

Tomas Andersson
Senior Research Officer
Telephone number: +46 8-546 44 173
Email: Tomas.Andersson@vr.se

Funding criteria and regulations

Funding commitment

Total: 1,000,000.00 EUR (approx. 10 M SEK)

Funding regulations

1. SRC funds Swedish partners within the sub-call **Quantum phenomena and resources**.
2. SRC funds basic research of the highest scientific quality, and promotes research collaboration and the exchange of experience.
3. Awarded grant: Maximum 1.5 M SEK per year for three years for the Swedish partner (Total 4.5 M SEK). Exchange rate by the deadline of the Call will be used.

Eligibility of a partner as a beneficiary institution

1. The investigators need to hold a PhD at the time of application.
2. **The grants distributed by the SRC must be administrated by a Swedish university, higher education institution (HEI) or other public organisation that fulfils the Swedish Research Councils criteria for an administrating organisation:** https://vr.se/english/calls-and-decisions/apply-for-a-grant/who-can-apply.html
3. A researcher may only apply for funds from the SRC in **one** application in the QuantERA III Call2025.

Submission of the proposal at the national level

Swedish applicants are encouraged to communicate with the national contact person regarding their intention to participate in the Call, before submission of the consortium application.

All Swedish project leaders participating in the Call for support from the Swedish Research Council shall also submit a parallel application using the Swedish Research Council's application system Prisma. The application form in Prisma can be reached from the Call text at the Council's website.

Parallel application is a mandatory eligibility criterion. Failure to submit the parallel application to the Swedish Research Council before the deadline of the Prisma call will result in the Swedish partner being declared ineligible.

Additional requirements

The project grant may be used to fund all types of project-related costs, such as salaries (including your own salary, however no more than corresponding to the person's activity level in the project), running costs (such as consumables, travel including stays at research facilities,

publication costs and minor equipment), premises and depreciation costs. Grants may not be used for scholarships. If a doctoral student participates, project funds may not be paid out as salary during teaching or other departmental duties. The minimum amount for which you may apply is SEK 400 000 per year, including indirect costs.

See national call texts in Swedish (https://www.vr.se/5.69a4b979196f6304be29df8.html) and English (https://www.vr.se/5.69a4b979196f6304be29e81.html) for all national requirements.

Participating organisation

Swiss National Science Foundation
Wildhainweg 3
P.O. Box 8232
CH-3001 Berne
http://www.snf.ch

Contact

Dr. Ahmad Zein Assi
Scientific Officer
Telephone number: +41 31 308 21 83
Email: quantera@snf.ch

Funding criteria and regulations

Funding commitment

2,000,000.00 EUR for both topics

Maximum awarded grant:

- For a 3-years project: 400‘000 CHF / partner
- For a 2-years project: 266‘667 CHF / partner

Funding regulations

There are no restrictions on the participation of researchers in Switzerland in this QuantERA Call. They can coordinate consortia and are even encouraged to do so.
Projects must comply with the SNSF Project Funding regulations and practices.
In particular, all Swiss applicants and co-applicants must be eligible for Project Funding, see also the Regulations on project funding. Applicants in Switzerland who have not previously obtained a Project Funding grant of the SNSF are encouraged to contact the national contact point.
Partners of an international project consortium located in a country of the QuantERA transnational consortium cannot be declared as project partners in the sense of article 11.2 of the SNSF Funding Regulations. They should be declared as consortium partner instead and apply for their funding at their respective research funding organisation. In case a consortium includes industrial or commercial partners (in or outside Switzerland), a confirmation of non-commercial goals must be provided in case the project is approved as a condition for the release of funds.
Article 17, cl. a, of the SNSF Funding Regulations only applies in the sense that proposals with overlapping funding periods are only approved if the research projects pursue different goals in the context of this European programme than any ongoing projects by the same applicant. Article 6.3 of the Regulations on project funding applies. Swiss applicants may participate in at most one QuantERA proposal.
Article 13 of the Regulations on project funding applies. In particular, Swiss applicants may receive a maximum of three grants from the SNSF for the same funding period, except for the instruments highlighted here.

Overhead costs cannot be requested since those are settled at the institutional level and do not count towards the overall budget.
Grants will be managed according to standard SNSF rules; reporting requirements duplicating those of QuantERA will not be enforced.

Eligibility of a partner as a beneficiary institution

Submission of the proposal at the national level

Applicants must provide basic administrative data by submitting administrative applications via the online submission system mySNF for the same deadlines as the consortium applications. Please select the instrument "Programmes/ERA-NET + EJP".

Additional requirements

Open Access to Publications and Research Data
Since October 2018, any publication based on research funded by the SNSF must be published in Open Access, with the possibility of adopting the Green Road (maximum embargo of 6 months for articles and 12 months for books/book chapters). The SNSF covers the costs for publishing in Open Access through a dedicated platform. Further details can be found at https://www.snf.ch/en/VyUvGzptStOEpUoC/topic/research-policies.
Since 2017 all funded proposals need to submit a DMP before the start of the project. In particular, research data is expected to be archived on FAIR data repositories. No funds will be transferred without submitting a consistent DMP. Further details can be found at https://www.snf.ch/en/dMILj9t4LNk8NwyR/topic/research-policies.

Participating organisation

The Scientific and Technological Research Council of Türkiye

Contact

Dr. Kıvılcım KÖSEOĞLU
Scientific Expert
Telephone number: 0090 312 298 1185
Email: kivilcim.koseoglu@tubitak.gov.tr

Funding criteria and regulations

Funding commitment

200,000.00 EUR

Funding regulations

https://tubitak.gov.tr/sites/default/files/2024-08/1071_ar-ge_ve_yenilik_projeleri_surec_dokumani.pdf

Eligibility of a partner as a beneficiary institution

Higher education institutions within the scope of the Higher Education Law, training and research hospitals, public institutions and organizations, and capital companies that are based in Türkiye, hold a trade registry certificate, and generate added value at the firm level regardless of their sector or size are eligible for support under the TÜBİTAK 1071 Support Program. Foundations, associations and their economic enterprises, cooperatives, unions, sole proprietorships, and ordinary partnerships are not eligible to apply. For further details, the national call text should be reviewed carefully.

Submission of the proposal at the national level

Yes. To be announced on TÜBİTAK website.

Additional requirements

Successful projects will be funded under TÜBİTAK 1071 Programme - Support Programme for Increasing Capacity to Benefit from International Research Funds and Participation in International R&D Cooperation.

United Kingdom UKRI-EPSRC

Participating organisation

UK Research and Innovation, Engineering and Physical Sciences Research Council

Contact

Liam Behbehani Boyle
Quantum Technologies Senior Manager
Telephone number: +447874795747
Email: quantumtechnologies@epsrc.ukri.org

Funding criteria and regulations

Funding commitment

Total: 5,000,000.00 EUR

Up to 5 M EUR is available to support projects within the *Applied Quantum Science* Call only involving researchers based in UK Higher Education Institutions or eligible research organisations.

Funding regulations

Funding is available for a maximum of £750,000 per consortium, £400,000 per partner at 80% full economic costs (fEC). All research grant proposals submitted must be costed on the basis fEC. If a grant is awarded, research councils will provide funding at 80% of the fEC. The applying organisation must agree to find the balance of fEC for the project from other resources.

Eligibility of a partner as a beneficiary institution

Before preparing a proposal, prospective applicants should familiarise themselves with:

1. The eligibility of investigators for EPSRC funding: https://www.ukri.org/councils/epsrc/guidance-for-applicants/check-if-you-are-eligible-for-funding/
2. The eligibility of organisations for EPSRC funding: https://www.ukri.org/apply-for-funding/before-you-apply/check-if-you-are-eligible-for-research-and-innovation-funding/eligibility-as-an-organisation/#contents-list
3. The eligibility requirements relating to resources: https://www.ukri.org/councils/epsrc/guidance-for-applicants/costs-you-can-apply-for/

All consortium 'partners' (research groups) that request funding from EPSRC must be represented by a Project Lead. The Project Lead must fulfil EPSRC's standard eligibility requirements (see above). Eligible Investigators can only request EPSRC funding on one proposal submitted to this Call.

Submission of the proposal at the national level

Project Leads of UK partners are strongly advised to register their intention to submit by emailing EPSRC at: quantumtechnologies@epsrc.ukri.org before the submission deadline.

Project Leads of successful UK partners will be invited to make a submission through the UK Research and Innovation (UKRI) Funding Service following the notification of successful proposals.

Additional requirements

All of the outlined Target Research areas are open for UK funding as described in the QuantERA call document. We are expecting this research to align with the research priorities of the UKRI EPSRC Quantum Technology theme. Only proposals in the Call topic “Applied Quantum Science” will be accepted.

Annex 2: QuantERA Gender Equality Statement

TOWARDS MORE GENDER-BALANCED RESEARCH IN QUANTUM TECHNOLOGIES

Since its inception, the QuantERA Consortium has been committed to addressing the gender imbalance in Quantum Technologies (QT) research and spreading the research excellence across the European Research Area (ERA). QuantERA III continues its strive to improve the gender equality measures at both the consortium as well as the project level, thus becoming an active agent of the changes encouraged and promoted by the European Commission, i.e. through its *Gender Equality Strategy*, 2020–2025[17], research performing institutions and Research Funding Organisations.

In 2022, female scientists and engineers in the EU accounted for 41 per cent of total employment in science and engineering with even lower figures in the quantum community [18].

However:

- Women's representation is considerably lower among all tertiary education levels and academic staff in STEM, compared to all fields. Among grade A staff[19] in Engineering and Technology women constitute 19 %[20].
- Women represent around 12% of quantum science positions across Europe, with minority groups facing additional barriers to entry and advancement[21].

The promotion of STEM vocations free from gender stereotypes should start at the level of schools, since this is a societal issue and constitutes a long-term objective.

Aware of the impact that QuantERA Research Funding Organisations can have in the quantum field, we encourage physics institutes and departments as well as researchers in physics and in Quantum Technologies:

- To create a gender-sensitive environment and organisational culture in the physics field that promotes equal opportunities between women and men **at all levels**.
- To create an equality standard regarding the management structure of their organisations.
- To acknowledge that diversity is beneficial for science and strengthens the quality of science.
- To encourage all women PhDs in physics and in Quantum Technologies and provide them with the adequate career support services and fair chances in promotion process.
- To acquaint STEM students with role models of women researchers in Quantum Technologies.

And thus, help us to have a more gender-balanced research in Quantum Technologies in the coming years. The QuantERA III evaluation process will consider gender balance in the evaluation panels as much as possible and will provide gender-sensitive peer review guidelines for evaluators. The success rates of this Call will be monitored by the QuantERA consortium.

Please visit our **Call for proposals** **site** and **join us in our desire to reinforce the participation of fairly composed consortia of researchers**.

The QuantERA Call 2025 Call Steering Committee

[17] See: https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/gender-equality/gender-equality-strategy_en

[18] See: https://ec.europa.eu/eurostat/web/products-eurostat-news/w/ddn-20240212-1

[19] the highest grade/post at which research is normally conducted

[20] European Commission, She Figures 2024 (Gender Equality in Research & Innovation). https://op.europa.eu/en/publication-detail/-/publication/7646222f-e82b-11ef-b5e9-01aa75ed71a1

[21] European Commission, She Figures 2021 (Gender Equality in Research & Innovation). https://op.europa.eu/en/publication-detail/-/publication/67d5a207-4da1-11ec-91ac-01aa75ed71a1

Annex 3: QuantERA Guidelines for Responsible Research and Innovation (RRI)

The QuantERA consortium encourages incorporating Responsible Research and Innovation (RRI) into proposals submitted to QuantERA.

What is RRI?

Responsible Research and Innovation is an approach that anticipates and assesses potential implications and societal expectations regarding research and innovation, with the aim to foster the design of inclusive and sustainable research and innovation.

Responsible Research and Innovation implies that societal actors (researchers, citizens, policy makers, business, third sector organisations, etc.) work together during the whole research and innovation process in order to better align both the process and its outcomes with the values, needs and expectations of society.

RRI is a cross-cutting issue that is generally articulated around 5 main dimensions:

- Public engagement,
- Open science,
- Gender,
- Ethics,
- Science education.

QuantERA and RRI

Provided below is a non-exhaustive list of possible actions that can be put in place to make RRI and integral part of your research. You may choose those that are relevant to your project.

- Involve all partners and participants in ongoing considerations of RRI throughout the project life.
- Draft and regularly update your data management plan outlining the procedures for data collection, storage, sharing, and archiving in alignment with the FAIR principles (Findable, Accessible, Interoperable, Reusable).
- Deposit a machine-readable electronic copy of the published version or the final peer-reviewed manuscript in a trusted repository at the time of publication.
- Disseminate results/outcomes of your research on open access platforms and repositories (e.g. arXiv, Software Heritage, ORE, Zenodo, etc.).
- Consider licensing publications under the Creative Commons Attribution International Public Licence (CC BY) to maximise accessibility and reuse. For monographs and other long-text formats, CC BY-NC or CC BY-ND licenses may be used to exclude commercial uses and derivative works where appropriate.
- Make metadata for all deposited publications openly available under a Creative Commons Public Domain Dedication (CC0) and including comprehensive information, i.e. author details, publication venue, date, and Horizon Europe funding details.
- Provide persistent identifiers (e.g. DOI) for publications, authors, institutions, and all related research outputs.
- Disseminate your research results/outcomes towards the general public (scientific outreach).
- Reflect on your research with regards to ethical issues, in particular with regards to privacy and data protection issues, IP protection, etc.
- Encourage the implication of early career researchers in your research projects.

- Provide equal opportunities for researchers of all genders (see Annex 2).
- Involve relevant stakeholders in your project at the earliest possible stage and consider the involvement of RRI experts in your project implementation.

QuantERA Proposal Template

QuantERA Proposal Template

Project Acronym

Project Title

Topic: ☐ Quantum Phenomena and Resources ☐ Applied Quantum Science

Coordinator contact point for the proposal

Name	
Institution/Department	
Address	
Country	
Phone	
E-mail	

People involved in the realisation of the project **(please enter these names and their email addresses in the financial template as well)**

Partner Number	***Countr y***	***Institution/ Department***	***Name of the Principal Investigator (PI)***[3]	***Name of the co-Investigator s***[4]	***Name of the other personnel participating in the project***[5]
1 *Coordinato r*					
2					
3					
4					
5					
6					

(Use as many lines as needed)

---

[3] *The Principal Investigator (PI) is the point of contact of the partner for the corresponding Research Funding Organisation.*
[4] *A co-investigator is a known scientist and/or group leader making a substantial contribution to the project.*
[5] *If the name is for the moment unknown, specify the level of expertise sought (PhD, post-doc, engineer, professor…).*

Duration: months

Summary of the project (publishable abstract, max. 1/2 page):

Be precise and concise. This summary will be used to select suited reviewers for the proposal.

Relevance to the topic addressed in the call (in particular specify here which part of the call text is addressed by your project, max. 1/4 page):

Be precise and concise.

Detailed project information

General recommendations:
1. The same font and style should be used for the whole proposal (Arial, 11pt, single spaced).
2. Please complete all sections.
3. Please adhere to the given page limits.
4. For the evaluation criteria, please refer to the Call Announcement. Your proposal should include all details required.

1. Excellence (max. 6 pages)

1.1 Targeted breakthrough, baseline of knowledge and skills

Describe the targeted breakthroughs of the project.
Describe how the science and technology contribute to the establishment of a solid baseline of knowledge and skills for the specific theme addressed.
Describe the specific objectives of the project, which should be clear, measurable, realistic and achievable within the duration of the project.

1.2 Novelty, level of ambition and foundational character

Describe the advance your proposal would provide beyond the state-of-the-art, and to what extent the proposed work is ambitious, novel and of a foundational nature. Your answer could refer to the ground-breaking nature of the objectives, concepts involved, issues and problems to be addressed, and approaches and methods to be used.

1.3 Concept and methodology

Describe and explain the overall concept and research approach underpinning the project. Describe the main ideas, models or assumptions involved. Identify any interdisciplinary considerations and, where relevant, use of stakeholder knowledge.
Describe any national or international research and innovation activities which will be linked with the project, especially where the outputs from these will feed into the project.

Describe the methodology and explain its relevance to the objectives.
Describe the appropriateness of the methodology to narrow down multiple options and to address high scientific and technological risks.

1.4 Interdisciplinary nature

Describe the research disciplines involved and the range of added value from interdisciplinarity, including measures for exchange, cross-fertilisation and synergy.

1.5 Gender dimension and open science practices

Describe how gender dimension in research content is addressed.
Describe how open science practices including sharing and management of research outputs are implemented.

2. Impact (max. 3 pages)

2.1 Expected impacts

Be specific, and provide only information that applies to the proposal and its objectives. Wherever possible, use quantified indicators and targets.
Describe how the project will contribute to the expected impacts (see 'Research Targeted in the Call' of the Call Announcement).
Describe the importance of the technological outcome with regard to its transformational impact on technology and/or society.

2.2 Dissemination, exploitation of results, communication

Provide a plan for disseminating and exploiting the project results beyond the project itself (scientifically and/or economically).
Results include any data produced in the framework of the project. If applicable, describe how data curation and distribution can be ensured beyond the project duration.

When applicable, describe how subsequent translation of research results into innovations and development of new products and/or services are facilitated

Describe the proposed communication measures for promoting the project and its findings during the period of the project. Measures should be with clear objectives. They should be tailored to the needs of different target audiences, including groups beyond the project's own community.
Where relevant, include measures for public/societal engagement on issues related to the project.

3. Implementation

3.1 Work plan

(max. 2 pages)

Provide a brief presentation of the overall structure of the work plan.
Clearly define the intermediate targets.
Provide a timing of the different work packages and their components (Gantt chart or similar).
Provide a graphical representation of the work packages components showing how they inter-relate (Pert chart or similar).

3.2 Work packages

(max. 1 page per WP)

For the description of each work package, please use the template provided. Use as many templates as needed.

<table>
<tr><td>WP 1</td><td colspan="6">WP Title</td><td>Start month</td><td>End month</td></tr>
<tr><td colspan="9">Contribution of project partners</td></tr>
<tr><td>Partner number[6]</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>Total effort per partner (person*months)</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td colspan="9">Aim of the WP
Description of the objectives of the WP and the interrelation with other WPs.</td></tr>
<tr><td colspan="9">Tasks</td></tr>
<tr><td>T1.1</td><td colspan="8">Task title (start month – end month: responsible partner; involved partners)[7]
Description of work and role of participants</td></tr>
<tr><td>T1.2</td><td colspan="8">Task title (start month – end month: responsible partner; involved partners)
Description of work and role of participants</td></tr>
<tr><td></td><td colspan="8">Add tasks as needed</td></tr>
<tr><td>Deliverabl e</td><td colspan="2">Month of delivery</td><td colspan="6">Title of deliverable</td></tr>
<tr><td>D1.1</td><td colspan="2"></td><td colspan="6"></td></tr>
<tr><td>D1.2</td><td colspan="2"></td><td colspan="6"></td></tr>
<tr><td></td><td colspan="2"></td><td colspan="6">Add deliverables as needed</td></tr>
</table>

[6] ***Bold*** *the partner number of the work package leader*

[7] *For instance: T1.1 Development of something (M3-M6; responsible: 3; involved: 1, 4)*

Work package overview (total effort per WP and partner in person.months)

Use as many lines and columns as needed.

Partner	WP1	WP2	WP3	WP4	WP5	WP6	Total
1							
2							
3							
4							
5							
6							
Total							

3.3 Management structure, milestones, risk assessment

(max. 2 pages)

Describe the organisational structure and the decision-making including a list of milestones (template provided). A milestone is a major and visible achievement. It should be SMART: Specific, Measurable, Attainable, Relevant, Time-bound.
Explain why the organisational structure and decision-making mechanisms are appropriate to the complexity and scale of the project.

List of milestones

Use as many lines as needed but try to limit the number of milestones.

Milestone	**Delivery month**	**WP involved**	**Title**
M1			

Describe any critical risks, relating to project implementation, that the stated project's objectives may not be achieved. Detail any risk mitigation measures. Please provide a table with critical risks identified and mitigating actions (template provided).

Critical risks for implementation

Use as many lines as needed.

Description of risk (indicate level of likelihood: Low/Medium/high)	**WP(s) involved**	**Proposed risk-mitigation measures**

3.4 Consortium as a whole

(max. 1 page)

The individual members are described in section 3.5, there is no need to repeat that information there.
Describe the consortium. How will it match the project's objectives and bring together the necessary expertise? How do the members complement one another?
In what way does each of them contribute to the project? Show that each has a valid role and adequate resources in the project to fulfil that role.
If applicable, describe the industrial/commercial involvement in the project and explain why this is consistent with and will help to achieve the specific measures which are proposed for exploitation of the results of the project.

3.5 Description of the consortium

(max. 1 page per partner)

Describe expertise and role in the project for each partner (templates provided). The information provided here will be used to judge the operational capacity. Use the following templates for the coordinator, the other partners requesting funding, and partners not requesting funding if any. If the project relies on input to be provided by a third party, append a letter of commitment at the end of the proposal.

<table>
<tr><td>Partner 1
Project Coordinator</td><td>Organisation name / Department</td></tr>
<tr><td colspan="2">Expertise:
Expertise of the organisation related to the project objectives.

For the principal investigators give a brief CV, including gender, highlighting research experience; and list up to 5 relevant publications, and/or products, services (incl. widely-used datasets or software), or other achievements relevant to the call content.</td></tr>
<tr><td colspan="2">Role in project:</td></tr>
</table>

Use this template if this partner is requesting funding

Partner n	**Organisation name / Department**
Expertise: Expertise of the organisation related to the project objectives. For the principal investigators give a brief CV, including gender, highlighting research experience; and list up to 5 relevant publications, and/or products, services (incl. widely-used datasets or software), or other achievements relevant to the call content.	
Role in project:	

Use this template if this partner is <u>not requesting funding</u>

Partner n	**Organisation Full name / Department**
Expertise: Expertise of the organisation related to the project objectives. For the principal investigators give a brief CV, including gender, highlighting research experience; and list up to 5 relevant publications, and/or products, services (incl. widely-used datasets or software), or other achievements relevant to the call content.	
Role in project:	
Please explain how the partner is able to secure its own funding:	

3.6 Consortium agreement principles (partner's rights and duties, IPR management, data management, open access, GDPR)

(max. ½ page)

3.7 Significant facilities and large equipment available to the consortium to perform the project

(max. ½ page)

3.8 Link with ongoing projects

(max. ½ page)

For each partner indicate (if applicable) the ongoing projects linked to the proposal topic, and their funding sources.

3.9 Financial plan

(max. 1 page)

The resources to be committed for each project partner have to be described in the Electronic Submission System by the coordinator. These resources include: Personnel, Consumables, Equipment, Travel, Subcontracting, Provisions, Licensing fees, other. Justify them here. Both the justification and the information in the system will be communicated to the Evaluation Panel.

4. Ethical issues (max. ½ page)

Describe any foreseeable ethical issue that may arise during the course of the research project. Describe all mitigation strategies employed to reduce ethical risk, and justify the research methodology with respect to ethical issues.

5. References (max. 30 references)

Provide references of articles and publicly available documents directly supporting the proposal.

공고문 (QuantERA Notice in Korean)

과학기술정보통신부 공고 제2025 - 0961호

양자정보과학 인적기반조성사업 리더급연구역량강화(연구혁신형) 2026년도 QuantERA 양자기술연구지원사업 신규과제 공모

과학기술정보통신부에서는 양자기술 분야 범유럽 연구지원기관 네트워크인 “QuantERA”에서 운영하는 다자간 국제협력 양자기술 프로그램을 지원하기 위해 「QuantERA 양자기술연구지원사업」의 2026년도 신규과제를 다음과 같이 공모하오니 연구자분들의 많은 관심과 참여 바랍니다.

2025년 10월 27일

<주무부처> 과학기술정보통신부 장관 배 경 훈

<전문기관> 한국연구재단 이사장 홍 원 화

※ 본 공고는 QuantERA측 공고('25.9.4.)에 대하여 한국연구자를 위한 안내사항임.

[illegible]

1 사업개요

※ "___" : 이번 공모과제 관련 내용

□ **목 적**

◦ 폭발적 파급 잠재력을 지닌 양자정보과학 분야에 우수 인재가 유입되고 고급인력으로 커나갈 수 있도록 성장 경로를 구축하고 연구 저변을 확대

□ **주요 내용**

◦ (**양자대학원 운영**) 국내 대학(원)이 연합하여 대학‧학제 간 개방형 교육‧연구 기반의 양자과학기술 핵심인재 양성체계(양자대학원) 구축‧운영 지원

◦ (**리더급 연구역량 강화**) 신규 진입 교수 등을 대상으로 해외 유수 연구소‧대학과의 인력교류 중심 공동연구를 통해 미래 연구주제 발굴 및 글로벌 네트워크 구축 등 지원(체질강화형, 연구혁신형, 전략기술형)

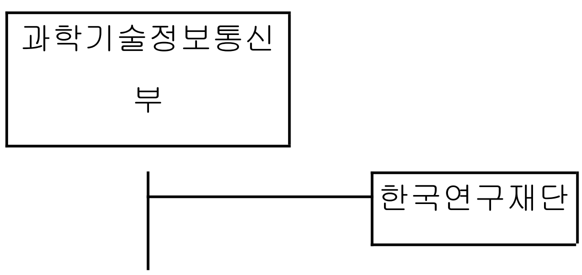
◦ (**장기연수 지원**) 석박사, 포스닥 등을 대상으로 국내외 유수 대학‧기업‧연구소의 관련 연구 프로젝트 및 전문화된 교육과정 참여 지원

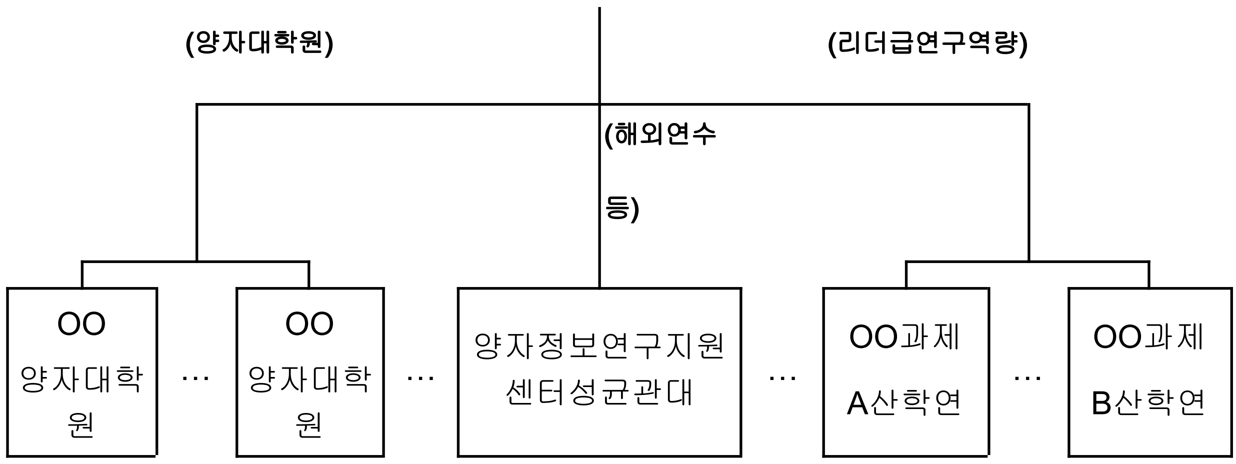
※ 연수 대상자 선발 등에 관한 사항은 양자정보연구지원센터 홈페이지 참조

◦ (**미래인재유입 촉진**) 초중고, 대학생, 일반인 대상으로 양자정보과학 이해도 제고 및 관심 유도 등을 위한 맞춤형 교육프로그램 운영

◦ (**저변확대 등**) 양자정보과학 종합정보시스템 구축‧운영 및 교육‧홍보 콘텐츠 개발‧활용, 국제 컨퍼런스 개최, 정책 지원 등 추진

< 사업 수행 체계 >




(양자대학원)
(리더급연구역량)
(해외연수
등)
OO
양자대학
원
…
OO
양자대학
원
…
양자정보연구지원
센터성균관대
…
OO과제
A산학연
…
OO과제
B산학연

2 지원내용

😪 지원 목적

◦ 범유럽 양자기술 네트워크인 QuantERA* 운영 프로그램 지원으로 유럽 양자기술 연구 협력 생태계에 본격적으로 참여, 장기 협력할 수 있는 계기 마련

* QuantERA(퀀테라) : '16년 결성된 양자기술 분야 범유럽 공공 펀딩기관(RFO) 네트워크(30개 회원국, 40개 펀딩기관)로, 대한민국(한국연구재단)은 아시아 국가 최초로 가입함('25.5.)

◦ 다자간 연구협력을 통해 국제 양자기술 연구생태계를 활성화시키고, 실질적인 연구 성과확산 도모

😒 지원 분야

◦ **지원 구분**

1) **(QPR*, 기초)** 양자 현상 및 자원 *Quantum Phenomena and Resources

- 미래 양자기술의 기초 원리 확립, 새로운 현상, 자원, 프로토콜, 알고리즘 탐구, 응용 확산을 막는 핵심 과제 해결

2) **(AQS*, 응용)** 응용 양자 과학 *Applied Quantum Science

- 기존 양자 개념을 기술 응용으로 전환, 새로운 제품·장치·시스템 개발, 산업 응용 촉진

◦ **세부 연구분야**(5개 분야)

- ① 양자통신, ② 양자컴퓨팅, ③ 양자 시뮬레이션, ④ 양자센싱 및 계측, ⑤ 일반 양자과학(기초 이론, 원천 기술 등)

□ **지원조건**

◦ **(컨소시엄 구성)** 우리나라 포함 퀀테라 회원국(붙임4 참조) **최소 3개국(3개 기관)이상의 각국 연구팀**으로 구성된 **다국가 컨소시엄**

- 한 컨소시엄에 연구팀(참여국가) 수에 제한은 없으며, 컨소시엄 내 1명의 연구자가 코디네이터 역할 수행

- 컨소시엄 내 성별 균형, 지역 다양성, 연령 균형, 산업계 참여 등 권장

- 한국측 연구팀의 경우, 산학연 구성이 가능하며 연구자의 국적 제한은 없음

◦ **(연구방법)** 양자기술 협력연구를 통해 연구협력 생태계를 구축, 상호 인력교류 등

◦ **(필수성과물)** 연구기간 동안 국제학술대회 2편, SCI급 논문 2편

※ 필수성과물은 한국측 연구팀만 해당

□ ’26년도 지원 규모 및 선정 주안점

◦ 지원 규모 및 방식

- 선정과제 수 : 가변적 ※ 예산상황 및 QuantERA 선정여부 등에 따라 다름.

- 지원기간 : **총 3년**(36개월, ’26.9~’29.8)

※ 과제 개시일은 QuantERA 및 각국 사정에 따라 변동될 수 있음.

- 한국측연구비 : 1) **(기초)** 과제당 **연간 300백만원** 이내 2) **(응용)** 과제당 **연간 500백만원** 이내

※ 각국 정부(연구관리전문기관)가 과제에 참여하는 자국 연구진에게 각각 지원하므로, 컨소시엄에 참여하는 국가의 지원금에 따라 컨소시엄별 총 연구비가 달라질 수 있음.

※ 컨소시엄별 총 연구비 구성 시 개별 지원금액이 단일국가 60%, 단일 연구기관 40%를 초과할 수 없으며, 기관별로는 최대 100만 유로까지 지원이 가능함.

< (한국측) 연차별 과제기간 및 과제비 예시 >

보안등급		**일반**	
과제 형태		**일반과제**	
예산지원 (안) - 기초 -	총 연구기간	**36**개월 **(2026. 9. 1. ~ 2029. 8. 31.)**	**900**백만원 이내
	1차년도	3개월 (2026. 9. 1. ~ 2026. 12. 31.)	100백만원 이내
	2차년도	12개월 (2027. 1. 1. ~ 2027. 12. 31.)	300백만원 이내
	3차년도	12개월 (2028. 1. 1. ~ 2028. 12. 31.)	300백만원 이내
	4차년도	9개월 (2029. 1. 1. ~ 2029. 8. 31.)	200백만원 이내
예산지원 (안) - 응용 -	총 연구기간	**36**개월 **(2026. 9. 1. ~ 2029. 8. 31.)**	**1,500**백만원 이내
	1차년도	3개월 (2026. 9. 1. ~ 2026. 12. 31.)	167백만원 이내
	2차년도	12개월 (2027. 1. 1. ~ 2027. 12. 31.)	500백만원 이내
	3차년도	12개월 (2028. 1. 1. ~ 2028. 12. 31.)	500백만원 이내
	4차년도	9개월 (2029. 1. 1. ~ 2029. 8. 31.)	333백만원 이내

※ 지원기간 및 예산은 매년 예산 확보 상황 등에 따라 변경 가능

◦ **선정평가항목**

평가항목	배점	
	기초	응용
• Excellence	50	25
• Impact	25	50
• Implementation	25	20

※ 평가방법은 QuantERA측 기준에 따름.

3 신청자격 및 신청제한

※ 동 사항은 한국측 연구자에만 해당되며,

각국 연구자의 자격 등은 QuantERA 및 각국

□ 신청자격

○ 연구개발기관의 자격

-「국가연구개발혁신법」제2조 제3호 및 동법 시행령 제2조에서 정하는 기관 및 단체

▪「국가연구개발혁신법」제2조 제3호

3. "연구개발기관"이란 다음 각 목의 기관·단체 중 국가연구개발사업을 수행하는 기관·단체를 말한다.

가. 국가 또는 지방자치단체가 직접 설치하여 운영하는 연구기관

나.「고등교육법」제2조에 따른 학교(이하 "대학"이라 한다)

다.「정부출연연구기관 등의 설립·운영 및 육성에 관한 법률」제2조에 따른 정부출연연구기관

라.「과학기술분야 정부출연연구기관 등의 설립·운영 및 육성에 관한 법률」제2조에 따른 과학기술분야 정부출연연구기관

마.「지방자치단체출연 연구원의 설립 및 운영에 관한 법률」제2조에 따른 지방자치단체출연 연구원

바.「특정연구기관 육성법」제2조에 따른 특정연구기관

사.「상법」제169조에 따른 회사

아. 그 밖에 대통령령으로 정하는 기관·단체

▪「국가연구개발혁신법」시행령 제2조 제①항

① 「국가연구개발혁신법」(이하 "법"이라 한다) 제2조제3호아목에서 "대통령령으로 정하는 기관·단체"란 다음 각 호의 기관·단체를 말한다.

1.「중소기업기본법」제2조에 따른 중소기업

2.「민법」또는 다른 법률에 따라 설립된 비영리법인

3. 외국에서 외국 법령에 따라 설립된 외국법인(국내 연구개발기관과 연구개발과제를 공동으로 수행하는 경우로 한정한다)

-「기초연구진흥 및 기술개발지원에 관한 법률」제14조 제1항에서 정하는 기관 및 단체

제14조(특정연구개발사업의 추진) ① 과학기술정보통신부장관은 기초연구의 성과 등을 바탕으로 하여 국가 미래 유망기술과 융합기술을 중점적으로 개발하기 위한 연구개발사업(이하 "특정연구개발사업"이라 한다)에 대하여 계획을 수립하고, 연도별로 연구과제를 선정하여 이를 다음 각 호의 기관 또는 단체와 협약을 맺어 연구하게 할 수 있다. 이 경우 제2호의 기관 중 대표권이 없는 기관에 대하여는 그 기관이 속한 법인의 대표자와 협약할 수 있다.

1. 제6조제1항 각 호에 해당하는 기관

2. 제14조의2제1항에 따라 인정받은 기업부설연구소 또는 연구개발전담부서

3.「산업기술연구조합 육성법」에 따른 산업기술연구조합

3의2.「협동연구개발촉진법」제2조제3호에 따른 과학기술인 협동조합

4.「나노기술개발 촉진법」제7조에 따른 나노기술연구협의회

5.「민법」또는 다른 법률에 따라 설립된 과학기술분야 비영리법인 중 연구 인력・시설 등 대통령령으로 정하는 기준에 해당하는 비영리법인

6.「의료법」에 따라 설립된 의료법인 중 연구 인력・시설 등 대통령령으로 정하는 기준에 해당하는 의료법인

6의2.「1인 창조기업 육성에 관한 법률」제2조에 따른 1인 창조기업으로서 연구 인력 및 시설 등 대통령령으로 정하는 기준을 충족하는 기업

7. 그 밖에 연구 인력・시설 등 대통령령으로 정하는 기준에 해당하는 국내외 연구 기관 또는 단체 및 영리를 목적으로 하는 법인

ㅇ 연구책임자의 자격

-「국가연구개발혁신법」제2조제3호에서 정하는 기관 및 단체에 소속된 연구자

※ 단, 기업의 경우 「기초연구진흥 및 기술개발지원에 관한 법률」제14조의2 제1항에 따라 인정받은 기업부설 연구소 또는 연구개발전담부서를 보유한 기관 및 단체에 소속된 연구자

-「국가연구개발혁신법」제6조 및 제7조에 따른 요건을 갖춘 자

제6조(연구개발기관의 책임과 역할) 연구개발기관은 이 법의 목적을 달성하기 위하여 다음 각 호의 사항을 성실히 이행하여야 한다.

1. 연구개발 역량 강화 및 연구개발의 효율적인 추진을 위하여 노력할 것
2. 소속 연구자가 우수한 연구개발성과를 창출할 수 있도록 연구지원에 최선을 다할 것
3. 소속 연구자의 고유의 연구개발 외 업무 부담이 과중하지 아니하도록 배려할 것
4. 소유하고 있는 연구개발성과가 신속·정확하게 권리로 확정되고 효과적으로 보호될 수 있도록 노력할 것
5. 소유하고 있는 연구개발성과가 경제적·사회적으로 널리 활용될 수 있도록 노력할 것
6. 연구개발성과 창출·활용에 기여한 소속 연구자에게 보상하도록 노력할 것
7. 소속 연구자가 제7조에 따른 책임과 역할을 다할 수 있도록 필요한 조치를 할 것

제7조(연구자의 책임과 역할) ① 연구자는 이 법의 목적을 달성하기 위하여 다음 각 호의 사항을 성실히 이행하여야 한다.

1. 자율과 책임을 바탕으로 성실하게 국가연구개발 활동을 수행할 것
2. 국가연구개발 활동을 수행할 때 도전적으로 자신의 능력과 창의력을 발휘하되, 그 경제적·사회적 영향을 고려할 것
3. 연구윤리를 준수하고 진실하고 투명하게 국가연구개발 활동을 수행할 것

② 연구개발과제를 총괄하는 연구자(이하 "연구책임자"라 한다)는 그 연구개발에 참여하는 연구자가 연구개발 활동에 전념할 수 있도록 배려하여야 한다.

ㅇ 국가연구개발사업 지원 제외 조건

- 연구에 참여하는 기관이 다음 각 호에 해당하는 경우 지원 대상에서 제외

① 주관연구개발기관, 공동연구개발기관, 위탁연구개발기관의 부도

② 국세 또는 지방세 등의 체납처분을 받은 경우(단, 중소기업진흥공단 및 신용회복위원회(재창업지원위원회)를 통해 재창업자금을 지원받은 경우와 신용보증기금 및 기술신용보증기금으로부터 재도전기업주 재기지원 보증을 받은 경우는 예외)

③ 민사집행법, 신용정보집중기관에 의한 채무불이행자경우(단, 중소기업진흥공단 및 신용회복위원회(재창업지원위원회)를 통해 재창업자금을 지원받은 경우와 신용보증기금 및 기술신용보증기금으로부터 재도전 기업주 재기지원보증을 받은 경우는 예외)

④ 파산·회생절차·개인회생절차의 개시 신청이 이루어진 경우(단, 법원의 인가를 받은 회생계획 또는 변제계획에 따른 채무변제를 정상적으로 이행하고 있는 경우는 예외)

⑤ 결산 기준 사업개시일 또는 법인설립일이 3년 이상이고 최근 2년 결산 재무제표 상 부채비율(벤처캐피탈협회 회원사로부터 대출형 투자유치(CB, BW)를 통한 신규차입금은 부채총액에서 제외 가능)이 연속 500% 이상인 기업 또는 유동비율이 연속 50% 이하인 기업(단, 기업신용평가등급 중 종합신용등급이‘BBB’ 이상인 경우, 기술신용평가기관(TCB)의 기술신용평가등급이 “BBB”이상인 경우 또는 「외국인투자 촉진법」에 따른 외국인투자기업 중 외국인투자비율이 50% 이상이며, 기업설립일로부터 5년이 경과되지 않은 외국인투자기업은 예외). 사업개시일로부터 접수마감일까지 3년 미만인 기업의 경우 적용 예외

⑥ 최근 결산 기준 자본전액잠식

※ 한국채택국제회계기준(K-IFRS)을 적용함에 따라 자본전액잠식이 발생한 경우에는 일반기업회계기준(K-GAAP)을 적용하여 자본전액잠식 여부 판단 가능. 이 경우, 연구개발기관은 자본잠식 여부 판단을 위해 추가적인 회계기준에 따른 자료를 전문기관에 제출하여야 하며, 한국채택국제회계기준과 일반기업회계기준을 혼용할 수 없음.

※ 단, 전년도 자본잠식에 따른 자격제한 요건에 해당하더라도 당해연도에 자본금 이상의 투자유치를 달성하여 자본전액잠식에 해당하지 않는 경우*, 재무건전성을 확보한 것으로 보아 연구개발기관 자격 제한에서 제외

* 최근 회계연도말 결산기준 자본총계 + 당해연도 자본금 증가분이 “0” 이상인 경우(단, 자본금 증가분은 상환의무가 없는 보통주(우선주) 자본금에 한함)

⑦ 외부감사 기업의 경우 최근년도 결산감사 의견이 “의견거절” 또는 “부적정”

※ 단, 비영리 기관 및 공기업(공사)은 적용하지 않음

ㅁ 신청 제한사항

<연구자에 대한 제한사항>

ㅇ (참여제한 중인 자) 국가연구개발사업 참여제한 중인 자는 신청할 수 없음. (단, 연구개발계획서 제출마감일 전일에 참여제한이 종료된 자는 과제 신청 및 수행 가능)

※ 관련: 「국가연구개발혁신법」제32조 및 「국가연구개발현신법 시행령」제59조 제1항

ㅇ (연구개발과제 수의 제한) 동 사업은 인력 양성 사업으로 「국가연구개발혁신법 시행령」 제64조(연구개발과제 수의 제한) 제3항 제4호에 따라, **3책5공에서 제외**

ㅇ (인건비 계상률*) 연구책임자를 포함하여 모든 연구자는 수행 중인 국가연구개발사업 과제의 인건비 계상률 총합이 100%**를 초과하여 신청할 수 없음

* 인건비 계상률은 실제로 과제에 참여하는 정도가 아닌 인건비 및 연구수당 계상을 위한 용도로만 사용하고, 종전의 참여율 개념은 폐지됨(산식 = 해당연도에 해당과제 연구개발비에서 인건비로 계상한 금액 / 연 급여)

** 정부출연기관 및 전문생산기술연구소는 130%(출연연 기본사업 인건비계상률 포함), 국가연구개발사업으로는 100%만 참여 가능

<연구기관에 대한 제한사항>

ㅇ (연구기관 구성 제한) 동일한 연구개발과제 내 주관연구개발기관, 공동연구개발기관, 위탁연구개발기관은 모두 다른 기관*으로 구성해야 함

* 동일기관 여부: 법인등록번호 기준으로 판단(협약시, 법인인증서 사용)

※ 사업자등록번호는 다르나 법인등록번호가 같은 기관의 경우, 동일기관으로 협약체결

불가

- 주관.공동.위탁 연구개발기관 중 동일 기관으로 구성된 모든 과제는 상위 주관연구개발기관을 포함하여 평가대상에서 제외함

※ 컨소시엄연구개발과제의 경우 컨소시엄을 구성하는 각 ‘연구개발과제’별 적용

< 동일기관 과제 구성 제한 사례별 신청 가능 여부 >

연구개발기관 구분	사례1	사례2	사례3	사례4	사례5
주관연구개발기관	A기관	A기관	A기관	A기관	A기관
공동연구개발기관1	B기관	A기관	B기관	B기관	B기관
공동연구개발기관2	C기관	B기관	B기관	C기관	C기관
위탁연구개발기관	D기관	D기관	C기관	A기관	B기관
사례별 신청 가능 여부	가능	불가능	불가능	불가능	불가능

<과제 신청 및 구성에 대한 제한사항>

○ (중복 신청 제한) 본 공고에 대해 1개 과제의 주관연구책임자로만 신청 가능하며, **동일 내내역사업*의 과제를 주관연구책임자로 기 신청하였거나 기존에 수행 중인 경우, 본 공고에 신청 불가** * 양자인적기반조성사업(리더급 연구역량 강화)

- 중복 신청자가 포함된 모든 과제는 평가 대상에서 제외되며, 공동연구개발기관 과제에서 중복신청자가 확인된 경우에도 상위과제인 주관연구개발기관 과제까지 평가에서 제외

※ 공동연구책임자 및 참여연구원은 중복 신청 시 평가 제외하지 않으나 지양할 것을 권고

- 단, 동일 내내역사업에서 수행 중인 과제가 선정연도 내(2026.12.31.) 최종종료(연차/단계종료 제외)되는 경우, 예외적으로 신청 및 수행 가능

○ (유사과제 신청 제한) 기존에 유사과제를 수행하거나 참여하고 있는 경우는 신청을 지양하고, 신청하고자 하는 연구계획과 기 지원된 국가연구개발과제(타부처 포함)와의 차별성을 과제 신청 전 반드시 개별 확인

※ 차별성 검토 방법: www.ntis.go.kr 로그인 → 과제참여·관리 → 차별성검토

- 기존 국가연구개발사업 과제와 중복 과제로 판명 시 선정에서 제외될 수 있음

4 신청방법 및 제출서류

ㅁ 신청방법 및 절차

○ 범부처통합연구지원시스템(IRIS, https://www.iris.go.kr)에 연구책임자가 로그인하여 온라인 입력정보 작성 및 연구계획서 등 탑재 후 주관연구기관 확인·승인

※ <u>(1) 연구개발계획서 1개 파일(HWP)</u>과 <u>(2) 기타증빙 1개 파일(PDF)</u>로 각각 업로드(일반연구개발과제인 경우 **<u>공동/위탁 기관은 주관연구개발기관 연구개발계획서에 포함하여 작성, 별도 계획서 제출 불필요</u>**)

접수 전 필수 이행사항			신청			
연구(책임)자가 NRI 가입, 연구자전환 동의 및 정보 등록.갱신 (학력, 경력 등)	IRIS 기관등록, 기관총괄담당자 신청(기관담당자 권한 부여), 기관대표자 등록 등	⇨	온라인 입력정보 작성 및 연구계획서 등 등록	⇨	주관연구기관의 온라인 등록사항 확인·승인	⇨ 연구과제 **신청완료**
😪 **연구자**	😒 **주관연구기관**		**연구책임자**		**주관연구기관장**	

※ 연구책임자가 연구개발계획서 신청을 시작하기 전에 기관 대표자/담당자 정보가 입력되어 있어야 연구책임자의 과제 신청이 완료 가능. 온라인 신청 전 기관 담당자에게 확인

※ 세부사항은 별첨 세부사업 신청요강을 반드시 확인

▸IRIS를 통한 과제신청을 위해 접수 전 필수 이행사항이 있으니 과제신청에 문제가 없도록 사전에 준비하여 주시기 바랍니다.

※ 세부내용은 [별첨] 연구개발과제 접수 전 필수 이행사항(KISTEP IRIS운영단), IRIS 회원가입(연구자 전환) 및 연구자정보 등록 매뉴얼 참조

😪 **(연구자)** ① IRIS 회원가입, ② IRIS 내 NRI(국가연구자정보시스템)로 이동하여 연구자전환 동의(국가연구자번호 발급), ③ NRI 내 학력/경력* 및 주요 연구수행 실적** 정보 등록 필수

* 경력정보에서 근무(소속)부서 등록 필수

** 최근 5년간 수행완료 과제, 수행 중/신청 중 과제 목록 작성

※ **① and ②**: 연구책임자 포함 참여연구자 전원 필수(학생인건비 통합관리 기관의 학생연구자는 제외), **③**: 연구책임자만 필수

😒 **(주관연구기관)** IRIS 기관등록, 기관총괄담당자 신청(기관담당자 권한부여), 기관대표자 등록 등

※ 기관대표자 및 기관(총괄)담당자도 IRIS 회원가입 및 연구자전환 동의(국가연구자번호 발급)가 필수이며, 대표자 정보 미등록 시 연구자가 과제접수를 완료할 수 없으므로, **반드시 신청기간 시작 전까지 필수 이행사항 조치 필요**

※ 기관보유 시설장비 입력: NRI에 등록된 시설장비를 선택하여 추가

- 시설장비 등록방법: 기관총괄담당자 로그인 >[R&D고객센터]>[보유장비정보] 메뉴에서 등록

< 주관연구기관 선택 시 유의 사항 >

- 과제신청 시 주관연구기관은 <00대학교 산학협력단>이 아닌, <00대학교>로 신청요망

- <00대학교>의 기관정보(기관대표자 등록, 기관총괄담당자 신청, 기관담당자 승인권한 부여 등) 등록 필수

- 승인권한은 산학협력단 기관담당자가 산학협력단 과제뿐만 아니라 본교명(00대학교)으로 신청한 과제까지 모두 승인 가능

※ 현재 <00대학교 산학협력단>으로만 기관정보(대표자 및 기관총괄담당자 등)가 등록되어 있고, 접수마감까지 시간이 촉박하여 <00대학교>로 정보를 변경하여 신청하기가 어려울 경우 <00대학교 산학협력단>으로도 신청가능

▸**IRIS 문의처: IRIS 콜센터 1877-2041 또는 IRIS 홈페이지 사용문의 게시판 활용**

○ 과제 접수 매뉴얼 참조: 범부처통합연구지원시스템(http://www.iris.go.kr)

로그인 → 우측의 ‘R&D 업무포탈’ 클릭 및 접속 → R&D 고객센터 → IRIS 사용 매뉴얼 → [IRIS R&D 통합업무포털-연구자용] 접수 매뉴얼 다운로드

ㅇ 연구개발계획서의 작성이 완료되면, 화면 우측상단의 ‘최종확인’ 완료 이후 ‘제출’이 가능함. 제출된 연구개발계획서는 추가 수정 또는 삭제 불가

※ 연구개발계획서 수정은 연구책임자 접수마감일까지만 가능

ㅇ 과제 신청 시 연구계획에 부합하는 ‘국책연구_기획평가전문분야(국책연구사업)’를 선택하여야 함

ㅁ 제출서류

ㅇ QauntERA 양자기술연구지원사업 연구개발계획서(영문) 1부

※ QuantERA에 제출하는 영문계획서와 동일 서류 제출

※ 분량 : 25~35페이지 이내(예산서 별도 첨부)

ㅇ 별첨 서류 각 1부

※ 모든 별첨 서류는 한국측 연구기관만 해당되며, 주관연구개발기관이 취합하여 시스템에 일괄 제출

※ 필수(해당 시 필수 포함) 문서 미제출 시 평가 제외됨

<별첨 서류 목록>

구분	목록	제출주체		
		주관	공동	위탁
별첨1	[필수] 신청 자격의 적정성 확인서	○	○	○
별첨2	[필수] 개인정보·과세정보 제공•활용 동의서/연구윤리 준수서약서	○	○	○
별첨3	[필수] 기관 참여 확약서	○	○	○
별첨4	[영리기관 참여 시 필수] 기업 참여 의사 확인서 및 자격 증빙	○	○	○
별첨5	[해당 시 필수] 기관 지원확약서	○	○	○

5 신청기간 및 신청 시 유의사항

ㅁ 신청기간

구 분	내 용
연구책임자 신청 기간	**~ 2025.12.8.(월) 13:00까지** **※ QuantERA 접수기한은 2025.12.5.17:00(CET)임.**
주관연구기관 검토·승인기간	**~ 2025.12.8.(월) 13:00까지**
신청 절차	주관연구기관 연구책임자 접수 ▹ 주관연구기관 승인 ▹ 신청 완료

※ 연구책임자는 신청마감일까지 연구개발계획서 등록 및 기관검토 요청을 필히 완료해야 하며, 연구책임자의 신청사항에 대해 주관연구기관장의 승인이 완료되어야 신청접수가 최종 완료됨

○ 연구책임자: [연구책임자 신청 기간] 내에 연구개발계획서 등록(신청완료) 및 기관검토 요청까지 반드시 모두 완료하는 것을 원칙으로 함

○ 연구개발기관: [연구수행기관 검토·승인 기간] 내에 연구자가 신청 완료한 연구개발계획서에 대한 검토 및 승인을 완료해야 함 (단, 연구책임자 신청 기간에도 미리 검토.승인 가능함)

○ 기간 내에 신청 완료되지 않은 과제에 대한 구제는 절대 불가하며, 계획서업로드 시 작성 오류가 빈번하므로(유효성검증 오류 등) 연구자 신청마감 3일 전까지 업로드를 권장

ㅁ 신청 시 유의사항

○ **컨소시엄 코디네이터가 QuantERA에 접수*하며 각국 연구팀은 자국측 안내에 따라 신청 및 서류제출**

- 컨소시엄 참여 각국 연구팀은 자국의 안내사항 확인 필요

* **QuantERA 접수기한 : 2025. 12. 5.(Fri) 17:00 CET(한국시간 : 2025. 12. 6.(Sat) 01:00 KST)**

※ QuantERA 공고 사이트 : https://quantera.eu/call-2025/

○ **한국측 연구팀의 주관연구책임자는 QuantERA에 제출하는 동일서류(연구개발계획서)와 별첨 서류를 IRIS에 제출하며, 한쪽이라도 신청 기간 내 접수되지 않은 경우 미신청한 것으로 간주**

- **한국측 신청기간과 QuantERA측 신청기간이 상이한 점을 유의**해야함

○ 연구개발계획서는 QuantERA 제공 영문양식으로만 작성·제출하며, 컨소시엄 참여 연구진 간 협의를 통해 해당 연구자의 연구주안점, 연구목표 등의 업무영역 등을 명확히 구별하여 작성

○ 한국측 예산편성은 국가연구개발혁신법 매뉴얼에 의거하여 작성

○ 해당 시 필수자료를 포함한 필수 제출 별첨자료를 미작성, 미제출 시 평가에서 제외되는 등의 불이익이 있을 수 있음

○ 공고내용에 충족하는 과제가 없을 경우에 선정하지 않을 수 있음

○ 신청마감일 이후 신청서 제출, 제출서류 미비, 타 과제와의 연구내용 중복, 신청자격 부적격 등의 경우에 평가에서 제외 가능

○ 공고 내 명시된 연구개발과제 지원 조건(QuantERA 회원국 3개국(3개기관) 이상의 연구팀 구성)을 반드시 준수하여야 하며, 미 준수 시 평가 제외(요건탈락)

○ 평가위원회·추진위원회 의견 등에 따라서 과제 목표 및 내용, 과제 구성, 연구비, 연구기간 등의 조정이 가능하며, 정부 예산 상황에 따라 선정과제 수 조정 예정

○ 각종 증빙자료의 기산일은 신청 마감일 기준을 원칙으로 함(단, 참여제한의 경우 신청마감일 전일을 기준으로 함)

○ 사실과 다른 내용을 연구계획서, 별첨자료 등에 기재한 경우 제재(선정 취소 등) 가능

○ 선정 후 주요사항(주관·공동·위탁 연구개발기관 및 책임자 등) 변경을 원칙적으로 금지함

○ 본 공고문에서 정하지 않은 사항은 관련 법령 및 규정, QuantERA 기준에 의함

※ 「과학기술기본법」, 「기초연구진흥 및 기술개발지원에 관한 법률」, 「

국가연구개발혁신법」, 「국가연구개발혁신법 시행령」, 「과학기술정보통신부 소관 과학기술분야 연구개발사업 처리규정」 등

○ 본 공고문은 추후 공고 기간 내 수정사항이 발생할 수 있으며 수정사항이 발생할 경우, 별도 공지 예정

○ 연구책임자는 「국가연구개발혁신법」 제7조, 동법 시행령제9조제3항에 따라 협약 시 제출하는 연구개발계획서 내 국외수혜* 현황정보**를 포함하여야 하고, 과제 수행 중 국외 수혜 발생일로부터 30일 이내(권고) 현행화(IRIS 활용)

* 국외수혜: 외국 정부·기관·단체 등으로부터 재정적·행정적(연구과제, 인력, 장비, 시설) 지원 및 강의·자문·겸직 등으로 대가를 받는 사항(단순 문의·제안·논의 및 종료사항 미포함)

** 지원·지급 출처, 사유, 기간, 내용, 현재 수행중인 연구개발과제와의 관련 여부

시행령 제9조(연구개발과제 및 연구개발기관의 공모 절차) ③ 연구개발계획서에는 다음 각 호의 사항이 포함돼야 한다. … 8. 연구책임자가 연구개발기간 동안 외국의 정부・기관・단체 등으로부터 받는 행정적・재정적 지원이나 노무 또는 자문 등을 제공하고 받는 대가에 관한 사항

○ 생성형 인공지능(AI)을 활용할 경우, 신청자 및 수행자는 아래 사항 수행

※ 상세내용은「생성형 인공지능(AI) 도구의 책임 있는 사용을 위한 권고사항」('25.9, 한국연구재단) 참조

- 연구개발계획서 및 단계/최종보고서 작성 과정에서 생성형 AI 도구를 작성한 경우, 해당 계획서 및 보고서에 AI 도구 사용 내역을 기술(권장)

- 연구의 전 단계에서 AI 도구를 활용할 경우, 연구자가 직접 출처를 검증하여 자료의 신뢰성 및 타당성을 확인하고 연구윤리 위반 가능성을 점검하여 필요시 수정·보완 과정을 거쳐야함

※ AI 사용 시 발생할 수 있는 표절, 위조, 변조 등 다양한 연구진실성 위반 사례 예방 위함

- 연구의 전 단계에서 AI 도구를 활용할 경우, 중요한 연구정보*가 유출되지 않도록 주의

* 연구정보 : 연구데이터, 과제정보 및 수행자의 개인 정보 등

※ 연구자료를 AI 도구에 업로드하여 연구정보 유출이 발생할 경우, AI 도구를 활용한 연구자에게 책임이 있음.

ㅇ (해당 시) 주관기관 또는 공동연구개발기관이 기업인 경우 「국가연구개발혁신법 시행령」제19조제3항에 따라 기관부담연구개발비를 부담하여야 함

※ 단, 연구개발성과를 국가 소유로 하거나, 연구개발서비스업자가 시험·분석 등 연구개발서비스의 제공만을 목적으로 하는 공동연구개발기관으로서 참여하는 경우는 제외

6 선정평가

□ 평가방법

○ 국내 관련 법령(국가연구개발혁신법 및 동법 시행령 등) 및 QuantERA 규정 등에 근거하여 추진

○ 한국연구재단은 평가 전 사전검토를 통해 국내법에 따른 한국측 연구팀의 적격여부 등을 확인하며, 선정평가는 QuantERA에서 시행

○ 평가항목 및 배점

평가항목	배점	
	기초	응용
• Excellence	50	25
• Impact	25	50
• Implementation	25	20

□ 평가절차

접수과제 교환	①사전검토	②선정평가	③사업추진위 심의	④최종선정 및 이의신청
접수여부 확인 (한쪽이라도 미접수 시 미신청으로 간주)	한국측 연구자 응모자격 및 제출서류의 적절성 검토·보완	QuantERA 기준에 따라 선정평가 실시	지원과제 심의 및 조정/보고	평가결과 선정 공고 및 이의신청 접수
NRF, QuantERA	NRF	QuantERA	과기정통부 (NRF)	과기정통부

ㅇ **(사전검토)** 각국 기준에 따라 연구개발과제에 지원한 기관·단체·연구자에 대한 신청자격, 참여제한 대상 여부 등을 선정평가에 앞서 검토

한국측 연구자에 대한 사전검토 항목(국가연구개발혁신법 제10조, 동법 시행령 제11조)
ㅇ 국가연구개발혁신법 제32조제1항에 따른 참여제한 해당 여부 ㅇ 국가연구개발혁신법 시행령 제9조제1항제2호가목에 따른 신청 자격의 적합 여부 ※ 연구개발계획서 등 신청서류에 허위사실을 기재하거나 각종 제출자료 및 정보를 조작한 경우 선정 대상에서 제외하며, 선정된 이후 이러한 사실이 발견되면 선정 취소, 연구개발비 환수 등 제재처분이 적용될 수 있음

- 한국측의 경우, 국가과학기술지식정보서비스(NTIS, www.ntis.go.kr) 차별성검토 시스템 및 평가위원회에서 신청과제에 대한 차별성 검토를 실시할 수 있음

※ 타 국가연구개발사업 과제와 차별성이 없는 과제로 판명된 경우, 선정 제외될 수 있음

ㅇ **(선정평가)** QuantERA 기준에 따라 선정평가 실시

- **(선정평가 세부절차)** 2단계 적격성 심사(국가 및 국제 기준 동시 적용) → 국제 전문가 개별 평가(3인) → 합의평가 → 분야별 랭킹리스트 작성 → CSC(국가조정위원회) 승인

- **(평가지표 외 추가 고려사항)** 자금 집행 극대화, Widening 국가 참여 여부, 컨소시엄 내 성별 균형, 국가별 예산 분배 균형 등

ㅇ **(이의신청)** 연구개발기관의 장은 평가결과를 통보받은 후 10일 이내에 전문기관의 장에게 이의신청서를 제출할 수 있음

- **(인정 범위)** 통보된 평가결과에 대해서만 이의신청을 받으며, 평가위원, 평가 방법 및 절차 등에 관한 사항은 제외

7 기타사항

※ 동 사항은 한국측 연구자에만 해당됨

<필수사항>

ㅁ 연구개발과제의 성실 수행

○ 연구개발과제에 참여하는 연구자는 연구노트(연구개발과제 수행 과정과 연구개발성과를 기록한 자료를 말한다)를 작성하고 관리하여야 함

국가연구개발사업 연구노트 지침 **제8조(연구노트의 작성)** ① 연구개발기관의 장은 소속 연구자가 연구노트를 작성하도록 관리하여야 한다. ② 연구노트의 작성에 관한 사항은 연구개발기관의 장이 자체규정으로 정한다. ③ 제1항에도 불구하고 연구개발과제의 협약 당사자(법 제4조제1호에 따른 다른 법률에 따라 직접 설립된 기관의 기본사업의 경우에는 해당 기관의 장을 말한다)는 개인사업자, 창업초기기업 등 연구노트를 관리하기 어렵다고 인정하는 연구개발기관의 경우나, 사전조사・기획평가, 연구개발과제의 조정・관리, 인문・사회분야, 인력양성, 기반구축 등 연구노트 작성의 필요성이 크지 아니하다고 인정하는 연구개발과제의 경우에 법 제12조제4항에 따른 연차보고서 또는 제12조제5항에 따른 최종보고서(같은 항에 따른 단계보고서를 포함한다) 등의 작성을 연구노트 작성으로 볼 수 있다. ④ 하나의 연구개발과제에 다수의 연구개발기관이 참여하는 경우에는 연구개발기관마다 연구노트를 각각 작성하는 것을 원칙으로 한다. ⑤ 연구개발기관의 장은 자체규정으로 정하는 바에 따라 연구자별로 연구노트를 각각 작성하게 하거나, 하나의 연구노트를 다수의 연구자가 공동으로 작성하게 할 수 있다. 이 경우 모든 연구자는 연구노트를 작성하는 것을 원칙으로 한다. ⑥ 기록자는 연구노트를 작성할 때에 내용의 위조・변조 없이 객관적인 사실을 기록하고, 제3자가 연구개발 수행 과정과 결과를 재현하는데 활용할 수 있도록 노력하여야 한다.

ㅁ 특수관계자(배우자, 직계존·비속 등)가 연구과제에 참여하고자 하는 경우 전문기관의 승인을 받아야 함

ㅁ 향후 소관 부처 및 연구관리전문기관에서 연구책임자에게 사업 기획·평가 참여 요청 시 적극 협조할 것을 권장함

ㅁ 연구개발과제 보고 및 평가 관련 안내사항

ㅇ 연구책임자는 외국정부·기관·단체 등으로부터 행정적· 재정적 지원을 받거나 노무 또는 자문 등을 제공하고 받는 대가에 관한 사항을 연구개발계획서 내 포함하여 제출하여야 함

※ 관련: 국가연구개발혁신법 시행령 제9조 제③항

○ 연차.최종보고서 제출의무 준수

※ 관련: 국가연구개발혁신법 제12조 및 국가연구개발혁신법 시행령 제18조

* 협약의 내용을 변경하려는 경우 변경사유와 내용을 사전에 문서로 명확히 알리고 상호 협의를 거쳐야 함(국가연구개발혁신법 제11조 및 국가연구개발혁신법 시행령 제14조)

○ 최종평가 시 연구개발과제의 수행과정과 연구개발성과 등에 대하여 평가함

※ 관련: 국가연구개발혁신법 제12조 및 동법 시행령 제16~17조

○ 특정 사유가 발생하는 경우 특별평가를 통해 연구개발과제의 변경 및 중단 여부를 결정

※ 관련: 국가연구개발혁신법 제15조 및 동법 시행령 제31조

□ 기술료 납부에 관한 사항

○ 연구개발성과소유기관은 국가연구개발혁신법 제18조제1항에 따라 기술실시계약을 체결하고 기술료를 징수하는 경우 기술료 징수 결과 보고서를 제출해야 함

○ 기술료에 관한 세부사항은 관련 국가연구개발혁신법 등 관련 법령과 규정에 따름

<주관연구개발기관 또는 공동연구개발기관이 기업인 경우>

□ 중소·중견·대기업의 기관부담연구개발비

○ 신규과제의 연구개발기관 중 중소·중견·대기업에 대하여 「국가연구개발혁신법 시행령」 별표1에 따라 산정하여야 함

< 총 연구개발비 중 정부지원 연구개발비 지원 기준 >

중소기업인 경우	국제공동연구개발비를 제외한 연구개발비의 75% 이하
중견기업인 경우	국제공동연구개발비를 제외한 연구개발비의 70% 이하
공기업, 대기업인 경우	국제공동연구개발비를 제외한 연구개발비의 50% 이하

< 총 연구개발비 중 기관부담 연구개발비 현금부담 비율 >

중소기업인 경우	기관부담 연구개발비의 10% 이상
중견기업인 경우	기관부담 연구개발비의 13% 이상
공기업, 대기업인 경우	기관부담 연구개발비의 15% 이상

8 향후일정

일정	내용
~2025. 12. 8.(월) 13시까지	신청 마감일 ※ QuantERA 신청기한은 2025. 12. 5. 17:00(CET)임.
~2025. 12. 8.(월) 13시까지	주관연구기관 검토·승인기간
2026. 상반기	선정평가 실시
2026. 9. 1. 이후	연구개시

※ 상기 일정은 추진 상황에 따라 향후 변동될 수 있음.

9 적용 법령 및 규정

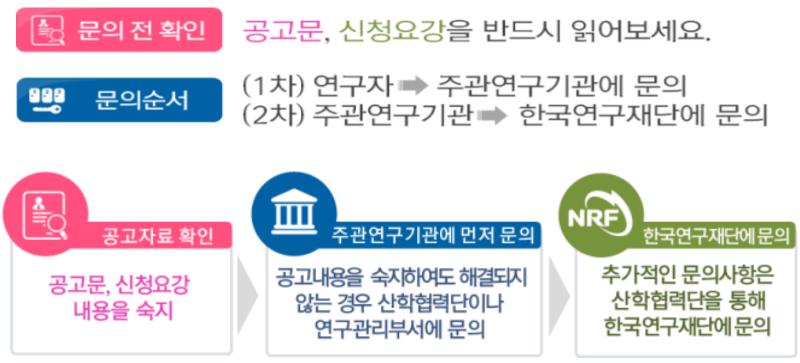
ㅁ 공고에서 정하지 않은 사항은 아래 법령 및 관련 규정을 따름

근거법령	ㅇ 국가연구개발혁신법, 동법 시행령·시행규칙 및 관련 행정규칙 - 국가연구개발사업 연구개발비 사용기준, 국가연구개발사업 연구노트 지침, 국가연구개발사업 동시수행 연구개발과제 수 제한기준, 국가연구개발사업 보안대책, 연구지원기준, 국가연구개발정보처리기준, 국가연구개발 시설.장비의 관리 등에 관한 표준지침 등 ㅇ 과학기술기본법 및 하위법령, 시행규칙 ㅇ 기초연구진흥 및 기술개발지원에 관한 법률

	ㅇ 기타: 국가연구개발혁신법 매뉴얼, 국가연구개발 연구윤리 길잡이 등 ㅇ 양자과학기술 및 양자산업 육성에 관한 법률
관련규정	ㅇ 과학기술정보통신부 소관 과학기술분야 연구개발사업 처리규정 등
기타	ㅇ 과학기술정보통신부 연구개발사업 종합시행계획 ㅇ 양자과학기술 연구개발사업 시행계획 등

10 문의처

□ **문의 절차** ※ 문의 전화 전 반드시 공고문 및 FAQ 확인 후 연구 수행기관을 통하여 질의 요망



※ 본 공모내용에 명시되지 않은 세부사항은 QuantERA 공고 확인 후 문의

□ **공고문 및 양식 확인 방법**

○ 공고내용이 수정되는 경우에도 아래 사업 공지사항 메뉴를 통해 수정사항이 게시되므로 지속 확인 필수 (수정사항을 개별 안내하지 않음)

○ 과학기술정보통신부 (https://msit.go.kr) → 소식 → 사업공고

○ 한국연구재단 (https://www.nrf.re.kr) → 사업안내 → 사업공고→ 사업공지

□ **관련 법령, 규칙, 매뉴얼 등 조회 방법**

○ 범부처 연구비통합관리시스템 홈페이지(http://gaia.go.kr) 접속

- 법,규정,규칙: 「자료실」→「국가R&D연구비관련 법·규정」→ '공통 법·규정', '과학기술정보통신부(한국연구재단)' 관련 규정 확인

□ **문의처: 문의 사항별 담당 부서가 다르므로 문의처 구분 확인 요망**

※ 담당 PM(단장)에게는 신청 완료 전까지만 문의 가능(신청마감 후에는 기술단 담당 직원(PO, 연구원 등)에게만 문의 가능하며, 단장에게는 문의 불가)

구분	담당부서 / 기관명	연락처	이메일
IRIS 시스템 관련 문의	IRIS 콜센터(1877-2041) 또는 IRIS 홈페이지 사용문의 게시판 활용		
RFP, 접수, 양식 관련 문의	양자기술단	042-869-7837	star0rin@nrf.re.kr
정책 관련 문의	과학기술정보통신부 양자혁신기술개발과	044-202-6875	-
QuantERA측 문의	QuantERA Coordination Office (Dr. Maurice Tia)	+33 (0)1 72 73 06 90	Maurice.Tia@anr.fr

붙임 1. 연구개발계획서 양식

2. 별첨자료 양식

3. IRIS 관련 매뉴얼

4. QuantERA Call 2025

요약문 (Summary in Korean)

■ 국가연구개발혁신법 시행규칙 [별지 제1호서식]

연구개발계획서	[ ] 신청용 [ ] 협약용	보안등급 일반[ ], 보안[ ]

중앙행정기관명		사업명	사업명	
전문기관명(해당 시 작성)			내역사업명 (해당 시 작성)	
공고번호		총괄연구개발 식별번호 (해당 시 작성)		
		연구개발과제번호		
선정방식	정책지정[ ] 공모: 지정공모[ ] 품목공모[ ] 분야공모[ ] 자유공모[ ]			

기술분류	국가과학기술표준분류	1순위 소분류 코드명	%	2순위 소분류 코드명	%	3순위 소분류 코드명	%
	부처기술분류 (해당 시 작성)	1순위 소분류 코드명	%	2순위 소분류 코드명	%	3순위 소분류 코드명	%

총괄연구개발명 (해당 시 작성)	국문			
	영문			
연구개발과제명	국문			
	영문			
주관연구개발기관	기관명		사업자등록번호	
	주소	(우)	법인등록번호	

연구책임자	성명			직위	
	연락처	직장전화		휴대전화	
		전자우편		국가연구자번호	

연구개발기간	전체			YYYY. MM. DD – YYYY. MM. DD(년 개월)
	단계 (해당 시 작성)	1단계	1년차	YYYY. MM. DD – YYYY. MM. DD(년 개월)
			n년차	YYYY. MM. DD – YYYY. MM. DD(년 개월)
		n단계	1년차	YYYY. MM. DD – YYYY. MM. DD(년 개월)
			n년차	YYYY. MM. DD – YYYY. MM. DD(년 개월)

연구개발비 (단위: 천원)		정부지원 연구개발비	기관부담 연구개발비		그 외 기관 등의 지원금				합계			연구개발비 외 지원금
					지방자치단체		기타( )					
		현금	현금	현물	현금	현물	현금	현물	현금	현물	합계	
총계												
1단계	1년차											

	n년차										
n단계	1년차										
	n년차										

공동연구개발기관 등 (해당 시 작성)	기관명	책임자	직위	휴대전화	전자우편	비고	
						역할	기관유형
공동연구개발기관							
위탁연구개발기관							
연구개발기관 외 기관							

연구개발과제 실무담당자	성명			직위	
	연락처	직장전화		휴대전화	
		전자우편		국가연구자번호	

관련 법령 및 규정과 모든 의무사항을 준수하면서 이 연구개발과제를 성실하게 수행하기 위하여 연구개발계획서를 제출합니다. 아울러 이 연구개발계획서에 기재된 내용이 사실임을 확인하며, 만약 사실이 아닌 경우 연구개발과제 선정 취소, 협약 해약 등의 불이익도 감수하겠습니다.

년 월 일

연구책임자: (인)
주관연구개발기관의 장: (직인)
공동연구개발기관의 장: (직인) (신청시 제외)
위탁연구개발기관의 장: (직인) (신청시 제외)

중앙행정기관의 장 귀하

앞표지 작성 요령(작성 요령은 제출하지 않습니다)

1. 보안등급: 법 제21조제2항에 따른 보안과제에 해당하는 경우 '보안'에, 그 외의 경우 '일반'에 [√] 표시합니다(연구자 직접 기재 불필요).
2. 중앙행정기관명: 연구개발과제를 공고한 중앙행정기관의 명칭을 기재합니다(중앙행정기관이 복수인 경우에는 모든 해당 중앙행정기관의 명칭).
3. 전문기관명: 연구개발과제를 관리하는 전문기관명을 기재합니다(연구자 직접 기재 불필요).
4. 사업명: 해당 연구개발과제의 사업명을 기재합니다(연구자 직접 기재 불필요).
5. 내역사업명: 해당 연구개발과제의 내역사업명을 기재합니다(연구자 직접 기재 불필요).
6. 공고번호: 연구개발과제 공고문 상단의 공고번호를 기재합니다(연구자 직접 기재 불필요).
7. 총괄연구개발 식별번호: 총괄연구개발명에 부여되는 번호를 기재합니다(연구자 직접 기재 불필요).
8. 연구개발과제번호: 연구개발과제 선정 시 부여되는 번호를 기재합니다(연구자 직접 기재 불필요).
9. 선정방식: 공고문에서 제시한 선정방식을 기재합니다(연구자 직접 기재 불필요).
10. 국가과학기술표준분류: 「과학기술기본법」 제27조제1항에 따른 국가과학기술표준분류표 중 연구개발과제에 해당하는 소분류를 우선순위에 따라 그 코드명과 비중을 기재합니다.
11. 부처기술분류: 중앙행정기관에서 소관 법령에 따라 입력을 요청하는 과학기술분류 중 연구개발과제에 해당하는 소분류를 우선순위에 따라 그 코드명과 비중을 기재합니다.
12. 총괄연구개발명: 2개 이상의 연구개발과제가 서로 연관되어 추진되는 경우에 이를 총괄하는 연구개발 명칭을 기재합니다.
13. 연구개발과제명: 연구개발기관이 수행하는 연구개발과제의 명칭을 기재합니다.
14. 연구개발기간: 연구개발과제가 단계로 구분되지 않는 경우에는 연구개발기간 전체를 1단계로 간주합니다.
   1) 전체: 연구개발과제의 전체 연구개발기간으로서 협약기간을 기재합니다.
   2) 단계: 연구개발과제가 단계로 구분된 경우에 해당 단계의 연구개발기간을 기재합니다.
15. 연구개발비: 연구개발과제가 단계로 구분되지 않는 경우에는 연구개발기간 전체를 1단계로 간주합니다.
   1) 정부지원연구개발비: 중앙행정기관이 지원하는 연구개발비를 기재합니다.
   2) 기관부담연구개발비: 시행령 제19조 및 시행령 [별표 1]에 따라 연구개발기관이 부담하는 연구개발비를 현금과 현물로 구분하여 기재합니다.
   3) 그 외 기관 등의 지원금: 1) 또는 2)에 해당하지 않는 연구개발비를 지원하는 기관이거나, 연구개발성과를 활용・구매 등을 목적으로 하는 기관 등이 지원하는 연구개발비로서 현금과 현물로 구분하여 기재합니다.
   4) 연구개발비 외 지원금: 국제기구, 외국의 정부・기관・단체 등이 지원・부담하는 금액이거나, 중앙행정기관(소속기관 포함)이 소관 업무를 위하여 직접 수행하는 사업의 금액으로 「국가연구개발혁신법」에 따른 연구개발비에 포함하지 않는 금액을 기재합니다.
16. 공동연구개발기관의 역할
   1) 공동연구개발기관으로서 연구개발성과를 활용・구매 등을 목적으로 하는 기업(수요기업)인 경우에 "수요"로 기재합니다.
   2) 공동연구개발기관이 수요기업이 아닌 경우에 "공동"으로 기재합니다.
17. 위탁연구개발기관의 역할 : "위탁"으로 기재합니다.
18. 연구개발기관 외 기관의 역할(공모 시 요구한 경우에 한하여 기재)
   1) 해당 기관이 지방자치단체인 경우에 "지자체"로 기재합니다.
   2) 해당 기관이 국외 연구개발기관인 경우에 "국협"으로 기재합니다.
   3) 해당 기관이 연구개발성과를 활용하는 기관인 경우에 "수혜"로 기재합니다.
   4) 해당 기관이 연구개발과제와 관련된 컨설팅을 하는 기관인 경우에 "컨설팅"으로 기재합니다.
   5) 그 외는 "기타"로 기재합니다.
19. 기관유형
   1) 국가가 직접 설치하여 운영하는 연구기관인 경우에 "국립연"으로 기재합니다(중앙행정기관(소속기관을 제외)이 직접 연구개발과제를 수행하는 경우에는 "정부부처").
   2) 지방자치단체가 직접 설치하여 운영하는 연구기관인 경우에 "공립연"으로 기재합니다(지방자치단체(소속기관을 제외)가 직접 연구개발과제를 수행하는 경우에는 "지자체").
   3) 「고등교육법」 제2조에 따른 학교인 경우에 "대학"으로 기재합니다.
   4) 다음의 어느 하나에 해당하는 기관인 경우에 "정부출연연"으로 기재합니다.

앞표지 작성 요령(작성 요령은 제출하지 않습니다)

(1)「정부출연연구기관 등의 설립·운영 및 육성에 관한 법률」제2조에 따른 정부출연연구기관

(2)「과학기술분야 정부출연연구기관 등의 설립·운영 및 육성에 관한 법률」제2조에 따른 과학기술분야 정부출연연구기관

(3)「특정연구기관육성법」제2조에 따른 특정연구기관

(4)「한국해양과학기술원법」 제3조에 따라 설립된 한국해양과학기술원

(5)「국방과학연구소법」 제3조에 따라 설립된 국방과학연구소

5)「지방자치단체출연 연구원의 설립 및 운영에 관한 법률」제2조에 따른 지방자치단체출연연구원인 경우에 "지자체출연연"으로 기재합니다.

6)「중소기업기본법」제2조에 따른 기업인 경우에 "중소기업"으로 기재합니다.

7)「중견기업 성장촉진 및 경쟁력 강화에 관한 특별법」제2조제1호에 따른 기업인 경우에 "중견기업"으로 기재합니다.

8)「상법」제169조에 따른 회사로서 중소기업 또는 중견기업이 아닌 경우에 "대기업"으로 기재합니다.

9)「공공기관의 운영에 관한 법률」제5조제4항제1호에 따른 공기업인 경우 "공기업"으로 기재합니다.

10)「의료법」제3조제2항제3호에 따른 병원급 의료기관인 경우 "병원"으로 기재합니다.

11)「산업기술혁신 촉진법」 제42조제1항에 따른 전문생산기술연구소인 경우 "전문연"으로 기재합니다.

12) 1)부터 11)까지에 해당하지 않는 기관인 경우에 "기타"로 기재합니다.

20. 연구개발과제 실무담당자: 연구개발과제에 참여하여 연구개발내용에 이해도가 높고 전문기관과 연구개발내용에 대한 실무적인 협의가 가능한 주관연구개발기관 담당자를 기재합니다.

21. 기관장 서명: 전자서명으로 하고, 신청서 작성·제출 시에는 주관연구개발기관의 장, 협약 시에는 주관연구개발기관의 장과 공동연구개발기관의 장, 위탁연구개발기관의 장의 전자서명을 날인합니다.

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2)

< 요 약 문 >

※ 요약문은 5쪽 이내로 작성합니다.

<table>
<tr><td colspan="2">사업명</td><td colspan="3"></td><td colspan="2">총괄연구개발 식별번호
(해당 시 작성)</td><td colspan="2"></td></tr>
<tr><td colspan="2">내역사업명
(해당 시 작성)</td><td colspan="3"></td><td colspan="2">연구개발과제번호</td><td colspan="2"></td></tr>
<tr><td rowspan="2">기술분류</td><td>국가과학기술
표준분류</td><td>1순위 소분류 코드명</td><td>%</td><td>2순위 소분류 코드명</td><td>%</td><td>3순위 소분류 코드명</td><td>%</td></tr>
<tr><td>부처기술분류
(해당 시 작성</td><td>1순위 소분류 코드명</td><td>%</td><td>2순위 소분류 코드명</td><td>%</td><td>3순위 소분류 코드명</td><td>%</td></tr>
<tr><td colspan="2">총괄연구개발명
(해당 시 작성)</td><td colspan="6"></td></tr>
<tr><td colspan="2">연구개발과제명</td><td colspan="6"></td></tr>
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<table>
<tr><td>전체 연구개발기간</td><td colspan="5"></td></tr>
<tr><td>총 연구개발비</td><td colspan="5">총 천원
(정부지원연구개발비: 천원, 기관부담연구개발비 : 천원,
지방자치단체지원연구개발비: 천원, 그 외 지원연구개발비: 천원)</td></tr>
<tr><td>연구개발단계</td><td>기초[ ] 응용[ ] 개발[ ]
기타(위 3가지에 해당되지 않는 경우)[ ]</td><td>기술성숙도
(해당 시 작성)</td><td colspan="3">착수시점 기준( )
종료시점 목표( )</td></tr>
<tr><td>연구개발과제 유형
(해당 시 작성)</td><td colspan="5"></td></tr>
<tr><td>연구개발과제 특성
(해당 시 작성)</td><td colspan="5"></td></tr>
</table>

<table>
<tr><td rowspan="6">연구개발
목표 및 내용</td><td colspan="2">최종 목표</td><td></td></tr>
<tr><td colspan="2">전체 내용</td><td></td></tr>
<tr><td rowspan="2">1단계
(해당 시 작성)</td><td>목표</td><td></td></tr>
<tr><td>내용</td><td></td></tr>
<tr><td rowspan="2">n단계
(해당 시 작성)</td><td>목표</td><td></td></tr>
<tr><td>내용</td><td></td></tr>
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<table>
<tr><td>연구개발성과
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기대 효과</td><td colspan="5"></td></tr>
<tr><td>국문핵심어
(5개 이내)</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>영문핵심어
(5개 이내)</td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2)

요약문 작성 요령(작성 요령은 제출하지 않습니다)

1. 사업명: 해당 연구개발과제의 사업명을 기재합니다(연구자 직접 기입 불필요).

2. 내역사업명: 해당 연구개발과제의 내역사업명을 기재합니다(연구자 직접 기입 불필요).

3. 총괄연구개발 식별번호: 총괄연구개발명에 부여되는 번호를 기재합니다(연구자 직접 기입 불필요).

4. 연구개발과제번호: 연구개발과제 선정 시 부여되는 번호를 기재합니다(연구자 직접 기입 불필요).

5. 기술분류: 연구개발계획서 표지에 기재한 기술분류를 기재합니다.

6. 총괄연구개발명: 연구개발계획서 표지에 기재한 총괄연구개발명을 기재합니다.

7. 연구개발과제명: 연구개발계획서 표지에 기재한 연구개발과제명을 기재합니다.

8. 전체 연구개발기간: 연구개발계획서 표지에 기재한 연구개발과제의 전체 연구개발기간을 기재합니다.

9. 총 연구개발비: 연구개발계획서 표지에 기재한 연구개발과제의 총 연구개발비를 기재합니다.

10. 연구개발단계: 해당되는 연구개발과제의 연구개발단계 유형에 [√] 표시합니다.

1) 기초연구단계란 특수한 응용 또는 사업을 직접적 목표로 하지 아니하고 현상 및 관찰 가능한 사실에 대한 새로운 지식을 얻기 위하여 수행하는 이론적 또는 실험적 연구단계를 의미합니다.

2) 응용연구단계란 기초연구단계에서 얻어진 지식을 이용하여 주로 실용적인 목적으로 새로운 과학적 지식을 얻기 위하여 수행하는 독창적인 연구단계를 의미합니다.

3) 개발연구단계란 기초연구단계, 응용연구단계 및 실제 경험에서 얻어진 지식을 이용하여 새로운 제품, 장치 및 서비스를 생산하거나 이미 생산되거나 설치된 것을 실질적으로 개선하기 위하여 수행하는 체계적 연구단계를 의미합니다.

4) 기타는 기초, 응용, 개발 등 3가지 단계에 해당하지 않는 경우를 의미합니다.

11. 기술성숙도: 특정기술(재료, 부품, 소자, 시스템 등)의 성숙도로서 최종 연구개발 목표, 내용, 최종 결과물 등을 고려하여 아래의 9단계 중 해당하는 단계를 선택합니다(특정기술의 개발을 목적으로 하는 연구개발과제의 경우에만 작성).

1) 기초연구단계: 1단계(기초 이론・실험), 2단계(실용 목적의 아이디어, 특허 등 개념 정립)

요약문 작성 요령(작성 요령은 제출하지 않습니다)
2) 실험단계: 3단계(연구실 규모의 기본성능 검증), 4단계(연구실 규모의 소재・부품・시스템 핵심성능 평가) 3) 시작품단계: 5단계(확정된 소재・부품・시스템 시작품 제작 및 성능 평가), 6단계(시범규모의 시작품 제작 및 성능 평가) 4) 제품화단계: 7단계(신뢰성평가 및 수요기업 평가), 8단계(시제품 인증 및 표준화) 5) 사업화단계: 9단계(사업화) 12. 연구개발과제 유형: 중앙행정기관이 연구개발과제 공고 시 자율적으로 구분한 유형을 기재합니다(연구자 직접 기입 불필요). 13. 연구개발과제 특성: 중앙행정기관이 연구개발과제 공고 시 기재한 연구개발과제의 특성을 기재합니다(연구자 직접 기입 불필요). 14. 연구개발 목표: 연구개발과제의 목표를 500자 내외로 기재합니다. 15. 연구개발 내용: 연구개발과제의 내용을 1,000자 내외로 기재합니다. 16. 연구개발성과 활용계획 및 기대효과: 연구개발성과의 수요처, 활용내용, 경제적 파급효과 등을 500자 내외로 기재합니다(연구시설・장비 구축을 목적으로 하는 연구개발과제의 경우에 연구시설・장비를 활용한 성과관리 및 자립운영계획, 수입금 관리 및 운영계획 등).

연구제안서 본문 **(Proposal in Korean)**

< 본문 2 >

1. 연구개발기관 현황

1) 연구책임자 등 현황

(1) 주관연구개발기관 연구책임자

가. 인적사항

개인	국문		국적	
	영문		국가연구자번호	
직장	기관명		전화번호	
	부서		휴대전화	
	직위		전자우편	
	주소	(우:)		

나. 학력(연구개발과제 특성에 따라 선택적으로 적용이 가능합니다)

취득연월(최근 순으로 작성)	학교명	전공	학위	지도교수
yy.mm~yy.mm				
yy.mm~yy.mm				
최종학위 논문명(해당 시):				

다. 경력(연구개발과제 특성에 따라 선택적으로 적용이 가능합니다)

기간	기관명	직위	비고
yy.mm~yy.mm			
yy.mm~yy.mm			

라. 주요 연구개발 실적(최근 5년간 5개 이내의 실적으로 작성하되, 신청 중이거나 수행 중인 연구개발과제는 필수적으로 작성해야 합니다)

중앙행정기관	세부사업명	연구개발과제명	주관연구개발기관	연구개발기간 (참여한 기간)	역할: 연구책임자/	비고 (신청/수행중

(전문기관)			당시 소속기관		연구자	/완료)
				yy.mm.dd~yy.mm.dd (yy.mm.dd~yy.mm.dd)		
				yy.mm.dd~yy.mm.dd (yy.mm.dd~yy.mm.dd)		

마. 대표적 논문/저서 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분 (논문/저서)	논문명/저서명	게재지 (권, 쪽)	게재연도 (발표연도)	역할	등록번호 (ISSN)	비고 (피인용 지수)
			yy			
			yy			

바. 지식재산권 출원·등록 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분 (특허/프로그램 등)	지식재산권명	국가명	출원·등록일	출원·등록번호/ 출원·등록자 수	비고

사. 그 밖의 대표적 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분	실적명	내용요약	실적연도
			yy
			yy

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2))

(2) 공동연구개발기관 책임자(해당 시 작성합니다)

가. 인적사항

개인	국문		국적	
	영문		국가연구자번호	
직장	기관명		전화번호	
	부서		휴대전화	
	직위		전자우편	
	주소	(우:)		

나. 학력(연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

취득연월(최근 순으로 작성)	학교명	전공	학위	지도교수
yy.mm~yy.mm				
yy.mm~yy.mm				
최종학위 논문명(해당 시):				

다. 경력(연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

기간	기관명	직위	비고
yy.mm~yy.mm			
yy.mm~yy.mm			

라. 주요 연구개발 실적(최근 5년간 5개 이내의 실적으로 작성하되, 신청 중이거나 수행 중인 연구개발과제는 필수적으로 작성해야 합니다)

중앙행정기관 (전문기관)	세부사업명	연구개발과제명	주관연구개발기관 / 당시 소속기관	연구개발기간 (참여한 기간)	역할: 연구책임자/연구자	비고 (신청/수행중/완료)
				yy.mm.dd~yy.mm.dd (yy.mm.dd~yy.mm.dd)		
				yy.mm.dd~yy.mm.dd (yy.mm.dd~yy.mm.dd)		

마. 대표적 논문/저서 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분 (논문/저서)	논문명/저서명	게재지 (권, 쪽)	게재연도 (발표연도)	역할	등록번호 (ISSN)	비고 (피인용 지수)
			yy			
			yy			

바. 지식재산권 출원·등록 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분 (특허/프로그램 등)	지식재산권명	국가명	출원·등록일	출원·등록번호/ 출원·등록자 수	비고

사. 그 밖의 대표적 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분	실적명	내용요약	실적연도
			yy
			yy

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2)

(3) 위탁연구개발기관 책임자(해당 시 작성합니다)

가. 인적사항

개인	국문		국적	
	영문		국가연구자번호	
직장	기관명		전화번호	
	부서		휴대전화	
	직위		전자우편	

	주소	(우:　　　)

나. 학력(연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

취득연월(최근 순으로 작성)	학교명	전공	학위	지도교수
yy.mm~yy.mm				
yy.mm~yy.mm				
최종학위 논문명(해당 시):				

다. 경력(연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

기간	기관명	직위	비고
yy.mm~yy.mm			
yy.mm~yy.mm			

라. 주요 연구개발 실적(최근 5년간 5개 이내의 실적으로 작성하되, 신청 중이거나 수행 중인 연구개발과제는 필수적으로 작성해야 합니다)

중앙행정기관 (전문기관)	세부사업명	연구개발과제명	주관연구개발기관 당시 소속기관	연구개발기간 (참여한 기간)	역할: 연구책임자/ 연구자	비고 (신청/수행중/완료)
				yy.mm.dd~yy.mm.dd (yy.mm.dd~yy.mm.dd)		
				yy.mm.dd~yy.mm.dd (yy.mm.dd~yy.mm.dd)		

마. 대표적 논문/저서 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분 (논문/저서)	논문명/저서명	게재지 (권, 쪽)	게재연도 (발표연도)	역할	ISSN	비고 (피인용 지수)
			yy			
			yy			

바. 지식재산권 출원·등록 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분 (특허/프로그램 등)	지식재산권명	국가명	출원·등록일	출원·등록번호/ 출원·등록자 수	비고

사. 그 밖의 대표적 실적(최근 5년간 5개 이내의 실적으로 작성하되, 연구개발과제 특성에 따라 선택적으로 작성이 가능합니다)

구분	실적명	내용요약	실적연도
			yy
			yy

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2)

(4) 참여연구자 및 연구지원인력

가. 참여연구자 현황

성명	국적	소속 기관	직위	국가 연구자 번호	학위 및 전공			학생연구자 학위 및 전공 (학생연구자인 경우만 기재)			담당역할	신규채용 구분 (해당 시 작성)	시간 선택제 근무 구분 (해당 시 작성)	참여연도				총 참여기간 (개월)
														1단계		n단계		
					최종 학위	전공	취득 년도	현 학위 과정	전공	입학 년도				1년	n년	1년	n년	

나. 연구지원인력 현황(직접비에서 인건비를 지급하는 경우에만 작성합니다)

성명	국적	소속 기관	직위	학위 및 전공			담당역할	신규채용 구분 (해당 시 작성)	시간 선택제 근무 구분 (해당 시 작성)	지원연도				총 지원기간 (개월)
										1단계		n단계		
				최종	전공	취득 년도				1년	n년	1년	n년	

				학위									

(5) 연구개발기관이 아닌 관계 기관(해당 시 작성합니다)

※ 연구개발비를 부담하나 사용하지 않는 기관(지방자치단체, 수혜기관 등) 또는 연구개발비를 사용하지 않으나 연구개발정보를 필요로 하는 기관에 한정하여 작성합니다.

가. 기관명: (역할:)

책임자	성명	국문		국적	
		영문			
	기관명			전화번호	
	부서			휴대전화	
	직위			전자우편	
실무 담당자	국문				
	영문				
	기관명			전화번호	
	부서			휴대전화	
	직위			전자우편	
주소			(우:)		

나. 기관명: (역할:)

책임자	성명	국문		국적	
		영문			
	기관명			전화번호	
	부서			휴대전화	
	직위			전자우편	
실무 담당자	국문				
	영문				
	기관명			전화번호	
	부서			휴대전화	
	직위			전자우편	
주소			(우:)		

2) 연구개발기관 연구개발 실적

(해당 시 작성하며, 작성 시에는 연구개발과제 특성에 따라 선택적으로 항목 적용이 가능합니다)

(1) 연구개발과제와 연관된 지식재산권 출원 및 등록 현황(최근 5년간의 실적을 기재합니다)

연구개발기관명 (소유권자)	지식재산권명	국가명	출원・등록번호 /출원・등록일

(2) 국가연구개발사업 주요 수행 실적(최근 5년간의 실적*을 기재합니다)

연구개발과제명	주관연구개발기관명 연구개발기관명 및 역할(주관/공동)	연구개발기간 (참여기간)	수행내용	중앙행정기관 (전문기관)	비고 (수행중/완료)
		yy.mm.dd~yy.mm.dd (yy.mm.dd ~yy.mm.dd)			
		yy.mm.dd~yy.mm.dd (yy.mm.dd ~yy.mm.dd)			

* 연구개발과제 종료 후 5년을 초과하더라도 (3) 국가연구개발사업 기술이전 실적 또는 (4) 국가연구개발사업 사업화 실적에 해당하는 연구개발과제는 기재해야 합니다.

(3) 국가연구개발사업 기술이전 실적(최근 5년간의 실적을 기재합니다)

(단위: 천원)

연구개발기관명	기술이전 유형	기술실시계약명	기술실시기관명	기술실시발생일	기술료	기술료 누적 징수액

(4) 국가연구개발사업 사업화 실적(최근 5년간의 실적을 기재합니다)

(단위: 천원, 달러)

연구개발기관명	사업화 방식1」	사업화 형태2」	지역3」	사업화명	내용	업체명	매출액		매출발생 연도	기술 수명
							국내	국외		

* 1」 기술이전 또는 자기실시 중 해당사항을 기재합니다.
* 2」 신제품 개발, 기존 제품 개선, 신공정 개발, 기존 공정 개선 등에서 해당하는 사항을 기재합니다.
* 3」 국내 또는 국외 중 해당사항을 기재합니다.
※ 기술이전 및 사업화 실적은 국가연구개발사업 조사ㆍ분석에 등록된 것이어야 합니다.

3) 연구시설ㆍ장비 보유현황(해당 시 작성합니다)

보유기관	연구시설ㆍ장비명	규격	수량	용도	활용시기	현물부담 반영여부 (해당 시 "○")

4) 연구개발기관 일반 현황(기업정보 데이터베이스와 연계가 가능합니다)

※ 비영리기관의 경우 순번 5부터 순번 15까지의 사항은 생략할 수 있습니다.

(단위: 천원, 백분율)

순번	구분 \ 기관명			
1	사업자등록번호			
2	법인등록번호			
3	대표자 성명/국적			

4	기관 유형 (대학, 정부출연연, 중소기업 등)					
5	최대 주주 성명/국적					
6	설립 연월일					
7	주생산 품목					
8	상시 종업원 수					
9	전년도 매출액					
10	매출액 대비 연구개발비 비율					
11	부채 비율 (최근 3년 간 결산 기준)		yyyy년			
			yyyy년			
			yyyy년			
12	유동 비율 (최근 3년 간 결산 기준)		yyyy년			
			yyyy년			
			yyyy년			
13	자본잠식 현황 (최근 3년 간 결산 기준)	자본 총계	yyyy년			
			yyyy년			
			yyyy년			
		자본금	yyyy년			
			yyyy년			
			yyyy년			
14	이자 보상 비율 (최근 3년 간 결산 기준)		yyyy년			
			yyyy년			
			yyyy년			
15	영업 이익 (최근 3년 간 결산 기준)		yyyy년			
			yyyy년			
			yyyy년			
16	연구개발기관의 연구개발과제 지원 담당자 (※ 대학의 경우 산학협력단의 연구개발과제 지원 담당을 말하며, 표지의 "실무담당자"와 다름)		성명			
			부서			
			직위			
			직장전화			
			휴대전화			
			전자우편			

		팩스			

210mm×297mm[(백상지(80g/m²) 또는 중질지(80g/m²)

2. 연구개발비 사용에 관한 계획

1) 연구개발비 지원・부담계획

(단위: 천원)

구분			정부지원 연구개발비	기관부담 연구개발비			그 외 기관 등의 지원금						합 계		
							지방자치단체			기타( )					
단계	연차	연구개발기관명 (기관역할[1])	현금	현금	현물	소계	현금	현물	소계	현금	현물	소계	현금	현물	합계
1	1														
	n														
	소계														
n	1														
	n														
	소계														
총계															

* 1」 주관연구개발기관, 공동연구개발기관 등 연구개발과제 내 해당 연구개발기관의 역할을 기재합니다.

2) 연구개발비 사용계획

(1) 연구개발기관별 사용계획

(단위: 천원)

연구개발기관명		연구개발비															연구개발비 외 지원금[5]	연구수당 계상기준 금액[6]
		직접비													간접비	합계		
		인건비	학생인건비		연구시설・장비비		연구재료비	위탁연구개발비	국제공동연구개발비	연구개발부담비	연구활동비	연구수당	소계					
			일반[1]	특례[2]	일반[3]	특례[4]												
	현금																	
	현물																	
	소계																	
	현금																	
	현물																	
	소계																	

총계	현금																
	현물																
	합계																

* 1」 국가연구개발사업 연구개발비 사용기준 제6장에 따른 학생인건비 사용에 관한 특례를 적용하지 않는 학생인건비를 기재합니다.

2」 국가연구개발사업 연구개발비 사용기준 제6장에 따른 학생인건비 사용에 관한 특례를 적용하는 학생인건비를 기재합니다.

3」 국가연구개발사업 연구개발비 사용기준 제7장에 따른 연구시설・장비비 사용에 관한 특례를 적용하지 않는 연구시설・장비비를 기재합니다.

4」 국가연구개발사업 연구개발비 사용기준 제7장에 따른 연구시설・장비비 사용에 관한 특례를 적용하는 연구시설・장비비를 기재합니다.

5」 국제기구, 외국의 정부・기관・단체 등이 지원・부담하는 금액이거나, 중앙행정기관(소속기관 포함)이 소관 업무를 위하여 직접 수행하는 사업의 금액으로 「국가연구개발혁신법」에 따른 연구개발비에 포함하지 않는 금액을 기재합니다.

6」 대학, 기업 등 참여연구자가 소속된 연구개발기관으로부터 연구개발과제와 별도로 인건비를 지급받는 연구개발기관에 한해 참여연구자들의 연구수당을 계상하기 위한 기준금액입니다. 해당 금액은 연구개발기관이 해당 연구개발과제의 연구개발기간 동안 참여연구자에게 지급하는 인건비를 같은 기간 동안 해당 참여연구자가 실제 해당 연구개발과제에 참여한 정도로 곱한 금액 중 해당 연구개발과제의 연구개발비에서 계상하지 아니한 금액을 기재합니다.

(2) 연차별 사용계획

(단위: 천원)

연차		연구개발비														연구개발비 외 지원금5」	연구수당 계상 기준금액6」
		직접비												간접비	합계		
		인건비	학생인건비		연구시설・장비비		연구재료비	위탁연구개발비	국제공동연구개발비	연구개발부담비	연구활동비	연구수당	소계				
			일반	특례	일반	특례											
1	현금																
	현물																
	소계																
n	현금																
	현물																
	소계																
총계	현금																
	현물																
	합계																

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2)

(3) 연구개발기관별-연차별 사용계획

가. 주관연구개발기관명:

(단위: 천원)

연차		연구개발비														연구개발비 외 지원금 5」	연구수당 계상기준 금액 6」
		직접비												간접비	합계		
		인건비	학생인건비		연구시설 · 장비비		연구재료비	위탁연구개발비	국제공동연구개발비	연구개발부담비	연구활동비	연구수당	소계				
			일반	특례	일반	특례											
1	현금																
	현물																
	소계																
n	현금																
	현물																
	소계																
총계	현금																
	현물																
	합계																

나. 공동연구개발기관명(해당 시 작성합니다):

(단위: 천원)

연차		연구개발비														연구개발비 외 지원금 5」	연구수당 계상기준 금액 6」
		직접비												간접비	합계		
		인건비	학생인건비		연구시설 · 장비비		연구재료비	위탁연구개발비	국제공동연구개발비	연구개발부담비	연구활동비	연구수당	소계				
			일반	특례	일반	특례											
1	현금																
	현물																
	소계																
n	현금																
	현물																
	소계																
총계	현금																
	현물																
	합계																

다. 위탁연구개발기관명(해당 시 작성합니다):

(단위: 천원)

연차		연구개발비															연구개발비 외 지원금[5]	연구수당 계상기준 금액[6]
		직접비													간접비	합계		
		인건비	학생인건비		연구시설・장비비		연구재료비	위탁연구개발비	국제공동연구개발비	연구개발부담비	연구활동비	연구수당	소계					
			일반	특례	일반	특례												
1	현금																	
	현물																	
	소계																	
n	현금																	
	현물																	
	소계																	
총계	현금																	
	현물																	
	합계																	

3) 연구시설・장비 구축・운영계획(해당 시 작성합니다)

(1) 연구시설・장비 구축계획(구축비용이 3천만원 이상인 경우에는 필수로 작성합니다)

(단위: 천원)

연구개발기관명	연구시설・장비명	현금/현물 구분	구축방식*	규격	수량	구축비용	구축기간	설치장소

* 개발, 구매, 임대, 용역 등 해당하는 사항을 기재합니다.

* **특히, 신규로 구축하는 3,000천만원 이상의 모든 연구시설장비 반드시 기재요망**

(2) 연구시설・장비 운영・활용계획

(단위: 천원)

연구개발기관명	연구시설명	기존/신규 구분	운영기간	비용			전담인력 수	활용계획	설치장소
				연간운영	과제반영	현금/현물			

				비용	비용	구분[1]			
			yy-yy						
			yy-yy						

* 1」협약기간 내 운영・활용하는 연구시설・장비에 소요되는 현금 또는 현물을 기재합니다.

3. 평가기준 및 평가방법

1) 성과지표 및 목표치

성과지표명 \ 단계		1단계(yy~yy)	n단계(yy~yy)	계	가중치(%)
전담기관 등록・기탁지표					
연구개발과제 특성 반영 지표					
계					100

2) 성능지표 및 측정방법

(1) 결과물의 성능지표

평가 항목 (주요성능[1])	단위	전체 항목에서 차지하는 비중[2] (%)	세계 최고수준 보유국/보유기관	연구개발 전 국내 수준	연구개발 목표치		목표 설정 근거
			성능수준	성능수준	1단계(yy~yy)	n단계(yy~yy)	

* 1」 정밀도, 인장강도, 내충격성, 작동전압, 응답시간 등 기술적 성능판단기준이 되는 것을 의미합니다.
* 2」 비중은 각 구성성능 사양의 최종목표에 대한 상대적 중요도를 말하며 합계는 100%이어야 합니다.

(2) 평가방법 및 평가환경

순번	평가항목 (성능지표)	평가방법	평가환경
1			
2			

< 별첨 자료 >

중앙행정기관 요구사항	별첨 자료
1.	1)
	2)
2.	1)
	2)

210mm×297mm[(백상지(80g/m^2) 또는 중질지(80g/m^2)

본문 2 작성 요령(작성 요령은 제출하지 않습니다)

1. 연구개발기관 현황

1) 연구책임자 등 현황

(1) 주관연구개발기관 연구책임자: 연구개발과제 연구책임자의 인적사항, 학력(최근 순으로 작성), 경력, 주요 연구개발 실적, 대표 논문/저서 실적, 지식재산권 출원·등록 실적을 기재합니다.

(2) 공동연구개발기관 책임자(해당 시 작성): 연구개발과제에 참여하는 공동연구개발기관의 수행내용을 총괄하는 연구자의 인적사항, 학력(최근 순으로 작성), 경력, 주요 연구개발 실적, 대표 논문/저서 실적, 지식재산권 출원·등록 실적을 기재합니다.

(3) 위탁연구개발기관 책임자(해당 시 작성): 연구개발과제에 참여하는 위탁연구개발기관의 수행내용을 총괄하는 연구자의 인적사항, 학력(최근 순으로 작성), 경력, 주요 연구개발 실적, 대표 논문/저서 실적, 지식재산권 출원·등록 실적을 기재합니다.

(4) 참여연구자 및 연구지원인력

가. 참여연구자 현황: 연구개발과제에 참여하는 연구자(이하 "참여연구자"라 한다)의 성명, 국적, 소속기관, 직위, 국가연구자번호, 학위 및 전공, 담당역할, 신규채용 구분(해당 시 작성), 시간 선택제 근무 구분(해당 시 작성), 참여연도, 총 참여기간을 기재합니다.

가) 신규채용 구분: 신규 전담연구인력인 경우 "신규(전담)", 정부지원연구개발비에 비례한 청년 신규채용인 경우 "신규(청년의무)", 연구개발기관 현금부담 감면을 위한 청년 신규채용'인 경우 "신규(청년추가)", 기타 신규채용인 경우 "신규(기타), 신규채용이 아닌 기존 인력의 경우 "기존"으로 기재합니다.

나) 시간선택제근무 구분: 시간선택제근무(육아부담으로 인한 경력단절 문제를 예방하기 위해 통상적인 근무 시간보다 짧은 '주당 15~35시간 범위에서 시간선택제로 근무)의 경우 "시간," 실습연구자(공동연구개발기관인 대학의 학사과정 중에 있는 학생으로서 방학기간 중 중소기업·중견기업이 주관연구개발기관인 연구개발과제에 참여하는 연구자)의 경우 "실습"으로 기재합니다.

다) 참여연도(지원 연도): 연구개발과제에 1개월이라도 참여 시 해당연도에 "○" 표시합니다.

나. 연구지원인력 현황(직접비에서 인건비를 지급하는 경우에만 작성): 연구개발과제를 지원함으로써 해당 연구개발과제의 직접비에서 인건비를 지급받는 연구지원인력의 성명, 국적, 소속기관, 직위, 학위 및 전공, 담당역할, 지원연도, 총 지원기간을 기재합니다.

(5) 연구개발기관이 아닌 관계 기관(해당 시 작성): 연구개발비를 부담하나 사용하지 않는 기관(지방자치단체, 수혜기관 등) 또는 연구개발비를 사용하지 아니하나 연구개발정보를 필요로 하는 기관에 한하여 작성합니다.

2) 연구개발기관 연구개발 실적(해당 시 작성, 작성 시 연구개발과제 특성에 따라 항목을 선택적으로 적용 가능)

(1) 연구개발과제와 연관된 지식재산권 출원 및 등록 현황(최근 5년간 실적): 연구개발과제와 연관된 지식재산권의 소유기관,

본문 2 작성 요령(작성 요령은 제출하지 않습니다)

해당 지식재산권명, 출원・등록 국가, 출원・등록번호, 출원・등록일을 기재합니다.

(2) 국가연구개발사업 주요 수행 실적(최근 5년간 실적): 국가연구개발사업의 연구개발과제를 수행한 실적을 기재합니다.

(3) 국가연구개발사업 기술이전 실적(최근 5년간 실적): 국가연구개발사업의 연구개발과제 수행에 따른 연구개발성과를 이전한 실적을 기재합니다.

(4) 국가연구개발사업 사업화 실적(최근 5년간 실적): 국가연구개발사업의 연구개발과제 수행에 따른 연구개발성과를 사업화한 실적을 기재합니다.

특허실적	○ 특허 실적 : 연구책임자의 최근 5년간(해당 과제 접수마감일 기준) 대표적 특허실적 총 5건 이내 작성 - 실적인정기준 : 특허는 등록된 것만 인정 - 특허명 및 등록 번호 : 특허 등록서에 명시된 공식 명칭 기재 - 출원자명 : 특허 출원자로 기재된 정보를 기재 - 발명자명 : 특허 발명자로 기재된 연구책임자 성명 기재 - 비고 : 특허 관련 비고자료(예시 : 삼극특허 등) 기재
논문실적	○ 논문 실적 : 연구책임자의 최근 5년간(해당 과제 접수마감일 기준) 대표적 논문실적 총 5건 이내 작성 - 실적인정기준 : 논문은 게재가 확정된 것(accepted, in press)만 인정 - 저널명은 반드시 정식명칭(full name)을 기재하도록 함(약어 사용 금지) - ISSN 번호는 ‘http://apps.webofknowledge.com’에서 논문 타이틀을 검색하거나 해당 저널 홈페이지에서 확인 가능(ISSN 예시 : ○○○○-○○○○) - ISSN이 없는 경우 ISBN을 기재하고, paper version을 우선적으로 기재 - 역할(제1,교신,공동): 논문에 참여한 연구책임자의 역할(제1, 교신, 공동 중 택1)을 기재 - 논문에 역할 구분이 명시되어 있지 않는 경우, 이메일 주소가 들어간 저자를 주저자(교신 또는 제1)로 기재
기타실적	○ 기타실적 : 주관연구책임자의 기술이전실적, 수상, 저서, 강연 등의 기타성과 총 5건 이내 작성 - 최근 5년간(해당 과제 접수마감일 기준) 논문/특허 이외의 연구실적(국제학회 초청강연, 국제학술지 편집위원 참여, 기술이전, 저서 등) 에 대하여 관련 내용 기술

3) 연구시설・장비 보유현황(해당 시 작성): 연구개발과제 수행에 활용할 연구시설・장비 보유 현황을 기재합니다.

4) 연구개발기관 일반현황: 기업정보 데이터베이스와 연계하여 작성 가능하며, 비영리기관의 경우에는 순번 5부터 순번 15까지는 생략하여 기재합니다.

본문 2 작성 요령(작성 요령은 제출하지 않습니다)

2. 연구개발비

1) 연구개발비 지원ㆍ부담계획: 정부가 지원하는 연구개발비와 연구개발기관이 부담하는 연구개발비 등을 현금과 현물로 구분하여 기재, 기관역할은 ‘주관’, ‘공동’, ‘위탁’ 중 선택하여 기재합니다.

2) 연구개발비 사용계획

(1) 연구개발기관별 사용계획: 연구개발기관별로 구분하여 연구개발비 항목별 총액을 기재합니다.

(2) 연차별 사용계획: 연차별로 구분하여 연구개발비 항목별 총액을 기재합니다.

(3) 연구개발기관별-연차별 사용계획: 연구개발기관별로 연차별로 구분하여 연구개발비 항목별 총액을 기재합니다.

3) 연구시설장비 구축ㆍ운영계획(해당 시 작성)

(1) 연구시설ㆍ장비 구축계획: 연구개발과제 수행에 활용할 연구시설ㆍ장비의 구축계획을 기재합니다.

(2) 연구시설 운영ㆍ활용계획: 연구개발과제 수행에 따라 구축될 연구시설의 활용계획을 기재합니다. 이 때 기존/신규 구분은 연구개발기간 시작 전에 구축이 완료된 경우 ‘기존’으로, 연구개발기간 중에 구축이 완료되는 경우 ‘신규’로 입력합니다.

3. 평가기준 및 평가방법(정성, 정량적 평가항목을 합하여 5개 내외로 반드시 설정해야함)

1) 성과지표 및 목표치: 영 별표 3에 따라 전담기관에 등록ㆍ기탁하는 연구개발성과와 그 밖에 연구개발과제의 특성에 따른 연구개발성과와 관련된 성과지표와 그 목표치를 기재합니다.

2) 성능지표 및 측정방법

(1) 결과물의 성능지표 : 연구개발과제 성격 및 분야별 특성을 고려하여 주요성능을 수치적으로 작성합니다.

(2) 평가방법 및 평가환경: 신뢰성이 전제되어야 하며, 공인기관 시험성적서 또는 확인서, 수요기업 평가 등을 활용하되, 부득이하게 자체평가인 경우 신뢰성을 입증할 수 있는 객관적 자료의 제시가 필요합니다.